

# Solutions

NCERT Class XII Chemistry · Chapter 1 · Solutions

*15 marks on the blueprint — the heaviest chapter in the paper. Theory you know; the marks live in the numericals. Every formula below is followed by the method that actually gets used in the exam.*

## Types of solution — *nine combinations, one table*

A solution is a homogeneous mixture of two or more components. Composition and properties are uniform throughout. Two components only = **binary solution**.

The component in the largest quantity is the **solvent** — and the solvent fixes the physical state of the solution. Everything else is **solute**.

| Solvent | Solute                 | Example                                                 |
|---------|------------------------|---------------------------------------------------------|
| Gas     | Gas                    | Air (O <sub>2</sub> in N <sub>2</sub> )                 |
| Liquid  | Chloroform in nitrogen |                                                         |
| Solid   | Camphor in nitrogen    |                                                         |
| Liquid  | Gas                    | O <sub>2</sub> in water; CO <sub>2</sub> in soft drinks |
| Liquid  | Ethanol in water       |                                                         |
| Solid   | Glucose in water       |                                                         |
| Solid   | Gas                    | H <sub>2</sub> in palladium                             |
| Liquid  | Amalgam (Na in Hg)     |                                                         |
| Solid   | Alloys — Cu in Au      |                                                         |

**Examiner asks:** one-markers naming the solute/solvent type for a given example. Camphor-in-nitrogen and hydrogen-in-palladium are the two people forget.

## Concentration terms — *seven of them — and which one the question wants*

The whole game is reading which term the question gives you and which it wants. Get that right and the arithmetic is trivial.

### Mass percentage (w/w)

$$\text{mass \%} = \frac{\text{mass of component}}{\text{mass of solution}} \times 100$$

"10% glucose by mass" means **10 g glucose in 100 g solution** — so 90 g water. Used for industrial chemicals.

### **Volume percentage (v/v)**

$$\text{volume \%} = \frac{\text{volume of component}}{\text{volume of solution}} \times 100$$

Used when both components are liquids. 35% v/v ethylene glycol is car antifreeze — it drops water's freezing point to 255.4 K.

### **Mass by volume (w/v)**

Mass of solute in 100 mL of solution. Medicine and pharmacy. **0.9% w/v NaCl = normal saline** — remember this one, it reappears in the osmosis questions.

### **Parts per million**

$$\text{ppm} = \frac{\text{parts of component}}{\text{total parts}} \times 10^6$$

For trace quantities — pollutants, dissolved oxygen in seawater.

### **Mole fraction (x)**

$$x_A = \frac{n_A}{n_A + n_B} \quad x_A + x_B = 1$$

No units. This is the one Raoult's and Henry's laws are written in.

### **Molarity (M)**

$$M = \frac{\text{moles of solute}}{\text{volume of solution in L}}$$

**Depends on temperature** — volume expands on heating, so molarity falls.

### **Molality (m)**

$$m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}}$$

**Independent of temperature** — mass doesn't change with T. This is why every colligative-property formula uses molality, not molarity.

**Trap:** molarity uses *volume of solution*; molality uses *mass of solvent*. Not solution. Losing the solute mass from the denominator is the single most common arithmetic slip in this chapter.

Worked · 2014 — molality and molarity from mass %

10% by mass glucose ( $M = 180 \text{ g/mol}$ ) in water, density  $1.2 \text{ g/mL}$ . Find molality and molarity.

**Set up:** 10% by mass  $\rightarrow$  10 g glucose in 100 g solution  $\rightarrow$  water = 90 g.

**Molality:**

$$m = \frac{10/180}{90/1000} = 0.62 \text{ mol kg}^{-1}$$

**Molarity:** need volume. From  $d = m/V$ :  $V = 100/1.2 = 83.3 \text{ mL} = 0.0833 \text{ L}$ .

$$M = \frac{10/180}{0.0833} = 0.67 \text{ mol L}^{-1}$$

**Examiner asks:** "given mass % and density, find molality and molarity" is a standing 3-marker. The density is only ever there to get you the volume.

### **Solubility — *what dissolves in what, and what temperature does to it***

**Solubility** = maximum amount of solute that dissolves in a specified amount of solvent at a specified temperature.

#### ***Solid in liquid***

**Like dissolves like.** Polar solutes (NaCl, sugar) dissolve in polar solvents (water); non-polar solutes (naphthalene, anthracene) dissolve in non-polar solvents (benzene).

Add solute and two opposing processes run: **dissolution** and **crystallisation**. When their rates become equal, dynamic equilibrium is reached, concentration goes constant, and the solution is **saturated**.

- **Temperature:** if dissolution is endothermic ( $\Delta H > 0$ ), solubility rises with T. If exothermic ( $\Delta H < 0$ ), solubility falls with T. (Le Chatelier.)
- **Pressure:** no significant effect — solids and liquids are incompressible.

#### ***Gas in liquid***

Solubility **increases with pressure**: more gas particles per unit volume above the solution → more strike the surface → more dissolve. Solubility **decreases with temperature** (dissolution of a gas is exothermic).

**Examiner asks:** "aquatic species are more comfortable in cold water" (2019) — because oxygen's solubility falls as water warms. Straight 1-2 marker.

### **Henry's law — *the gas-solubility law and its three applications***

At constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of that gas above the liquid. Equivalently:

$$p = K_H x$$

where  $p$  is partial pressure of the gas,  $x$  its mole fraction in solution, and  $K_H$  the Henry's law constant.

**Read  $K_H$  backwards:** higher  $K_H$  means *lower* solubility.  $K_H$  rises with temperature — which is exactly why gases get less soluble in warm water.

Worked · 2020, repeated 2026 — solubility from Henry's law

Solubility of  $\text{CO}_2$  in water at 298 K under 760 mm Hg.  $K_H = 1.25 \times 10^6 \text{ mm Hg}$ .

Solubility here means mole fraction. From  $p = K_H x$ :

$$x = \frac{p}{K_H} = \frac{760}{1.25 \times 10^6} = 6.08 \times 10^{-4}$$

Mole fraction has no unit.

### ***The three applications***

- **Carbonated drinks** — bottles sealed under high pressure to force CO<sub>2</sub> into solution.
- **Deep-sea diving / bends** — high pressure underwater dissolves N<sub>2</sub> into blood; on ascent it comes out as bubbles that block capillaries. Divers' tanks use helium-diluted air (11.7% He, 56.2% N<sub>2</sub>, 32.1% O<sub>2</sub>) because helium's high  $K_H$  means it barely dissolves.
- **Anoxia at altitude** — low partial pressure of O<sub>2</sub> → low oxygen in blood and tissue → climbers feel weak and can't think clearly.

**Examiner asks:** "state Henry's law + calculate mole fraction" as one 2-3 marker, or the bends/anoxia reasoning as a standalone. Both are near-certain appearances.

## **Raoult's law and vapour pressure — *volatile pairs, then non-volatile solutes***

### ***Two volatile liquids***

Partial vapour pressure of each component is proportional to its mole fraction *in the liquid*:

$$p_1 = p_1^\circ x_1 \quad p_2 = p_2^\circ x_2$$

$$p_{\text{total}} = p_1^\circ x_1 + p_2^\circ x_2$$

For the mole fraction in the **vapour phase**, use pressures instead of moles:

$$y_1 = \frac{p_1}{p_{\text{total}}}$$

Worked · 2023 — total vapour pressure of an ideal solution

$p_X^\circ = 120$  mm Hg,  $p_Y^\circ = 160$  mm Hg, equal moles of X and Y mixed, ideal solution. Find  $p_{\text{total}}$ .

Equal moles →  $x_X = x_Y = 0.5$ .

$$p_{\text{total}} = (120)(0.5) + (160)(0.5) = 60 + 80 = 140 \text{ mm Hg}$$

### ***Non-volatile solute in a volatile solvent***

Only the solvent contributes to vapour pressure, so  $p_{\text{solution}} = p_1^\circ x_1$ . Adding a non-volatile solute always *lowers* vapour pressure — solute particles occupy surface area that solvent molecules would otherwise escape from.

### ***Raoult's law as a special case of Henry's law***

Henry's law is  $p = K_H x$ ; Raoult's is  $p = p^\circ x$ . When the solute-solvent interaction happens to equal the solute-solute interaction,  $K_H$  becomes equal to  $p^\circ$  and the two laws coincide.

## Ideal and non-ideal solutions — *two deviations, with the examples that prove them*

### Ideal solution

Obeys Raoult's law over the entire concentration range, and:

$$\Delta H_{\text{mix}} = 0 \quad \Delta V_{\text{mix}} = 0$$

Happens when A-A, B-B and A-B interactions are all nearly equal. Examples: n-hexane + n-heptane, bromoethane + chloroethane, benzene + toluene.

### Positive deviation

A-B interaction **weaker** than A-A and B-B. Molecules escape more easily → vapour pressure *higher* than Raoult predicts.

$$p_A > p_A^\circ x_A, \quad \Delta H_{\text{mix}} > 0, \quad \Delta V_{\text{mix}} > 0$$

Examples: ethanol + acetone (ethanol's own H-bonding is broken up), carbon disulphide + acetone.

### Negative deviation

A-B interaction **stronger** than A-A and B-B. Molecules held tighter → vapour pressure *lower* than predicted.

$$p_A < p_A^\circ x_A, \quad \Delta H_{\text{mix}} < 0, \quad \Delta V_{\text{mix}} < 0$$

Examples: phenol + aniline, chloroform + acetone (they form an H-bond between them), nitric acid + water.

**Trap:** chloroform + acetone is negative deviation because a *new* hydrogen bond forms *between* the two. Ethanol + acetone is positive because an *existing* hydrogen bond is broken. Both are asked; the reasoning direction is what earns the mark.

### Azeotropes

Binary mixtures with the **same composition in liquid and vapour phase**, boiling at constant temperature — so they cannot be separated by fractional distillation.

- **Minimum boiling azeotrope** ← large *positive* deviation. Ethanol + water, 95% ethanol by volume.
- **Maximum boiling azeotrope** ← large *negative* deviation. Nitric acid + water, 68% HNO<sub>3</sub> by mass, boils at 393.5 K.

**Examiner asks:** "define azeotrope + which type from negative deviation, with example" — a 2-3 marker that has run repeatedly.

## Colligative properties — *four of them, four formulas, one idea*

Properties that depend on the **number** of solute particles, not their identity. That one sentence is the reason every question in this section works.

### 1 • Relative lowering of vapour pressure

$$\frac{p_1^\circ - p_1}{p_1^\circ} = x_2 = \frac{n_2}{n_1 + n_2}$$

For a dilute solution  $n_1 \gg n_2$ , so this simplifies to the working form:

$$\frac{p_1^\circ - p_1}{p_1^\circ} = \frac{w_2/M_2}{w_1/M_1}$$

### 2 • Elevation of boiling point

$$\Delta T_b = K_b m = \frac{K_b w_2 \times 1000}{M_2 w_1}$$

$K_b$  = molal elevation constant (ebullioscopic constant), units K kg mol<sup>-1</sup>. For water,  $K_b = 0.52$ .

### 3 • Depression of freezing point

$$\Delta T_f = K_f m = \frac{K_f w_2 \times 1000}{M_2 w_1}$$

$K_f$  = molal depression constant (cryoscopic constant). For water,  $K_f = 1.86$ , freezing point 273 K.

Both constants depend only on the solvent, and can themselves be calculated:

$$K_f = \frac{R M_1 T_f^2}{1000 \Delta H_{\text{fus}}} \quad K_b = \frac{R M_1 T_b^2}{1000 \Delta H_{\text{vap}}}$$

### 4 • Osmotic pressure

**Osmosis:** solvent flows through a semi-permeable membrane from pure solvent into solution.

**Osmotic pressure** ( $\pi$ ) is the extra pressure applied on the solution side that just stops that flow.

$$\pi = CRT = \frac{n_2 RT}{V} = \frac{w_2 RT}{M_2 V}$$

**Why it's preferred for macromolecules** (proteins, polymers): works at room temperature, so heat-unstable biomolecules survive; uses molarity, which is easy to measure; and its magnitude is large even for dilute solutions, which is all you get from a poorly soluble polymer.

**Trap:**  $\pi$  uses *molarity* and volume in litres.  $\Delta T_b$  and  $\Delta T_f$  use *molality* and solvent mass in kg. Mixing these up is the standard way to lose an otherwise-correct 3-marker.

### Isotonic, hypertonic, hypotonic

- **Isotonic** — same osmotic pressure, no net osmosis. Blood cells are isotonic with 0.9% w/v NaCl.

- **Hypertonic** — higher concentration outside; water leaves the cell; cell **shrinks**.
- **Hypotonic** — lower concentration outside; water enters; cell **swells** and may burst.

**Reverse osmosis:** apply pressure *greater than*  $\pi$  on the solution side and solvent flows backwards, out of the solution. Used to desalinate seawater, with a cellulose acetate membrane that passes water but not ions.

## **Abnormal molar mass and van't Hoff factor — *when the measured mass is wrong, and by how much***

Colligative properties give a molar mass. When the solute associates or dissociates, that measured value differs from the true one — an **abnormal molar mass**.

### **Association**

Molecules combine → fewer particles → colligative property falls → measured molar mass comes out **too high**. Ethanoic acid in benzene dimerises through hydrogen bonds: measured 120 g/mol against a true 60.

### **Dissociation**

Electrolyte splits into ions → more particles → colligative property rises → measured molar mass comes out **too low**. KCl gives 37.25 g/mol against a true 74.5.

### **van't Hoff factor**

$$i = \frac{\text{normal molar mass}}{\text{abnormal molar mass}} = \frac{\text{observed colligative property}}{\text{calculated colligative property}}$$

Equivalently, particles after dissociation/association ÷ particles before.

- **Dissociation** →  $i > 1$ . KCl: 2.  $\text{MgSO}_4$ : 2.  $\text{K}_2\text{SO}_4$ : 3.  $\text{CaCl}_2$ : 3.
- **Association** →  $i < 1$ . Complete dimerisation gives  $i = 0.5$ .
- **Neither** →  $i = 1$ . Glucose, urea, sucrose.

Every colligative formula then carries  $i$ :

$$\frac{p_1^\circ - p_1}{p_1^\circ} = i x_2 \quad \Delta T_b = i K_b m \quad \Delta T_f = i K_f m \quad \pi = \frac{i n_2 R T}{V}$$

### **Partial dissociation or association**

$$\text{dissociation: } i = 1 + (n - 1)\alpha \quad \text{association: } i = 1 + \left(\frac{1}{n} - 1\right)\beta$$

$\alpha$  = degree of dissociation,  $\beta$  = degree of association,  $n$  = particles produced or combined.

**Trap:** a degree of dissociation cannot exceed 1. If your  $\alpha$  comes out at 1.056, the intended answer is "essentially complete dissociation" — say so rather than writing an impossible number.

## Numerical patterns, collected — every calculation type in this chapter, one model each

Six patterns cover essentially every numerical this chapter sets. Method first, then one worked model.

### A · Concentration conversion — 3 marks

*Recognise it: gives a mass %, asks for molality/molarity, and hands you a density you'd otherwise have no use for.*

1. Take 100 g of solution. Solute mass = the percentage; solvent mass = 100 – that.
2. Moles of solute =  $w_2/M_2$ .
3. Molality: divide by solvent mass in kg.
4. Molarity: get volume from  $V = \text{mass}/d$ , convert to litres, divide.

### B · Henry's law — 2 marks

*Recognise it: a  $K_H$  value appears.*

1.  $x = p/K_H$ . Answer is a mole fraction, unitless. That's the whole question.

### C · Total vapour pressure — 3 marks

*Recognise it: two  $p^\circ$  values for two volatile liquids.*

1. Find mole fractions in the liquid (equal moles  $\rightarrow$  0.5 each).
2.  $p_{\text{total}} = p_1^\circ x_1 + p_2^\circ x_2$ .
3. If it asks for vapour-phase composition:  $y_1 = p_1/p_{\text{total}}$ .

### D · $\Delta T_b / \Delta T_f \rightarrow$ molar mass or temperature — 3 marks

*Recognise it: a  $K_b$  or  $K_f$  is given. This is the most-set numerical in the chapter.*

1. Decide whether  $i$  is needed — is the solute an electrolyte, or does it dimerise?
2.  $\Delta T = iK \frac{w_2 \times 1000}{M_2 w_1}$ . Rearrange for whatever's missing.
3. If asked for the actual boiling/freezing point, *add*  $\Delta T_b$  to  $T_b^\circ$  or *subtract*  $\Delta T_f$  from  $T_f^\circ$ . Don't stop at  $\Delta T$ .

Worked · 2018 — freezing point of a glucose solution

60 g glucose ( $M = 180$ ) in 250 g water,  $K_f = 1.86$ . Find the freezing point.

Glucose doesn't dissociate, so  $i = 1$ .

$$\Delta T_f = \frac{1.86 \times 60 \times 1000}{180 \times 250} = 2.48 \text{ K}$$

$$T_f = 273 - 2.48 = 270.52 \text{ K}$$

### E · Osmotic pressure — 3 marks

*Recognise it:  $R$  is given, or the word isotonic appears.*

1.  $\pi = \frac{w_2 RT}{M_2 V}$ , volume in **litres**.

2. For isotonic pairs, set the two concentrations equal:  $\frac{w_1}{M_1} = \frac{w_2}{M_2}$  (volumes cancel) and solve for the unknown  $M$ .

F · van't Hoff factor  $\rightarrow \alpha$  or  $\beta$  — 3-5 marks

*Recognise it: asks for degree of dissociation/association, or "predict the nature of the solute".*

1. Get  $i$  from the colligative-property equation with  $i$  left in as the unknown.
2.  $i > 1 \rightarrow$  dissociation, use  $i = 1 + (n - 1)\alpha$ .  $i < 1 \rightarrow$  association, use the association form.
3. Read  $n$  off the actual dissociation equation — write it out, don't guess.

Worked · 2026 — degree of dissociation of fluoroacetic acid

19.5 g FCH<sub>2</sub>COOH ( $M = 78$ ) in 500 g water gives  $\Delta T_f = 1$  K.  $K_f = 1.86$ .

$$1 = i \times \frac{1.86 \times 19.5 \times 1000}{78 \times 500} \Rightarrow i = 1.075$$

FCH<sub>2</sub>COOH  $\longrightarrow$  FCH<sub>2</sub>COO<sup>−</sup> + H<sup>+</sup>, so  $n = 2$ :

$$1.075 = 1 + (2 - 1)\alpha \Rightarrow \alpha = 0.075 = 7.5\%$$

## Past year questions · question types — ranked by how often they turn up

*From the last 10 years of boards. 22 questions are captured below with their years; the source video states it covers 25, so treat this as the bulk of the set rather than all of it.*

1 · Colligative-property numerical — 3 marks

*Recognise it:  $K_b$ ,  $K_f$  or  $R$  is supplied.*

1. Pick the right formula from the constant given.
2. Decide on  $i$ .
3. Convert to the final temperature if asked.

**Trap:** stopping at  $\Delta T$  when the question asked for the boiling or freezing point.

2 · Deviation from Raoult's law, with reason — 2-3 marks

*Recognise it: names a specific liquid pair and asks "what type of deviation, why".*

1. Compare A-B interaction against A-A and B-B.
2. Stronger  $\rightarrow$  negative, vapour pressure falls. Weaker  $\rightarrow$  positive, vapour pressure rises.
3. Name the actual interaction — which H-bond forms or breaks.

**Trap:** answering "negative deviation" without saying *which* bond forms between the two. The reason carries the mark.

3 · van't Hoff factor / degree of dissociation — 3-5 marks

*Recognise it: asks for  $i$ ,  $\alpha$ ,  $\beta$ , or "predict association or dissociation".*

1. Solve the colligative equation for  $i$ .
2. Compare against 1 to decide association vs dissociation.
3. Apply the matching  $\alpha/\beta$  relation.

**Trap:** using  $n = 2$  automatically.  $\text{CaCl}_2$  gives 3 ions,  $\text{K}_2\text{SO}_4$  gives 3.

4 · Henry's law statement + calculation — 2-3 marks

*Recognise it:  $K_H$  appears, often paired with "state Henry's law".*

1. State the law in mole-fraction form.
2.  $x = p/K_H$ .
3. If asked for applications, give carbonated drinks and the bends.

5 · Osmosis reasoning (cells, meat, isotonic) — 2-3 marks

*Recognise it: blood cells, salting meat, or a % NaCl solution.*

1. Compare concentration inside vs outside.
2. Water moves toward the higher concentration.
3. State the consequence: shrink, swell, or preserved.

6 · Definitions and two-difference questions — 2 marks

*Recognise it: "define...", "write two differences between..."*

1. Give the definition in the book's own words.
2. For differences, answer in pairs — ideal vs non-ideal on both  $\Delta H_{\text{mix}}$  and  $\Delta V_{\text{mix}}$ .

### **Past year questions · mark slots — *what each type is worth, and the time that buys***

| Question type             | Slot                           | Time    |
|---------------------------|--------------------------------|---------|
| Colligative numerical     | 3-marker                       | 4-5 min |
| Deviation + reason        | 2-3 marker                     | 3 min   |
| van't Hoff / $\alpha$     | 3-marker, sometimes inside a 5 | 5 min   |
| Henry's law               | 2-3 marker                     | 3 min   |
| Osmosis reasoning         | 2-3 marker                     | 3 min   |
| Definitions / differences | 2-marker                       | 2 min   |
| Concentration conversion  | 3-marker                       | 4 min   |

*Slots marked from the question shapes in the paper — the video groups them as short-answer type I (2 marks) and type II (3 marks).*

### **Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter***

These are the reason the PYQ video is worth watching. Each has run at least twice.

2020 Q3 · 2026 Q1

Henry's law statement plus mole fraction of  $\text{CO}_2$  at 760 mm Hg — the identical numbers reappeared six years apart.

2019 Q11(a) · lecture PYQ

Aquatic species more comfortable in cold water — gas solubility falls as temperature rises.

2019 Q11(b) · 2019 Q12

Anoxia at high altitude, and chloroform + acetone negative deviation. Both recur as reason-type one/two markers.

2018 Q10 · 2026 Q18

Freezing-point depression with a given  $K_f$  — glucose in water, then fluoroacetic acid in water with a degree of dissociation on top.

2023 Q5 · lecture PYQ

Total vapour pressure of an ideal solution from two  $p^\circ$  values and equal moles. Same 120/160 mm Hg numbers in both.

2016 Q7 · 2025 Q6

van't Hoff factor: definition, and why osmotic pressure is preferred for macromolecular molar masses.

2017 Q13 · 2020 Q2

Blood cells in 1.2% and 0.4% NaCl, and why 0.1 M KCl boils higher than 0.1 M glucose. Both test "colligative = number of particles".

### **Past year questions · cold practice — *answers only* — *work them before you look***

One model per pattern is above. These are the same patterns with different numbers: work each one cold, then check.

#### ***Colligative numericals***

- 2026 Q14 — 8 g non-volatile solute in 100 g diethyl ether, bp rises  $35.60 \rightarrow 36.86$  °C,  $K_b = 2.02$ . Find  $M_2$ . 128.25 g/mol
- 2017 — 10 g  $\text{CaCl}_2$  ( $M = 111$ ) in 200 g water,  $K_b = 0.52$ . Find  $\Delta T_b$ . 0.702 K (remember  $i = 3$ )
- 2026 Q8 — 1 molal trichloroacetic acid boils at 100.18 °C,  $K_b = 0.512$ . Find  $i$ . 0.35
- 2021 Q15 — 2.56 g sulphur in 100 g  $\text{CS}_2$ ,  $\Delta T_f = 1.62$  K,  $K_f = 4.9$ ,  $M(\text{S}_8) = 256$ . Find % association.  $i = 0.846$ ; 15.4%

#### ***Vapour pressure***

- 2026 Q19 — 61 g benzoic acid in 500 g benzene,  $p^\circ = 51.2$  mm Hg, complete dimerisation. Find  $p_s$ . 50.42 mm Hg (use  $i = 0.5$ )
- Lecture PYQ — pure water  $p^\circ = 32$  mm Hg at 308 K falls to 31.84 with 10 g solute in 200 g water. Find  $M_2$ . 180 g/mol

#### ***Osmotic pressure***

- 2020 Q3 — 5% urea solution ( $M = 60$ ) at 300 K,  $R = 0.0821$ . Find  $\pi$ . 20.51 atm
- 2019 Q17 — 6% glucose ( $M = 180$ ) isotonic with 2.5% of an unknown. Find the unknown's  $M$ . 75 g/mol

**Freezing point, two-stage**

- 2021 Q21 — 5% cane sugar ( $M = 342$ ) freezes at 271 K. Find the freezing point of 5% glucose ( $M = 180$ ). 269.2 K — find  $K_f$  from the sugar first

Built from Sourabh Raina's Solutions one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 1 (Rationalised 2022-23). Constants verified: water  $K_b = 0.52$ ,  $K_f = 1.86 \text{ K kg mol}^{-1}$ .

# Electrochemistry

NCERT Class XII Chemistry · Chapter 2 · Electrochemistry

*14 marks — second heaviest on the blueprint. Two engines run this chapter: the Nernst equation and Faraday's laws. Nearly every numerical is one of those two wearing a different hat.*

## The two cell types — *the distinction every question hangs off*

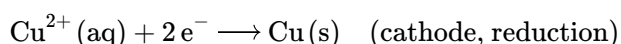
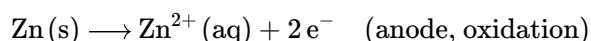
|            | Electrochemical (galvanic) | Electrolytic          |
|------------|----------------------------|-----------------------|
| Does       | Chemical → electrical      | Electrical → chemical |
| Reaction   | Spontaneous                | Non-spontaneous       |
| $\Delta G$ | Negative                   | Positive              |
| Anode      | Negative                   | Positive              |
| Cathode    | Positive                   | Negative              |

Constant across both: **oxidation at the anode, reduction at the cathode**. Only the charge signs flip.

**Examiner asks:** "two points of difference" (2020) is a standing 2-marker. Give the  $\Delta G$  and the anode-sign pair — they're the two that can't be waffled.

## Daniell cell and cell notation — *how to write a cell down, and why the salt bridge is there*

Zinc rod in 1 M  $\text{ZnSO}_4$ , copper rod in 1 M  $\text{CuSO}_4$ , joined by a salt bridge and an external wire. EMF = **1.1 V** at unit concentration.



The zinc rod thins; the copper rod thickens.

### Cell notation

Anode on the left, cathode on the right. Single line = phase boundary, double line = salt bridge:



### What the salt bridge actually does

It's a U-tube of electrolyte ( $\text{KCl}$ ,  $\text{KNO}_3$ ,  $\text{NH}_4\text{Cl}$ ) set in agar or gelatin. Without it,  $\text{Zn}^{2+}$  builds up in the anode compartment and  $\text{SO}_4^{2-}$  in the cathode compartment; the charge separation stalls the cell. The bridge's ions migrate in to neutralise both halves and complete the circuit.

**Examiner asks:** "why is a salt bridge necessary" (2026) — answer with both jobs: maintains electrical neutrality *and* completes the circuit.

## Electrode potential and SHE — *why everything is written as reduction potential*

Dip a metal in a solution of its own ions and charge separates until equilibrium. That potential difference between metal and solution is the **electrode potential**.

By IUPAC convention **every electrode potential is quoted as a reduction potential** — because you can't subtract an oxidation potential from a reduction potential meaningfully. Oxidation and reduction potentials of the same electrode are numerically equal and opposite in sign.

**Standard electrode potential ( $E^\circ$ ):** ion concentration 1 M, 298 K, and 1 bar for any gas.

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$$

Also written  $E_{\text{right}} - E_{\text{left}}$ , since cathode sits on the right in cell notation.  $E_{\text{cell}}$  for a working cell is always positive.

### Standard hydrogen electrode

A single half-cell's absolute potential can't be measured — you need a second electrode to get a reading at all. So SHE is defined as the reference with  $E^\circ = 0.00$  V exactly. It's a platinum foil coated with platinum black,  $\text{H}_2$  at 1 bar,  $\text{H}^+$  at 1 M.

Connect zinc to SHE: electrons flow zinc  $\rightarrow$  SHE, so zinc is the anode, reading 0.76 V. Then  $0.76 = 0 - E_{\text{Zn}^{2+}/\text{Zn}}^\circ$ , giving  $E_{\text{Zn}^{2+}/\text{Zn}}^\circ = -0.76$  V. Copper against SHE reads 0.34 V with electrons flowing the other way, giving +0.34 V.

## Electrochemical series — *reading a table of $E^\circ$ values*

Arrange standard electrode potentials in order and the series tells you three things at a glance.

- **High  $E^\circ$**   $\rightarrow$  reduces easily  $\rightarrow$  **strong oxidising agent**. Fluorine, +2.87 V, is the strongest.
- **Low (negative)  $E^\circ$**   $\rightarrow$  oxidises easily  $\rightarrow$  **strong reducing agent**. Lithium,  $-3.05$  V, is the strongest.
- **Feasibility:** the species actually being reduced must have the higher  $E^\circ$ . If a proposed reaction has it backwards, the reaction isn't feasible.

Worked · 2023 —  $E^\circ$  of a cell

$E_{\text{Ag}^+/\text{Ag}}^\circ = 0.80$  V,  $E_{\text{Fe}^{2+}/\text{Fe}}^\circ = -0.44$  V. Find  $E_{\text{cell}}^\circ$ .

Higher  $E^\circ$  becomes the cathode, so silver is cathode, iron is anode.

$$E_{\text{cell}}^\circ = 0.80 - (-0.44) = 1.24 \text{ V}$$

**Examiner asks:** "which of A and B liberates  $\text{H}_2$  from dilute  $\text{H}_2\text{SO}_4$ " (2026) — the one with the more negative  $E^\circ$ , because it oxidises more readily.

## Nernst equation — *the workhorse — for when concentrations aren't 1 M*

Standard potentials assume 1 M. Change the concentration and the potential changes. Nernst gives the relation:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \log Q$$

At 298 K, with  $R = 8.314$  and  $F = 96500$ , that collapses to the form you'll actually use:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{products}]}{[\text{reactants}]}$$

$n$  = electrons exchanged. **Solids and pure liquids are omitted** — their concentration is constant, taken as unity.

**Trap:**  $n$  is the electrons exchanged *after balancing*, not the charge on one ion. And stoichiometric coefficients become powers inside the log. Both are where marks vanish.

Worked · 2019 — EMF with unequal concentrations

$\text{Al} \mid \text{Al}^{3+} (0.001 \text{ M}) \parallel \text{Ni}^{2+} (0.1 \text{ M}) \mid \text{Ni}$  .  $E_{\text{Ni}^{2+}/\text{Ni}}^{\circ} = -0.25 \text{ V}$ ,  $E_{\text{Al}^{3+}/\text{Al}}^{\circ} = -1.66 \text{ V}$ .

Balance: Al loses  $3e^{-}$ ,  $\text{Ni}^{2+}$  gains  $2e^{-} \rightarrow$  multiply by 2 and 3  $\rightarrow n = 6$ .

$$E_{\text{cell}}^{\circ} = -0.25 - (-1.66) = 1.41 \text{ V}$$

$$E_{\text{cell}} = 1.41 - \frac{0.0591}{6} \log \frac{(10^{-3})^2}{(10^{-1})^3} = 1.41 - \frac{0.0591}{6} \log(10^{-3})$$

$$= 1.41 + \frac{0.0591 \times 3}{6} = 1.41 + 0.0295 = 1.44 \text{ V}$$

### Nernst at equilibrium $\rightarrow$ equilibrium constant

At equilibrium  $E_{\text{cell}} = 0$  and  $Q = K_c$ , so:

$$E_{\text{cell}}^{\circ} = \frac{0.0591}{n} \log K_c$$

**Examiner asks:** "why is  $K_c$  related to  $E_{\text{cell}}^{\circ}$  and not  $E_{\text{cell}}$ ?" (2026) — because at equilibrium  $E_{\text{cell}}$  is zero and generates no current, while  $E^{\circ}$  stays constant.

### Gibbs energy and cell EMF — one equation, two exam uses

$$\Delta_r G = -nFE_{\text{cell}} \quad \Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

The electrical work a cell does equals the fall in Gibbs energy.

**The intensive/extensive point** the examiner likes:  $E_{\text{cell}}$  is **intensive** — multiply the equation by 2 and it doesn't change.  $\Delta G$  is **extensive** — it depends on  $n$ , so doubling the equation doubles it.

Worked · 2025 —  $\Delta G^{\circ}$  and  $\log K_c$

$2 \text{ Cr} + 3 \text{ Cd}^{2+} \rightarrow 2 \text{ Cr}^{3+} + 3 \text{ Cd}$  .  $E_{\text{Cd}^{2+}/\text{Cd}}^{\circ} = -0.40 \text{ V}$ ,  $E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} = -0.74 \text{ V}$ ,  $F = 96500$ .

Cd is reduced → cathode.  $E_{\text{cell}}^{\circ} = -0.40 - (-0.74) = 0.34 \text{ V}$ . Cr loses  $3e^{-} \times 2 = 6$ , so  $n = 6$ .

$$\Delta_r G^{\circ} = -6 \times 96500 \times 0.34 = -196860 \text{ J mol}^{-1} = -196.86 \text{ kJ mol}^{-1}$$

## Conductance, conductivity, molar conductivity — *four quantities that are easy to confuse*

| Quantity           | Symbol      | Relation                                   | Unit                             |
|--------------------|-------------|--------------------------------------------|----------------------------------|
| Resistance         | $R$         | $R = \rho \frac{l}{A}$                     | $\Omega$                         |
| Resistivity        | $\rho$      | $\rho = R \frac{A}{l}$                     | $\Omega \text{ cm}$              |
| Conductance        | $G$         | $G = 1/R$                                  | S (siemens)                      |
| Conductivity       | $\kappa$    | $\kappa = 1/\rho$                          | $\text{S cm}^{-1}$               |
| Molar conductivity | $\Lambda_m$ | $\Lambda_m = \frac{\kappa \times 1000}{M}$ | $\text{S cm}^2 \text{ mol}^{-1}$ |

### Cell constant

$l/A$  is fixed for a given conductivity cell, so it's called the **cell constant**  $G^*$ :

$$G^* = \frac{l}{A} = \kappa \times R$$

Measuring  $l$  and  $A$  directly is unreliable, so  $G^*$  is found by filling the cell with a solution of known conductivity (usually KCl) and measuring resistance. Once known, the same cell gives  $\kappa$  for any solution:  $\kappa = G^*/R$ .

*Resistance is measured on a Wheatstone bridge with an AC source (DC would electrolyse the solution and change its composition) and a purpose-built conductivity cell (a solution can't be wired into the bridge directly).*

Worked · 2024 — resistivity, conductivity, molar conductivity

0.05 M NaOH, cell constant  $50 \text{ cm}^{-1}$ , resistance  $4.5 \times 10^3 \Omega$ .

$$\kappa = \frac{G^*}{R} = \frac{50}{4.5 \times 10^3} = 0.011 \text{ S cm}^{-1}$$

$$\rho = \frac{1}{\kappa} = 90 \Omega \text{ cm}$$

$$\Lambda_m = \frac{0.011 \times 1000}{0.05} = 220 \text{ S cm}^2 \text{ mol}^{-1}$$

### Variation with dilution — the two go opposite ways

- **Conductivity falls** on dilution: fewer ions per  $\text{cm}^3$ .
- **Molar conductivity rises** on dilution: the volume holding one mole of electrolyte increases, and interionic attraction weakens.

**Trap:** these two moving in opposite directions is itself an exam question. Conductivity is per unit *volume*; molar conductivity is per *mole*.

### **Strong vs weak electrolyte on the graph**

Plot  $\Lambda_m$  against  $\sqrt{c}$ :

- **Strong electrolyte** — starts high, rises gently and **linearly**. Extrapolate to zero concentration to read  $\Lambda_m^\circ$  straight off the graph.
- **Weak electrolyte** — starts low, then shoots up steeply near zero concentration, running almost **parallel to the y-axis**. Cannot be extrapolated; you must use Kohlrausch's law instead.

### **Kohlrausch's law — and its three applications**

**Kohlrausch law of independent migration of ions:** the limiting molar conductivity of an electrolyte is the sum of the individual contributions of its cation and anion.

$$\Lambda_m^\circ(\text{NaCl}) = \lambda_{\text{Na}^+}^\circ + \lambda_{\text{Cl}^-}^\circ$$

$$\Lambda_m^\circ(\text{BaCl}_2) = \lambda_{\text{Ba}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ$$

**Trap:** the stoichiometric multiplier.  $\text{Al}_2(\text{SO}_4)_3$  needs  $2\lambda_{\text{Al}^{3+}}^\circ + 3\lambda_{\text{SO}_4^{2-}}^\circ$ . Forgetting the 2 and 3 is the whole mistake.

#### **1 • $\Lambda^\circ$ of a weak electrolyte**

Can't be read off a graph, so build it from strong electrolytes that share ions:

$$\Lambda^\circ(\text{CH}_3\text{COOH}) = \Lambda^\circ(\text{CH}_3\text{COONa}) + \Lambda^\circ(\text{HCl}) - \Lambda^\circ(\text{NaCl})$$

The  $\text{Na}^+$  and  $\text{Cl}^-$  terms cancel, leaving exactly  $\lambda_{\text{CH}_3\text{COO}^-}^\circ + \lambda_{\text{H}^+}^\circ$ .

#### **2 • Degree of dissociation**

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

#### **3 • Dissociation constant**

$$K_a = \frac{c\alpha^2}{1-\alpha}$$

Worked • 2020 — limiting ionic conductivity from Kohlrausch

$\Lambda^\circ(\text{Al}_2(\text{SO}_4)_3) = 858$ ,  $\lambda_{\text{SO}_4^{2-}}^\circ = 160 \text{ S cm}^2 \text{ mol}^{-1}$ . Find  $\lambda_{\text{Al}^{3+}}^\circ$ .

$$858 = 2\lambda_{\text{Al}^{3+}}^\circ + 3(160) \Rightarrow 2\lambda_{\text{Al}^{3+}}^\circ = 858 - 480 = 378$$

$$\lambda_{\text{Al}^{3+}}^{\circ} = 189 \text{ S cm}^2 \text{ mol}^{-1}$$

## Faraday's laws of electrolysis — *the second engine of this chapter*

### First law

Mass deposited is proportional to the charge passed:

$$w = \frac{M I t}{n F}$$

$M$  = molar mass,  $I$  = current in amperes,  $t$  = time in **seconds**,  $n$  = electrons gained per ion,  $F = 96500 \text{ C mol}^{-1}$ .

### Second law

Same charge through different electrolytes deposits masses in the ratio of their **equivalent weights**:

$$\frac{w_1}{w_2} = \frac{E_1}{E_2}, \quad E = \frac{\text{atomic mass}}{n}$$

So Na (23/1), Mg (24/2 = 12), Al (27/3 = 9) deposit in the ratio 23 : 12 : 9 per faraday.

**Trap:** time in minutes. Convert to seconds first — this single slip has cost more marks in this chapter than any conceptual error.

Worked · 2017 — mass deposited

2 A through  $\text{AgNO}_3$  for 15 min.  $M(\text{Ag}) = 108$ .

$t = 15 \times 60 = 900 \text{ s}$ .  $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$ , so  $n = 1$ .

$$w = \frac{108 \times 2 \times 900}{1 \times 96500} = 2.014 \text{ g}$$

### Charge in faradays — without the formula

Often faster to reason directly. To reduce 1 mol  $\text{Zn}^{2+}$  you need 2 mol of electrons = **2 F** = 193000 C. To reduce 1 mol  $\text{MnO}_4^-$  to  $\text{Mn}^{2+}$ , Mn goes +7  $\rightarrow$  +2, so 5 mol electrons = **5 F**.

## Products of electrolysis — *the $E^\circ$ comparison, and the overpotential exception*

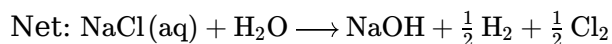
In aqueous solution, water competes with the dissolved ions at both electrodes. Two rules decide the winner:

- **At the cathode:** the species with the **higher**  $E^\circ$  is reduced.
- **At the anode:** the species with the **lower**  $E^\circ$  is oxidised.

### Aqueous NaCl — the standard case

**Cathode:**  $\text{Na}^+$  (−2.71 V) vs  $\text{H}^+$  (0.00 V). Hydrogen wins  $\rightarrow \text{H}_2$  gas.

**Anode:**  $\text{Cl}^-$  (1.36 V) vs water  $\rightarrow \text{O}_2$  (1.23 V). By the rule oxygen should win — **but it doesn't**. Oxygen evolution is kinetically slow and needs extra voltage (**overpotential**) to proceed at a useful rate, so chlorine is released instead.



**Trap:** the overpotential exception is the entire point of this question. Answering "oxygen, because  $1.23 < 1.36$ " is the trap being set. Name overpotential explicitly.

### ***Molten vs aqueous***

Molten NaCl has no water, so there's no competition: sodium at the cathode, chlorine at the anode.

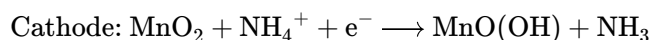
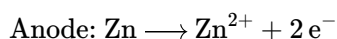
### ***Sulphuric acid***

Dilute  $\text{H}_2\text{SO}_4 \rightarrow$  oxygen at the anode. Concentrated  $\rightarrow$  peroxodisulphate,  $2\text{SO}_4^{2-} \longrightarrow \text{S}_2\text{O}_8^{2-} + 2\text{e}^-$  ( $E^\circ = 1.96 \text{ V}$ ).

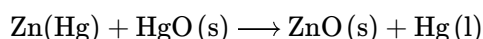
## **Batteries and fuel cells — *four devices, and the one fact asked about each***

### ***Primary — cannot be recharged***

**Dry cell (Leclanché):** zinc container = anode, graphite rod surrounded by  $\text{MnO}_2$  and carbon = cathode, moist  $\text{NH}_4\text{Cl}/\text{ZnCl}_2$  paste = electrolyte. Used in transistors and clocks.

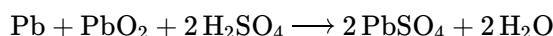


**Mercury cell:** used in hearing aids and watches. Its selling point — and the exam answer — is that **the potential stays constant through its life**, because no ions appear in the overall reaction, so no concentration changes:



### ***Secondary — rechargeable***

**Lead storage battery** (automobiles, inverters): Pb anode,  $\text{PbO}_2$  on a lead grid as cathode, 38%  $\text{H}_2\text{SO}_4$  electrolyte.

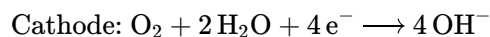
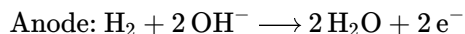


On **charging** the whole thing runs backwards:  $2\text{PbSO}_4 + 2\text{H}_2\text{O} \longrightarrow \text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4$ .

**Nickel-cadmium:** Cd anode,  $\text{NiO}_2$  cathode, KOH electrolyte. Longer life than lead storage but more expensive to make.

### ***Fuel cell***

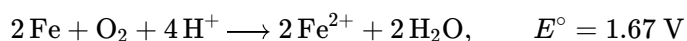
Converts a fuel's chemical energy *directly* to electricity, with reactants fed in continuously. The  $\text{H}_2$ - $\text{O}_2$  cell powered the **Apollo programme**, and its product water was condensed into the astronauts' drinking supply.



**Two advantages** (the standard 2-marker):  $\sim 70\%$  efficiency against  $\sim 40\%$  for a thermal plant, and it's pollution-free.

### **Corrosion — the electrochemical explanation, and two preventions**

Corrosion is chemical attack by atmospheric gases and moisture on a metal surface, giving oxides, sulphides and carbonates. Rusting is the electrochemical case:



$\text{Fe}^{2+}$  is then oxidised further by atmospheric oxygen to  $\text{Fe}^{3+}$ , and rust is **hydrated ferric oxide**,  $\text{Fe}_2\text{O}_3 \cdot x \text{H}_2\text{O}$ .

#### **Prevention**

- **Barrier protection** — paint, bisphenol, or a coat of another metal (Sn, Zn) between iron and the air.
- **Sacrificial protection** — attach a *more reactive* metal (Mg, Zn). It has the more negative  $E^\circ$ , so it oxidises in preference to the iron and corrodes away while the iron survives.

**Examiner asks:** "why are magnesium blocks fixed to iron pipelines" — sacrificial anode; magnesium is more reactive, more negative  $E^\circ$ , oxidises preferentially.

### **Numerical patterns, collected — five patterns, one model each**

A · Nernst / EMF — 3 marks

*Recognise it: concentrations given that aren't 1 M.*

1. Write both half-reactions; balance electrons to get  $n$ .
2.  $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$  .
3. Apply  $E = E^\circ - \frac{0.0591}{n} \log Q$  , coefficients as powers, solids omitted.

B ·  $\Delta G^\circ$  and  $K_c$  — 3 marks

*Recognise it:  $F$  is given, or the question says "at equilibrium".*

1. Get  $E^\circ_{\text{cell}}$  and  $n$ .
2.  $\Delta_r G^\circ = -nFE^\circ$  ; divide by 1000 for kJ.
3. For  $K_c$ :  $E^\circ = \frac{0.0591}{n} \log K_c$  .

C · Conductivity chain — 3-5 marks

*Recognise it: a resistance and either a cell constant or a known- $\kappa$  reference solution.*

1. Cell constant:  $G^* = \kappa \times R$  from the reference solution, or read it off directly.
2.  $\kappa = G^*/R$  for the unknown.

3.  $\Lambda_m = \kappa \times 1000/M$ .

4. If asked for  $\alpha$ : divide by  $\Lambda_m^\circ$  from Kohlrausch.

D · Faraday first law — 3 marks

*Recognise it: current in amperes and a time.*

1. Time  $\rightarrow$  seconds.
2. Find  $n$  from the ion's charge.
3.  $w = Mit/nF$ , rearranged for whatever's missing.

E · Faraday second law (cells in series) — 5 marks

*Recognise it: two electrolytic cells "connected in series".*

1. Series means identical charge through both.
2. Use the first law on the cell you have full data for, to get time.
3. Then  $w_1/w_2 = E_1/E_2$  with  $E$  = atomic mass/ $n$ .

Worked · 2020 — two cells in series

2 A through  $\text{ZnSO}_4$  (cell A) and  $\text{CuSO}_4$  (cell B) in series; 2 g Cu deposited.  $M(\text{Cu}) = 63.5$ ,  $M(\text{Zn}) = 65$ .

**Time:**  $2 = \frac{63.5 \times 2 \times t}{2 \times 96500} \Rightarrow t = \frac{2 \times 96500}{63.5} = 3039 \text{ s}$

**Zinc:** both gain 2 electrons, so

$$\frac{w_{\text{Zn}}}{2} = \frac{65/2}{63.5/2} \Rightarrow w_{\text{Zn}} = \frac{65 \times 2}{63.5} = 2.04 \text{ g}$$

## Past year questions · question types — ranked by how often they turn up

*From the last 10 years of boards — 25 questions, grouped by pattern.*

1 · Nernst / EMF calculation — 3 marks

*Recognise it: a cell notation with concentrations attached.*

1. Balance for  $n$ .
2.  $E^\circ_{\text{cell}}$  from the two  $E^\circ$  values.
3. Nernst, watching the powers in  $Q$ .

**Trap:** sign error when log of a negative power comes down.  $-\frac{0.0591}{n} \log 10^{-3}$  becomes *plus*.

2 · Conductivity / molar conductivity chain — 3-5 marks

*Recognise it: a resistance value in ohms.*

1. Cell constant.
2.  $\kappa$ , then  $\rho$  if asked.
3.  $\Lambda_m$ , then  $\alpha$  if asked.

**Trap:** forgetting the  $\times 1000$  in  $\Lambda_m$ , or dividing by the wrong concentration.

3 · Faraday's laws numerical — 3-5 marks

*Recognise it: amperes, minutes, or "how many faradays".*

1. Seconds.
  2.  $n$  from the ion.
  3.  $w = MIt/nF$ , or reason directly in moles of electrons.
- 4 · Electrolysis products with reasoning — 2-3 marks

*Recognise it: "predict the products", with  $E^\circ$  values supplied.*

1. Cathode: higher  $E^\circ$  reduced.
  2. Anode: lower  $E^\circ$  oxidised.
  3. Check for the overpotential exception before answering.
- 5 · Cell/battery definitions and comparisons — 2 marks

*Recognise it: "define", "name the cell used in...", "two differences".*

1. Name the device.
  2. Give both electrode reactions if asked.
  3. For differences, answer in matched pairs.
- 6 · Kohlrausch application — 2-3 marks

*Recognise it: limiting molar conductivities of ions given.*

1. Write the electrolyte as its ions with correct multipliers.
2. Sum or subtract as needed.
3. For weak electrolytes, build from three strong ones.

### **Past year questions · mark slots — *what each type is worth***

| Question type            | Slot                          | Time    |
|--------------------------|-------------------------------|---------|
| Nernst / EMF             | 3-marker                      | 5 min   |
| Conductivity chain       | 3-marker, or part of a 5      | 5-7 min |
| Faraday's laws           | 3-marker; series cells is a 5 | 4-7 min |
| Electrolysis products    | 2-3 marker                    | 3 min   |
| Definitions / cell names | 2-marker                      | 2 min   |
| Kohlrausch               | 2-3 marker                    | 3 min   |
| $\Delta G^\circ$ / $K_c$ | 3-marker                      | 4 min   |

*The video splits these as short-answer type I (2 marks), type II (3 marks), and long answer (5 marks) — the conductivity chain and series-cell problems are the ones that appear as 5-markers.*

### **Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter***

2019 Q6 · 2019 Q8 · lecture PYQ

Electrolysis of aqueous NaCl, and the overpotential reason chlorine appears instead of oxygen. Runs almost every year in some form.

2020 Q16 · lecture PYQ

Two electrolytic cells in series,  $\text{ZnSO}_4$  and  $\text{CuSO}_4$ , 2 g copper deposited — identical numbers both times.

2024 Q18 · 2026 Q22 · 2026 Q24

The conductivity  $\rightarrow$  resistivity  $\rightarrow$  molar conductivity chain, sometimes with degree of dissociation added. The single most reliable numerical in the chapter.

2025 Q1 · 2026 Q7(a)

Fuel cells: definition and two advantages, then the Apollo hydrogen-oxygen cell and its drinking-water by-product.

2022 Q13 · 2020 Q21 · 2026 Q22

Kohlrausch's law statement, then applied — degree of dissociation of acetic acid, limiting ionic conductivity of  $\text{Al}^{3+}$ ,  $\Lambda^\circ$  of  $\text{NaCl}$ .

2020 Q3 · 2026 Q23(b)

Electrochemical vs electrolytic cell differences, and primary vs secondary battery differences. Straight recall, always available.

2017 Q5 · 2020 Q21(a)

Which cell is used where — dry cell in transistors, mercury cell in hearing aids and watches, with electrode reactions.

## **Past year questions · cold practice — *answers only* — *work them before you look***

### **Nernst / EMF**

- 2020 Q2 — write cell notation and the Nernst equation for  $\text{Mg} + 2 \text{Ag}^+ \rightarrow \text{Mg}^{2+} + 2 \text{Ag}$ .  
 $\text{Mg}|\text{Mg}^{2+}||\text{Ag}^+|\text{Ag}; E = E^\circ - \frac{0.0591}{2} \log \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]^2}$
- 2026 Q19 — EMF of  $\text{Sn}|\text{Sn}^{2+}||\text{H}^+|\text{H}_2$  at 298 K with concentrations given. Apply Nernst with  $n = 2$ ;  $\text{H}_2$  omitted as a gas at 1 bar

### **$\Delta G^\circ$ and $K_c$**

- 2026 Q23 —  $E^\circ$  for  $\text{Cu} + 2 \text{Ag}^+ \rightleftharpoons \text{Cu}^{2+} + 2 \text{Ag}$  at equilibrium,  $K_c = 10^{15}$ .  $E^\circ = \frac{0.0591}{2} \times 15 = 0.44$  V

### **Conductivity**

- Lecture PYQ — 0.05 M  $\text{KCl}$ ,  $l = 50$  cm,  $A = 0.625$  cm<sup>2</sup>,  $R = 5 \times 10^3$   $\Omega$ . Find  $\rho$ ,  $\kappa$ ,  $\Lambda_m$ . 62.5  $\Omega$  cm; 0.016 S cm<sup>-1</sup>; 320 S cm<sup>2</sup> mol<sup>-1</sup>
- Lecture PYQ — cell gives 164  $\Omega$  with 0.02 M  $\text{KCl}$  ( $\kappa = 2.768 \times 10^{-3}$ ), then 78.5  $\Omega$  with 0.05 M  $\text{AgNO}_3$ . Find  $\kappa$  and  $\Lambda_m$  of  $\text{AgNO}_3$ .  $G^* = 0.4539$ ;  $\kappa = 5.78 \times 10^{-3}$ ;  $\Lambda_m = 115.6$  S cm<sup>2</sup> mol<sup>-1</sup>
- 2026 Q22 — 0.1 M  $\text{NaCl}$ ,  $\kappa = 1.06 \times 10^{-2}$  S cm<sup>-1</sup>,  $\lambda^\circ \text{Na}^+ = 50.1$ ,  $\text{Cl}^- = 76.5$ . Find  $\Lambda_m$  and  $\alpha$ . 106 S cm<sup>2</sup> mol<sup>-1</sup>;  $\Lambda^\circ = 126.6$ ;  $\alpha = 0.837$
- 2022 Q13(b) —  $\alpha$  of acetic acid if  $\Lambda_m = 48$  and  $\Lambda^\circ_m = 400$ . 0.12

### **Kohlrausch**

- 2021-22 —  $\Lambda^\circ$  of  $\text{MgCl}_2$  given  $\lambda^\circ \text{Mg}^{2+} = 106$ ,  $\text{Cl}^- = 76.3$ .  $285.6 \text{ S cm}^2 \text{ mol}^{-1}$  (remember the  $2\times$ )

### **Faraday's laws**

- 2014 — 5 A through  $\text{Ni}(\text{NO}_3)_2$  for 20 min,  $M = 58.7$ . Mass of Ni? 1.825 g
- 2015 — charge to reduce 1 mol  $\text{Zn}^{2+}$  to Zn.  $2 F = 193000 \text{ C}$
- 2026 Q25(b) — faradays to produce 40 g Al from molten  $\text{Al}_2\text{O}_3$ ,  $M = 27$ . 4.44 F

Built from Sourabh Raina's Electrochemistry one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 2 (Rationalised 2022-23). Verified against NCERT:

"Kohlrausch law of independent migration of ions"; Faraday constant  $96487 \text{ C mol}^{-1}$  (taken as 96500 in working); Daniell cell 1.1 V at unit concentration.

# Chemical Kinetics

NCERT Class XII Chemistry · Chapter 3 · Chemical Kinetics

*13 marks. Almost pure numerical territory — order from data, integrated rate equations, half-life, activation energy. Four formulas do nearly all the work; the marks go to knowing which one the question is pointing at.*

## What kinetics answers — and what it deliberately doesn't

Three different questions about a reaction, three different branches:

| Question                           | Answered by                  |
|------------------------------------|------------------------------|
| Is it feasible? ( $\Delta G < 0$ ) | Thermodynamics               |
| How far does it go?                | Chemical equilibrium ( $K$ ) |
| How fast, and by what mechanism?   | <b>Chemical kinetics</b>     |

The standard illustration: thermodynamics says diamond  $\rightarrow$  graphite is feasible. Kinetics explains why your ring is safe — the conversion is immeasurably slow.

## Rate of reaction — average, instantaneous, and the coefficient division

Rate = change in concentration per unit time. Reactant concentration falls, so its rate carries a minus sign to keep the rate positive:

$$\text{rate} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$

**Average rate** uses  $\Delta$  over a finite interval. **Instantaneous rate** is the slope of the tangent at one moment:  $-\frac{d[R]}{dt}$ .

### When coefficients differ — divide by them

For  $aA + bB \rightarrow cC + dD$ :

$$\text{rate} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

**Trap:** "rate of reaction" and "rate of disappearance of B" are different numbers whenever B's coefficient isn't 1. Read which one is asked — this is the entire content of a 2-marker that runs most years.

Worked · 2022 — rate of reaction vs rate of disappearance

$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ , rate of formation of  $\text{NH}_3 = 3.6 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$ . Find the rate of reaction and the rate of disappearance of  $\text{H}_2$ .

$$\text{rate} = \frac{1}{2} \frac{\Delta[\text{NH}_3]}{\Delta t} = \frac{1}{2}(3.6 \times 10^{-4}) = 1.8 \times 10^{-4}$$

$$-\frac{\Delta[\text{H}_2]}{\Delta t} = 3 \times \text{rate} = 5.4 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$$

**Examiner asks:** this exact shape, with  $\text{N}_2/\text{H}_2/\text{NH}_3$  or  $A + 3B \rightarrow 2C$ . Near-guaranteed 2-3 marker.

### Rate law and finding the order from data — *the standard 3-marker, step by step*

The rate law is **experimental** — powers are not read off the balanced equation:

$$\text{rate} = k[A]^x[B]^y$$

$x$  and  $y$  may or may not equal the stoichiometric coefficients. **Order** =  $x + y$ . Order can be zero, fractional, or negative; molecularity cannot.

#### Getting $x$ and $y$ from a data table — *the method*

1. Pick two experiments where **one** concentration is held constant and the other changes.
2. Write the rate law for each; divide one by the other.  $k$  and the constant term cancel.
3. Read off the power that makes the ratio work.
4. Repeat with a different pair for the other exponent.
5. Substitute any single experiment back to find  $k$ .

**Trap:** choosing two experiments where *both* concentrations change. Nothing cancels and the algebra becomes unsolvable. Scan the table for the constant column first.

#### Order vs molecularity — *the comparison they ask for*

| Order                                      | Molecularity                                       |
|--------------------------------------------|----------------------------------------------------|
| Sum of powers in the experimental rate law | Number of species colliding in one elementary step |
| Experimental                               | Theoretical                                        |
| Can be 0, fractional, negative             | Whole number only (1, 2, 3)                        |
| Applies to overall reaction                | Only defined for an elementary step                |

**Examiner asks:** "can this be an elementary reaction?" (2026) with a fractional order like  $3/2$  — **no**, because an elementary reaction's order equals its molecularity, which must be a whole number.

### Units of the rate constant — *one formula, every order*

From  $\text{rate} = k[\ ]^n$ , so  $k = \text{rate}/[\ ]^n$ :

$$\text{unit of } k = \text{mol}^{1-n} \text{ L}^{n-1} \text{ s}^{-1}$$

| Order  | Unit of k                           |
|--------|-------------------------------------|
| Zero   | mol L <sup>-1</sup> s <sup>-1</sup> |
| First  | s <sup>-1</sup>                     |
| Second | L mol <sup>-1</sup> s <sup>-1</sup> |

**Examiner asks:** given a rate law, state the units — or, run backwards, given the units, state the order. Both directions appear.

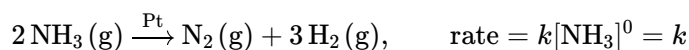
## Integrated rate equations — zero and first order, plus their graphs

### Zero order

$$[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$$

Plot  $[R]$  against  $t$ : straight line, **slope** =  $-k$ , intercept  $[R]_0$ .

**Example:** decomposition of gaseous ammonia on a hot platinum surface at 1130 K. At high pressure the metal surface saturates with ammonia, so adding more changes nothing — rate becomes independent of concentration:



### First order

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

Plot  $\log \frac{[R]_0}{[R]}$  against  $t$ : straight line through the origin, **slope** =  $k/2.303$ . Plot  $\ln[R]$  against  $t$  instead and the slope is  $-k$ , intercept  $\ln[R]_0$ .

**Examples:** hydrogenation of ethene; all natural and artificial radioactive decay; decomposition of  $\text{N}_2\text{O}_5$  and  $\text{N}_2\text{O}$ .

### First order in the gas phase — the closed form worth memorising

Concentration is proportional to partial pressure, so the same equation runs on pressures. For  $A(\text{g}) \rightarrow B(\text{g}) + C(\text{g})$  with all coefficients 1, let  $x$  be the pressure decomposed:

$$p_{\text{total}} = (p_i - x) + x + x = p_i + x \Rightarrow x = p_T - p_i$$

So the reactant's partial pressure is  $p_A = p_i - x = 2p_i - p_T$ , and:

$$k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_T}$$

Use this directly whenever the question gives you **initial and total** pressure. If it gives the reactant's partial pressure outright, the plain  $\log(p_i/p_A)$  form is enough.

Worked · 2016 — rate constant from total pressure

$\text{C}_2\text{H}_5\text{Cl}(\text{g}) \longrightarrow \text{C}_2\text{H}_4(\text{g}) + \text{HCl}(\text{g})$ . At  $t = 0$ ,  $p_i = 0.30$  atm; at  $t = 300$  s,  $p_T = 0.50$  atm. ( $\log 3 = 0.4771$ )

$$k = \frac{2.303}{300} \log \frac{0.30}{2(0.30) - 0.50} = \frac{2.303}{300} \log \frac{0.30}{0.10}$$

$$= \frac{2.303 \times 0.4771}{300} = 3.66 \times 10^{-3} \text{ s}^{-1}$$

The unit is  $\text{s}^{-1}$ , which itself confirms first order.

**Trap:** feeding the *total* pressure straight into the log. It must be  $2p_i - p_T$ , not  $p_T$ .

## Half-life — and the graph question that turns on it

Time for half the reactant to be consumed.

$$\text{First order: } t_{1/2} = \frac{0.693}{k} \quad \text{Zero order: } t_{1/2} = \frac{[R]_0}{2k}$$

**The distinction that gets examined:** first-order half-life is **independent of initial concentration**; zero-order half-life is **directly proportional** to it.

**Examiner asks:** "predict the order from the graph" (2019) — a flat  $t_{1/2}$  vs  $[R]_0$  line means first order; a straight rising line means zero order.

## Percentage-completion problems

Set  $[R]_0 = 100$  and subtract the percentage completed. 75% done  $\rightarrow [R] = 25$ ; 99% done  $\rightarrow [R] = 1$ .

Worked · 2026 — time for 3/4 decomposition

First order,  $k = 2.54 \times 10^{-3} \text{ s}^{-1}$ . Time for 3/4 of the reactant to decompose? ( $\log 4 = 0.6$ )

$[R]_0 = a$ , so  $[R] = a - \frac{3}{4}a = \frac{a}{4}$ .

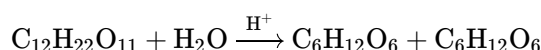
$$t = \frac{2.303}{k} \log \frac{a}{a/4} = \frac{2.303 \times 0.6}{2.54 \times 10^{-3}} = 544 \text{ s}$$

A neat consequence worth remembering: for first order,  $t_{99\%} = 2 \times t_{90\%}$  — because  $\log 100 = 2$  and  $\log 10 = 1$ .

## Pseudo first order — two examples, one idea

A reaction that is **bimolecular but follows first-order kinetics**, because one reactant is present in such large excess that its concentration doesn't measurably change.

**Hydrolysis of sucrose:**



Water is the solvent, so  $[\text{H}_2\text{O}]$  is effectively constant. Rate =  $k[\text{C}_{12}\text{H}_{22}\text{O}_{11}]$ : order 1, molecularity 2.

**Hydrolysis of an ester** in dilute acid behaves the same way.

**Examiner asks:** (2024) write the rate law, then state order *and* molecularity separately, then name the reaction type. The gap between order 1 and molecularity 2 is the whole point.

### Temperature dependence and Arrhenius — *the 5-marker's favourite*

Rate roughly **doubles for every 10 K rise** near room temperature. Quantitatively:

$$k = A e^{-E_a/RT}$$

$A$  = frequency factor,  $E_a$  = activation energy, the minimum energy colliding molecules need.

#### **The logarithmic form and its graph**

$$\ln k = -\frac{E_a}{RT} + \ln A$$

Against  $y = mx + c$  with  $y = \ln k$  and  $x = 1/T$ :

- **Slope** =  $-E_a/R$  (negative, so the line falls)
- **Intercept** =  $\ln A$

#### **Two-temperature form — the one used in numericals**

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[ \frac{T_2 - T_1}{T_1 T_2} \right]$$

Worked · 2025 —  $E_a$  when the rate doubles over 10 K

Rate doubles from 298 K to 308 K.  $2.303R = 19.15$ ,  $\log 2 = 0.3$ .

Concentrations are unchanged, so a doubled rate means a doubled  $k$ :  $k_2/k_1 = 2$ .

$$0.3 = \frac{E_a}{19.15} \left[ \frac{10}{308 \times 298} \right]$$

$$E_a = \frac{0.3 \times 19.15 \times 308 \times 298}{10} = 52729.9 \text{ J mol}^{-1} = 52.7 \text{ kJ mol}^{-1}$$

Worked · 2025 —  $E_a$  from two half-lives

First order, 50% complete in 20 min at 300 K and in 5 min at 350 K. ( $\log 4 = 0.602$ )

50% completion is the half-life, so  $k = 0.693/t_{1/2}$ :

$$\frac{k_2}{k_1} = \frac{0.693/5}{0.693/20} = 4$$

$$0.602 = \frac{E_a}{2.303 \times 8.314} \left[ \frac{50}{300 \times 350} \right] \Rightarrow E_a = 24205 \text{ J mol}^{-1} = 24.2 \text{ kJ mol}^{-1}$$

**Trap:** "50% completed in 20 minutes" is a half-life in disguise. Spot that and the question collapses into two lines.

## Collision theory and catalysis — *the definitions that carry marks*

### **Activated complex and the energy profile**

Colliding reactant molecules first form a short-lived **activated complex**. The energy needed to reach it is the **activation energy**. The reaction's own energy change is separate:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

On a potential-energy vs reaction-coordinate plot, the barrier height is  $E_a$  either way; what changes is where the products land. Products lower than reactants  $\rightarrow \Delta H$  negative  $\rightarrow$  **exothermic**. Products higher  $\rightarrow \Delta H$  positive  $\rightarrow$  **endothermic**.

### **Maxwell-Boltzmann distribution**

Plot the fraction of molecules ( $n_E/n_T$ ) against kinetic energy. Most molecules sit near the **most probable kinetic energy**; only a small tail — roughly 10–20% — carries energy above  $E_a$ , and only those can react.

**Raise the temperature by 10 K** and the curve flattens and shifts right: the fraction with energy at or above  $E_a$  roughly **doubles**. That is precisely why the rate doubles for a 10 K rise.

### **Collision theory**

Treats molecules as hard spheres that must collide to react.

**Collision frequency ( $Z$ ):** number of collisions per second per unit volume of reaction mixture.

$$\text{rate} = Z_{AB} e^{-E_a/RT}$$

Comparing this with Arrhenius shows the frequency factor  $A$  is essentially the **collision frequency**. The theory is accurate for atoms and simple molecules but deviates for complex ones — because not every collision works. Two conditions must both hold:

1. Colliding molecules have energy at least equal to the **threshold energy**.
2. They collide in the **proper orientation**, so old bonds can break and new ones form.

### **Catalysis**

A catalyst increases the rate **without itself undergoing any permanent chemical change**. By **intermediate complex theory**, it forms a temporary bond with the reactant, making a transitory intermediate that decomposes into product and releases the catalyst again — which is why a small amount catalyses a large amount of reactant.

The effect is an **alternative pathway of lower activation energy**, so more molecules clear the barrier and the rate rises.

**What a catalyst does not change:** the Gibbs energy of the reaction, the equilibrium constant, or the position of equilibrium. It catalyses forward and backward reactions *to the same extent*, so equilibrium is reached **earlier** but at the same place. And since it can't

change  $\Delta G$ , it cannot make a non-spontaneous reaction happen — only speed up one that already is.

**Examiner asks:** (2017) "effect of a catalyst on activation energy and on  $\Delta G$ " — lowers  $E_a$ , leaves  $\Delta G$  untouched. Answering only the first half loses the mark.

### **Reaction mechanism and the rate-determining step**

For a multi-step reaction, the **slowest step determines the rate** — write the rate law directly from that step's reactants.

Example (2025):  $2\text{H}_2\text{O}_2 \xrightarrow{\text{I}^-} 2\text{H}_2\text{O} + \text{O}_2$  via a slow step  $\text{H}_2\text{O}_2 + \text{I}^- \longrightarrow \text{H}_2\text{O} + \text{IO}^-$  and a fast step. So rate =  $k[\text{H}_2\text{O}_2][\text{I}^-]$ , overall order 2 — even though the balanced equation shows  $2\text{H}_2\text{O}_2$ .

### **Numerical patterns, collected — five patterns, one model each**

A · Rate from stoichiometry — 2 marks

*Recognise it: one species' rate is given, another is asked for.*

1. Write the full rate expression with 1/coefficient on every term.
2. Equate the known term to the wanted one.
3. Check whether "rate of reaction" or "rate of disappearance" is asked.

B · Order from a data table — 3 marks

*Recognise it: a table of concentrations and initial rates.*

1. Find two rows with one concentration constant.
2. Divide the two rate laws; solve for the exponent.
3. Repeat for the other exponent, then back-substitute for  $k$ .

C · Integrated first-order equation — 3 marks

*Recognise it: a percentage decomposed, or concentrations at two times.*

1. Set  $[R]_0 = 100$  and subtract the percentage.
2.  $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$ , rearranged for the unknown.
3. Gas phase → convert total pressure to reactant partial pressure first.

D · Half-life ↔ rate constant — 2-3 marks

*Recognise it: "half-life", or "50% completed".*

1. Identify the order from the rate law or the units of  $k$ .
2. First order:  $t_{1/2} = 0.693/k$ . Zero order:  $t_{1/2} = [R]_0/2k$ .

Worked · 2026 — half-life from percentage decomposition

First order, 25% decomposed in 25 min. Find  $t_{1/2}$ . ( $\log 4 = 0.6021$ ,  $\log 3 = 0.4771$ )

$$k = \frac{2.303}{25} \log \frac{100}{75} = \frac{2.303}{25} (0.6021 - 0.4771) = 0.0115 \text{ min}^{-1}$$

$$t_{1/2} = \frac{0.693}{0.0115} = 60.26 \text{ min}$$

E · Activation energy from two temperatures — 5 marks

*Recognise it: two temperatures, or "rate doubles when temperature rises by..."*

1. Get  $k_2/k_1$  — from rates directly, or from  $0.693/t_{1/2}$  at each temperature.
2.  $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[ \frac{T_2 - T_1}{T_1 T_2} \right]$  .
3. Answer in J mol<sup>-1</sup>, then divide by 1000 for kJ.

## Past year questions · question types — ranked by how often they turn up

*From the last 10 years of boards — 25 questions, grouped by pattern.*

### 1 · First-order integrated rate equation — 3 marks

*Recognise it: a percentage decomposed, or a concentration/pressure at two times.*

1.  $[R]_0 = 100$ , subtract the percentage.
2. Apply the 2.303 formula.
3. Convert total pressure to partial pressure if gaseous.

**Trap:** using the percentage *decomposed* as  $[R]$  instead of what remains.

### 2 · Activation energy (Arrhenius) — 5 marks

*Recognise it: two temperatures, or  $2.303R = 19.15$  handed to you.*

1. Get  $k_2/k_1$ .
2. Two-temperature Arrhenius form.
3. Convert to kJ if the answer is large.

**Trap:** inverting  $T_2 - T_1$  over  $T_1 T_2$ , or forgetting that the half-life ratio inverts the  $k$  ratio.

### 3 · Order from a data table — 3-5 marks

*Recognise it: a table of initial concentrations and rates.*

1. Pair rows sharing a constant concentration.
2. Divide; solve the exponent.
3. Back-substitute for  $k$  and state its units.

### 4 · Rate expression from stoichiometry — 2-3 marks

*Recognise it: one species' rate given, another asked.*

1. Full expression with 1/coefficient terms.
2. Equate and solve.

### 5 · Definitions and comparisons — 2 marks

*Recognise it: "define", "distinguish between", "write the unit of".*

1. Order vs molecularity: answer in matched pairs.
2. Half-life, collision frequency, effective collision, activation energy — book definitions.

### 6 · Graph interpretation — 3 marks

*Recognise it: a plot of  $\ln k$  vs  $1/T$ , or  $t_{1/2}$  vs  $[R]_0$ , or  $\log \frac{[R]_0}{[R]}$  vs  $t$ .*

1. Match the plot to its equation.
2. Read slope and intercept off  $y = mx + c$ .

3. State what each represents in symbols, not words alone.

### Past year questions · mark slots — *what each type is worth*

| Question type                   | Slot                      | Time    |
|---------------------------------|---------------------------|---------|
| First-order integrated equation | 3-marker                  | 4 min   |
| Activation energy               | 5-marker (or part of one) | 6–7 min |
| Order from data table           | 3-marker, sometimes a 5   | 5–6 min |
| Rate from stoichiometry         | 2–3 marker                | 3 min   |
| Definitions / comparisons       | 2-marker                  | 2 min   |
| Graph interpretation            | 3-marker                  | 4 min   |
| Pseudo first order              | 3-marker                  | 3 min   |

*The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — activation energy and the multi-part data questions are where the 5-markers sit.*

### Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2018 Q15 · lecture PYQ

"Second order in A, first order in B" — write the differential rate equation, then how the rate changes when concentrations are tripled or doubled. Identical wording both times.

2025 Q16 · 2025 Q23(a)

Activation energy from two temperatures — once from two half-lives, once from a rate that doubles over 10 K. Same formula, two disguises.

2019 Q5 · 2022 Q17

Graph questions:  $\ln k$  vs  $1/T$  (intercept and slope), and  $\log \frac{[R]_0}{[R]}$  vs  $t$  (order and slope). Both reduce to  $y = mx + c$ .

2022 Q8 · 2026 Q6

Order vs molecularity — as a straight comparison, and as "can this be an elementary reaction" with a fractional order.

2024 Q21 · lecture PYQ

Sucrose hydrolysis: rate law, order 1 but molecularity 2, and the name — pseudo first order.

2019 Q9 · 2019 Q10

Half-life: predicting order from a  $t_{1/2}$  vs  $[R]_0$  graph, and computing  $t_{1/2}$  from a first-order rate law.

2026 Q1 · 2024 Q3

Definitions: collision frequency, effective collisions, half-life period, and the units of  $k$  for zero and first order.

## Past year questions · cold practice — *answers only* — *work them before you look*

### Integrated rate equation

- 2017 Q11 —  $\text{N}_2\text{O}_5$  decomposition:  $1.6 \times 10^{-2} \text{ M}$  at  $t = 0$ ,  $0.8 \times 10^{-2} \text{ M}$  at 300 s. Show first order and find  $t_{1/2}$ .  $k = 2.3 \times 10^{-3} \text{ s}^{-1}$  at both times;  $t_{1/2} = 301 \text{ s}$
- 2026 Q22 —  $\text{C}_2\text{H}_5\text{Cl}$ :  $p_i = 0.30 \text{ atm}$ , total  $p = 0.50 \text{ atm}$  at 30 s. Find  $k$ .  $0.0368 \text{ s}^{-1}$
- 2026 Q4 —  $k = 2.54 \times 10^{-3} \text{ s}^{-1}$ , time for 3/4 decomposition. 544 s
- 2024 — first order,  $k = 1.25 \times 10^{-3} \text{ s}^{-1}$ . Time for 5 g to fall to 2.5 g. 554.6 s — note this is just  $t_{1/2}$  in disguise
- 2024 — show  $t_{99\%} = 2 t_{90\%}$  for first order.  $\log 100 = 2$  vs  $\log 10 = 1$
- 2021-22 — first order, 75% decomposed in 30 min. Find  $t_{1/2}$ .  $k = 0.046 \text{ min}^{-1}$ ;  $t_{1/2} = 15 \text{ min}$
- Lecture PYQ — zero order,  $[R]_0 = 0.1 \text{ M}$  falls to 0.064 M,  $k = 4 \times 10^{-3}$ . Find  $t$ . 9 s

### Half-life

- 2019 Q10 — rate  $= 5.5 \times 10^{-14} [\text{C}_2\text{H}_4]$ . Units of  $k$  and  $t_{1/2}$ .  $\text{s}^{-1}$ ;  $1.26 \times 10^{13} \text{ s}$
- 2026 Q24(b) — first order, 25% decomposed in 25 min. Find  $t_{1/2}$ .  $k = 0.0115 \text{ min}^{-1}$ ;  $t_{1/2} = 60.26 \text{ min}$

### Activation energy

- 2025 Q16 — 50% complete in 20 min at 300 K, 5 min at 350 K. Find  $E_a$ .  $24.2 \text{ kJ mol}^{-1}$
- 2025 Q23(a) — rate doubles from 298 K to 308 K. Find  $E_a$ .  $52.7 \text{ kJ mol}^{-1}$

### Rate and order

- 2020 —  $A + 3B \rightarrow 2C$ , rate of formation of C  $= 2.5 \times 10^{-4}$ . Find rate of reaction and rate of disappearance of B.  $1.25 \times 10^{-4}$ ;  $3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- 2026 Q6 — rate  $= k[A][B]^{3/2}$ . Overall order, and can it be elementary? 2.5; no — elementary reactions have whole-number order
- 2025 Q18 — order with respect to A and B from a rate table where doubling B leaves the rate unchanged. Order in B is zero

### Mechanism

- 2025 Q23(b) —  $2\text{H}_2\text{O}_2 \xrightarrow{\text{I}^-} 2\text{H}_2\text{O} + \text{O}_2$ , slow step  $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$ . Rate law and overall order. rate  $= k[\text{H}_2\text{O}_2][\text{I}^-]$ ; order 2

Built from Sourabh Raina's Chemical Kinetics one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 3 (Rationalised 2022-23). NCERT confirms the Arrhenius form  $k = Ae^{-E_a/RT}$  and the term "pseudo first order reaction".

# The d- and f-Block Elements

NCERT Class XII Chemistry · Chapter 4 · d and f Block Elements

*11 marks, and material you haven't met before. Everything here is written from scratch — terms tagged [exposure] are ones you're seeing for the first time, and each gets a proper explanation before it's used. Almost every answer in this chapter traces back to one idea: how many unpaired d electrons are there?\**

## What the d block is — *where these elements sit, and why they're called "transition"*

**[exposure] d-block elements** are the elements of groups 3 to 12 — the wide middle block of the periodic table, sitting between the s block on the left and the p block on the right. What defines them is that electrons are filling a **d orbital**. Each atom has an outermost shell; the shell just inside it is the **penultimate** one, and it's that shell's d orbital being filled. General configuration:

$$(n-1)d^{1-10} ns^{1-2}$$

So if the outermost shell is the 4th, the 3d orbital is filling. There are four such series — 3d, 4d, 5d, 6d.

**[exposure] Transition element** is a narrower term with a precise definition: an element with a **partially filled d orbital**, either in its atom *or* in any of its stable oxidation states. The word "transition" comes from their position, bridging the reactive metals of the s block and the non-metals of the p block. The definition matters because it excludes some d-block members — see zinc below. Transition elements share a family of properties: variable oxidation states, coloured ions, catalytic behaviour, complex formation, and hardness.

### The four series

| Series | Orbital | Runs from                         | Period |
|--------|---------|-----------------------------------|--------|
| First  | 3d      | Sc (21) → Zn (30)                 | 4th    |
| Second | 4d      | Y (39) → Cd (48)                  | 5th    |
| Third  | 5d      | La (57), then Hf (72) → Hg (80)   | 6th    |
| Fourth | 6d      | Ac (89), then Rf (104) → Cn (112) | 7th    |

The gaps in the third and fourth series (58-71 and 90-103) are where the f-block elements sit — pulled out and studied separately.

### 3d series configurations — *learn these, most questions start here*

All are  $[\text{Ar}] 3d^n 4s^2$  except two exceptions:

|                          |                       |                          |
|--------------------------|-----------------------|--------------------------|
| <b>Sc</b> $3d^1 4s^2$    | <b>Ti</b> $3d^2 4s^2$ | <b>V</b> $3d^3 4s^2$     |
| <b>Cr</b> $3d^5 4s^1$    | <b>Mn</b> $3d^5 4s^2$ | <b>Fe</b> $3d^6 4s^2$    |
| <b>Co</b> $3d^7 4s^2$    | <b>Ni</b> $3d^8 4s^2$ | <b>Cu</b> $3d^{10} 4s^1$ |
| <b>Zn</b> $3d^{10} 4s^2$ |                       |                          |

**Why Cr and Cu break the pattern:** a **half-filled** ( $d^5$ ) or **completely filled** ( $d^{10}$ ) d subshell is extra stable, so an electron shifts from 4s to 3d to reach it.

### ***Why Zn, Cd, Hg are not transition elements***

Their configuration is  $(n-1)d^{10}ns^2$  — the d orbital is **completely filled**, in the ground state and in their common +2 oxidation state ( $Zn^{2+}$  is  $3d^{10}$ ). No partially filled d orbital in any stable state means they fail the definition.

**Examiner asks:** the flip side — "why is Cu a transition element despite a filled d orbital in the ground state?" Because its common oxidation state  $Cu^{2+}$  is  $3d^9$ , which *is* partially filled. Same for Sc:  $3d^1$  in the ground state, so it qualifies even though  $Sc^{3+}$  is colourless.

### ***Enthalpy of atomization — the unpaired-electron idea, introduced properly***

**[exposure] Enthalpy of atomization** is the energy needed to take one mole of a solid metal and pull it apart into free gaseous atoms. It exists as a measure because it tells you directly how strong the metallic bonding is — strong bonds mean the atoms resist being separated, so the value is high. Transition metals have unusually high values: unpaired d electrons participate in metallic bonding, and more unpaired electrons means a stronger bond. Chromium, with six unpaired electrons ( $3d^5 4s^1$ ), sits near the top; zinc, with none ( $3d^{10} 4s^2$ ), is the lowest in the 3d series.

Two trends carry nearly all the marks:

- **Maximum in the middle of each series**, where unpaired electrons peak.
- **Second and third series > first series.** The 4d and 5d elements have poorly shielding inner electrons, so the effective nuclear charge is higher, metal-metal bonding is stronger, and atomization needs more energy.

**Examiner asks:** "zinc has the lowest enthalpy of atomization" (fully filled d, no unpaired electrons, weak metallic bond) and "why do 2nd/3rd series have greater enthalpy of atomization" (poor shielding → stronger metallic bond). Both recur.

### ***Melting and boiling points follow the same logic***

Highest in the middle: Cr in the 3d series, Mo in 4d, W in 5d. Mn and Tc are anomalously **low** — their  $d^5 s^2$  configuration is so stable that the electrons are less free to delocalise into metallic bonding.

Zn, Cd, Hg are **soft metals with low melting points** for the same reason as their low atomization enthalpy: all d electrons paired, weak metallic bond.

### ***Ionization enthalpy and the exchange-energy idea — why the trend is so irregular***

Ionization enthalpy depends on three competing things:

1. **Nuclear attraction** — higher nuclear charge holds electrons tighter, raising it.
2. **Electron repulsion (screening)** — electrons pushing each other out lowers it.
3. **Exchange energy** — see below.

**[exposure] Exchange energy** is a stabilising effect that appears when several electrons with **parallel spins** occupy orbitals of equal energy. Such electrons can swap positions with each other, and every possible swap releases a little energy. More possible swaps means more energy released, and losing energy means greater stability. This is the real reason  $d^5$  and  $d^{10}$  configurations are so stable — they maximise the number of parallel-spin exchanges. It's why chromium is  $3d^5 4s^1$  rather than  $3d^4 4s^2$ .

### ***The comparisons they set***

- **Mn<sup>+</sup> has lower IE than Cr<sup>+</sup>.** Mn<sup>+</sup> is  $3d^5 4s^1$ ; losing one electron gives the stable  $3d^5$  Mn<sup>2+</sup>, so it goes easily. Cr<sup>+</sup> is already  $3d^5$ ; removing an electron breaks that stability.
- **Fe<sup>2+</sup> has lower IE than Mn<sup>2+</sup>.** Fe<sup>2+</sup> ( $3d^6$ ) loses one to reach the stable  $3d^5$ ; Mn<sup>2+</sup> is already  $3d^5$  and resists.
- **Second IE of Cr and Cu is high.** Cr<sup>+</sup> ( $3d^5$ ) and Cu<sup>+</sup> ( $3d^{10}$ ) are both stable, so the next electron is hard to remove.
- **Third IE of Mn and Zn is high** — Mn<sup>2+</sup> ( $3d^5$ ) and Zn<sup>2+</sup> ( $3d^{10}$ ) are stable.

**Trap:** every one of these is answered by writing the configuration *before and after* the electron leaves, and pointing at whichever side is  $d^5$  or  $d^{10}$ . Don't reason in words alone — write both configurations.

### ***Variable oxidation states — the defining property, and why P-block differs***

**[exposure] Variable oxidation state** means one element can lose different numbers of electrons in different compounds — iron exists as both Fe<sup>2+</sup> and Fe<sup>3+</sup>, manganese runs from +2 all the way to +7. It happens because in transition elements the  $(n-1)d$  and  $ns$  orbitals are **very close in energy**. With so small a gap, electrons from both can be pulled into bonding, and the number that participates can vary. Main-group elements have no such near-degenerate pair, so their oxidation states are far more restricted.

- **Most common state is +2** across the 3d series (from losing the two 4s electrons) — except Sc, which is +3.
- **Maximum variety in the middle:** Mn shows +2 to +7, the most of any 3d element, because it has the most unpaired electrons available.
- **Fewest at the ends:** Sc and Ti have too few electrons to lose; Cu and Zn have too few vacant d orbitals to use.
- **Cu shows +1** ( $3d^{10}$ , stable) — the only 3d element with a common +1 state.

### ***Transition vs p-block variability — a standing question***

Transition metal oxidation states differ by **one** unit (Fe<sup>2+</sup>, Fe<sup>3+</sup>; V shows 2, 3, 4, 5) because both  $(n-1)d$  and  $ns$  contribute. P-block states differ by **two** (Sn<sup>2+</sup>, Sn<sup>4+</sup>) because there the p orbital dominates and electrons are lost in pairs.

### ***Zero oxidation state in carbonyls***

In Ni(CO)<sub>4</sub> and Fe(CO)<sub>5</sub> the metal is in oxidation state **zero**. CO is neutral and forms a sigma bond to the metal while accepting d-electron density back as a pi bond — so no charge is transferred overall.

## Standard electrode potentials — *reading the $E^\circ$ table for this series*

Across the 3d series,  $E^\circ(\text{M}^{2+}/\text{M})$  becomes **less negative** — the tendency to form the +2 ion decreases — because ionization enthalpies rise as nuclear charge increases across the row.

### **Copper is the exception — positive $E^\circ$**

$E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ , meaning copper prefers to stay as the solid metal. Converting  $\text{Cu(s)}$  to  $\text{Cu}^{2+}$  needs a high enthalpy of atomization *plus* a high sum of first and second ionization enthalpies, and the energy released on hydration doesn't compensate.

**Consequence asked directly:** copper does not displace hydrogen from acids. It lies below hydrogen in the electrochemical series with a positive reduction potential, so it can't reduce  $\text{H}^+$  to  $\text{H}_2$ . Only oxidising acids ( $\text{HNO}_3$ , hot conc.  $\text{H}_2\text{SO}_4$ ) dissolve it.

### **Three more anomalies, each with the same explanation shape**

- **Mn, Ni, Zn are more negative than the trend predicts.**  $\text{Mn}^{2+}$  is  $3d^5$  (half-filled, stable);  $\text{Zn}^{2+}$  is  $3d^{10}$  (filled, stable);  $\text{Ni}^{2+}$  is small so its hydration enthalpy is exceptionally large and negative.
- $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$  **is highly positive** —  $\text{Mn}^{3+}$  ( $3d^4$ ) readily gains an electron to become the stable  $3d^5$ . So  $\text{Mn}^{3+}$  is a **good oxidising agent**.
- $E^\circ(\text{Cr}^{3+}/\text{Cr}^{2+})$  **is negative** —  $\text{Cr}^{2+}$  ( $3d^4$ ) readily loses an electron to become  $3d^3$ , which is stable ( $t_{2g}^3$ , half-filled in the lower set). So  $\text{Cr}^{2+}$  is a **strong reducing agent**.

**Trap:**  $\text{Mn}^{3+}$  oxidising and  $\text{Cr}^{2+}$  reducing are a matched pair, both answered by "which side is the stable configuration". Learn them together — the exam often asks for both in one question.

$\text{Cu}^+$  **is unstable in aqueous solution** — it disproportionates to  $\text{Cu}^{2+}$  and  $\text{Cu}$ , because the much more negative hydration enthalpy of the smaller  $\text{Cu}^{2+}$  more than compensates for the second ionization enthalpy.

## **Magnetic properties — and the one formula you'll actually compute**

**[exposure] Paramagnetic** substances contain **unpaired electrons** and are weakly *attracted* into a magnetic field. **Diamagnetic** substances have all electrons paired and are weakly *repelled*. The distinction exists because an unpaired electron behaves like a tiny magnet whose field isn't cancelled by a partner. Most transition metal ions have unpaired d electrons, so most of their compounds are paramagnetic —  $\text{Mn}^{2+}$  ( $3d^5$ , five unpaired) is strongly so, while  $\text{Zn}^{2+}$  ( $3d^{10}$ ) is diamagnetic.

The strength is measured as **spin-only magnetic moment**:

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

$n$  = number of unpaired electrons; BM = Bohr magneton.

| Ion              | Config          | n | $\mu$ (BM) |
|------------------|-----------------|---|------------|
| Sc <sup>3+</sup> | 3d <sup>0</sup> | 0 | 0          |
| Ti <sup>3+</sup> | 3d <sup>1</sup> | 1 | 1.73       |
| V <sup>3+</sup>  | 3d <sup>2</sup> | 2 | 2.83       |
| Cr <sup>3+</sup> | 3d <sup>3</sup> | 3 | 3.87       |
| Fe <sup>2+</sup> | 3d <sup>6</sup> | 4 | 4.90       |
| Mn <sup>2+</sup> | 3d <sup>5</sup> | 5 | 5.92       |

Worked · 2020 — spin-only moment of Co<sup>2+</sup>

Co has  $Z = 27$ : [Ar] 3d<sup>7</sup> 4s<sup>2</sup>. Removing two electrons (the 4s pair) gives Co<sup>2+</sup> = [Ar] 3d<sup>7</sup>.

Filling five d orbitals with 7 electrons: two orbitals paired, three singly occupied →  $n = 3$ .

$$\mu = \sqrt{3(3 + 2)} = \sqrt{15} = 3.87 \text{ BM}$$

**Trap:** the 4s electrons come off **first**, before any 3d electron. Co<sup>2+</sup> is 3d<sup>7</sup>, not 3d<sup>5</sup>4s<sup>2</sup>.

## Why transition compounds are coloured — *d-d transition, and the one exception to it*

**[exposure]** A **d-d transition** is what happens when an electron absorbs light and jumps from a lower-energy d orbital to a higher-energy one. In a complex, the five d orbitals no longer all have the same energy — they split into two sets (called  $t_{2g}$  and  $e_g$ , covered properly in Coordination Compounds). The gap between them happens to match visible light, so the compound absorbs one colour and we see the **complementary** colour. Cu<sup>2+</sup> (3d<sup>9</sup>) has an unpaired electron available to jump, so its salts are blue; Zn<sup>2+</sup> (3d<sup>10</sup>) has no vacancy to jump into, so its salts are white.

**The rule:** colour requires **partially filled** d orbitals. Both d<sup>0</sup> and d<sup>10</sup> are colourless.

| Ion                                 | Config           | Colour     |
|-------------------------------------|------------------|------------|
| Sc <sup>3+</sup> , Ti <sup>4+</sup> | 3d <sup>0</sup>  | Colourless |
| Ti <sup>3+</sup>                    | 3d <sup>1</sup>  | Purple     |
| Cr <sup>3+</sup>                    | 3d <sup>3</sup>  | Green      |
| Mn <sup>2+</sup>                    | 3d <sup>5</sup>  | Pink       |
| Fe <sup>2+</sup>                    | 3d <sup>6</sup>  | Green      |
| Cu <sup>2+</sup>                    | 3d <sup>9</sup>  | Blue       |
| Zn <sup>2+</sup>                    | 3d <sup>10</sup> | Colourless |

### **The exception: charge transfer**

**[exposure]** **Ligand-to-metal charge transfer** is a second, different way a compound can be coloured — an electron jumps from the surrounding ligand onto the metal, rather than between two d orbitals. It matters because it explains the compounds that are intensely coloured despite having *no* d electrons at all. KMnO<sub>4</sub> is the standard case: Mn is +7, so 3d<sup>0</sup>,

and no d-d transition is possible — yet it's deep purple, because charge transfers from oxygen's 2p orbital to manganese's 3d. The same explains  $\text{K}_2\text{Cr}_2\text{O}_7$ 's orange.

**Examiner asks:** "why is  $\text{KMnO}_4$  coloured despite  $3d^0$ ?" — charge transfer, not d-d. Getting this distinction right is a whole mark.

## **Catalysis, interstitial compounds, alloys — *three properties, three clean explanations***

### ***Catalytic behaviour***

Two reasons, and good answers give both:

1. **Variable oxidation state** lets the metal form intermediates by changing state, then revert. The classic:  $\text{Fe}^{3+}$  catalysing the iodide-persulphate reaction —  $\text{Fe}^{3+}$  oxidises  $\text{I}^-$  to  $\text{I}_2$  and becomes  $\text{Fe}^{2+}$ , which is then re-oxidised by persulphate back to  $\text{Fe}^{3+}$ .
2. **Surface adsorption** — reactants stick to the metal surface, which raises their local concentration and weakens their bonds, making reaction easier.

$\text{V}_2\text{O}_5$  is asked specifically: large surface area, and it changes oxidation state easily to form an unstable intermediate, providing a lower-energy path.

### ***Interstitial compounds***

**[exposure]** An **interstitial compound** forms when small atoms — hydrogen, boron, carbon, nitrogen — get trapped in the gaps between metal atoms in a crystal lattice. Transition metals sit in three-dimensional lattices with spaces (**interstitial sites**) between the atoms, and small atoms slot into them without displacing anything. The result isn't a normal compound: the ratio isn't a whole number, so these are called **non-stoichiometric**.  $\text{VH}_{0.56}$  means 56 hydrogen atoms per 100 vanadium atoms;  $\text{TiH}_{1.7}$  means 170 per 100.

**Properties:** higher melting points than the pure metal, very hard (some borides approach diamond), retain metallic conductivity, and are chemically inert.

### ***Alloy formation***

**[exposure]** An **alloy** is a homogeneous solid solution of two or more metals — melt them together and let them solidify. It forms readily when the metals' atomic radii are within about **15%** of each other, because then one metal's atoms can simply take the other's positions in the crystal lattice. Transition metals have very similar atomic sizes, so they alloy easily. Ferrous alloys (Fe with Cr, V, W, Mo, Mn) give steel and stainless steel; brass is Cu-Zn and bronze is Cu-Sn, both pairing a transition with a non-transition metal.

**Examiner asks:** "why do transition metals form alloys" — similar atomic size, so one metal replaces the other in the lattice. Two lines, two marks.

## **Oxides and oxoanions — *the acid-base trend that gets asked every year***

The **highest oxidation number in an oxide coincides with the group number** from Sc to Mn:  $\text{Sc}_2\text{O}_3$  (+3, group 3) up to  $\text{Mn}_2\text{O}_7$  (+7, group 7). Beyond group 7 no higher oxide than  $\text{Fe}_2\text{O}_3$  is known.

### The acid-base rule

| Oxidation state of metal | Oxide is   | Chromium example                    |
|--------------------------|------------|-------------------------------------|
| Low                      | Basic      | CrO (+2)                            |
| Intermediate             | Amphoteric | Cr <sub>2</sub> O <sub>3</sub> (+3) |
| High                     | Acidic     | CrO <sub>3</sub> (+6)               |

**Why:** in a low oxidation state the metal has d electrons available to *donate*, so it behaves as a Lewis base. In a high oxidation state the large positive charge lets it *accept* an electron pair — a Lewis acid. As oxidation state rises, metallic character falls and covalent character rises.

**Examiner asks:** "Mn<sub>2</sub>O<sub>3</sub> is basic whereas Mn<sub>2</sub>O<sub>7</sub> is acidic" (2019) — a direct application. Give the oxidation numbers (+3 and +7) and the Lewis reasoning.

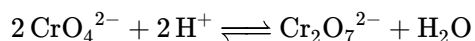
### Oxygen beats fluorine at stabilising high oxidation states

Manganese's highest fluoride is MnF<sub>4</sub> (+4), but its highest oxide is Mn<sub>2</sub>O<sub>7</sub> (+7) — even though fluorine is more electronegative. The reason: **oxygen can form multiple bonds** with the metal, fluorine only single bonds. More bonds per atom means a higher oxidation state can be supported.

### Potassium dichromate — preparation, the pH equilibrium, structures, oxidising action

#### The pH interconversion — the most-asked single fact

Chromate and dichromate exist in equilibrium, and pH decides which:

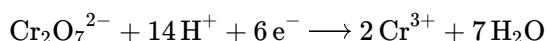


- **Acidic** (high H<sup>+</sup>) → equilibrium shifts right → **dichromate**, Cr<sub>2</sub>O<sub>7</sub><sup>2-</sup>, **orange**.
- **Alkaline** → shifts left → **chromate**, CrO<sub>4</sub><sup>2-</sup>, **yellow**.

#### Structures

**Chromate** CrO<sub>4</sub><sup>2-</sup>: **tetrahedral**, yellow. **Dichromate** Cr<sub>2</sub>O<sub>7</sub><sup>2-</sup>: **two tetrahedra sharing one corner** through a Cr-O-Cr bridge, bond angle **126°**, orange.

#### Oxidising action in acidic medium



Combine this reduction half with any oxidation half, matching electrons:

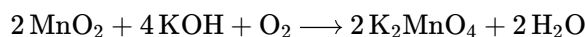
- $6 \text{Fe}^{2+} \longrightarrow 6 \text{Fe}^{3+} + 6 \text{e}^- \rightarrow \text{Cr}_2\text{O}_7^{2-} + 6 \text{Fe}^{2+} + 14 \text{H}^+ \longrightarrow 2 \text{Cr}^{3+} + 6 \text{Fe}^{3+} + 7 \text{H}_2\text{O}$
- I<sup>-</sup> → I<sub>2</sub>, H<sub>2</sub>S → S, Sn<sup>2+</sup> → Sn<sup>4+</sup> all work the same way.

Potassium dichromate is used as a **primary standard** in volumetric analysis; sodium dichromate is more soluble and preferred in organic oxidations.

## Potassium permanganate — *preparation, why it's purple, and both media*

### Preparation — two steps

**1. Fusion.**  $\text{MnO}_2$  fused with  $\text{KOH}$  in air (or with  $\text{KNO}_3$  as oxidant) gives green potassium manganate:

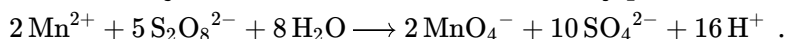


**2. Oxidation of manganate to permanganate**, either by disproportionation in acid or — commercially — by electrolytic oxidation in alkaline solution:



**[exposure]** That second reaction is a **disproportionation** — a reaction where *the same element* is simultaneously oxidised and reduced. It's a distinct category because normally one species oxidises while a different one reduces. Here manganese starts at +6 in  $\text{MnO}_4^{2-}$  and ends up both higher (+7 in  $\text{MnO}_4^-$ ) and lower (+4 in  $\text{MnO}_2$ ) in the same step.

**Laboratory route:** a  $\text{Mn}^{2+}$  salt oxidised by peroxodisulphate:



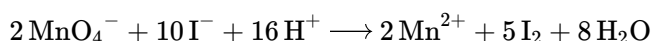
### Structure and colour

Both  $\text{MnO}_4^-$  and  $\text{MnO}_4^{2-}$  are **tetrahedral**. Manganate ( $\text{MnO}_4^{2-}$ , Mn is +6,  $3d^1$ ) is **green and paramagnetic** — one unpaired electron. Permanganate ( $\text{MnO}_4^-$ , Mn is +7,  $3d^0$ ) is **purple and diamagnetic**. Its colour comes from **charge transfer**, not d-d, since it has no d electrons.

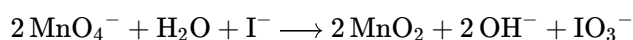
Heating above 513 K decomposes it:  $2 \text{KMnO}_4 \longrightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$ .

### Oxidising action — both media, both asked

**Acidic:**  $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$  (5 electrons)



**Alkaline/neutral:**  $\text{MnO}_4^- \rightarrow \text{MnO}_2$  (3 electrons), and iodide goes to **iodate**, not iodine:



**Trap:** the products differ by medium. Acidic gives  $\text{Mn}^{2+}$  and  $\text{I}_2$ ; alkaline gives  $\text{MnO}_2$  and  $\text{IO}_3^-$ . Writing the acidic products for an alkaline question loses the whole answer.

**Examiner asks:** "why is permanganate titration not carried out in presence of  $\text{HCl}$ ?" —  $\text{KMnO}_4$  oxidises  $\text{HCl}$  to  $\text{Cl}_2$ , a side reaction giving an inaccurate end point. Use  $\text{H}_2\text{SO}_4$  instead.

## Lanthanoids — *the f block, and the contraction that explains everything*

**[exposure]** The **lanthanoids** are the 14 elements following lanthanum (Ce,  $Z = 58$ , to Lu,  $Z = 71$ ) in which the **4f orbital** progressively fills. They're pulled out of group 3 and printed as a separate row because they're so chemically alike that keeping them in the main table would

be unwieldy. General configuration:  $4f^{1-14} 5d^{0-1} 6s^2$ . Their dominant oxidation state is +3, with a few showing +2 or +4 where that gets them closer to an empty, half-filled or full f subshell.

**[exposure] Lanthanoid contraction** is the steady decrease in atomic and ionic radii across the lanthanoid series, from lanthanum to lutetium, as atomic number rises. It happens because 4f electrons **shield poorly** — an f electron is diffuse and doesn't screen the outer electrons from the nucleus effectively. So each added proton pulls the whole electron cloud in a little more, and the atoms shrink steadily instead of staying the same size. The effect is small per element but accumulates over 14 elements into something chemically decisive.

### ***Its three consequences — the reason it's examined so heavily***

1. **Lanthanoids are hard to separate.** Similar radii → similar chemical properties → separating a mixture is difficult.
2. **4d and 5d elements have nearly identical sizes** (Zr and Hf, Nb and Ta), because the contraction cancels the expected size increase.
3. **Second and third transition series resemble each other** far more than either resembles the first.

$\text{Eu}^{2+}$  **is a strong reducing agent** — +3 is the stable lanthanoid state, so  $\text{Eu}^{2+}$  readily loses an electron to reach it, reducing something else in the process.

**[exposure] Misch metal** is a commercially useful lanthanoid alloy — roughly **95% lanthanoid, 5% iron**, with traces of S, C, Ca and Al. It exists because separating individual lanthanoids is expensive and often unnecessary; the mixture works fine. It's used in magnesium-based alloys and to make **bullets, shells and lighter flints**, since it sparks when struck.

### ***Actinoids — and the three-way comparison with lanthanoids***

**[exposure]** The **actinoids** are the 14 elements after actinium (Th,  $Z = 90$ , to Lr,  $Z = 103$ ) in which the **5f orbital** fills. What sets them apart from lanthanoids is that **all of them are radioactive**, and their 5f, 6d and 7s orbitals are so close in energy that electrons from all three join in bonding — giving a much wider range of oxidation states (+3 through +6 and beyond) than the lanthanoids' near-uniform +3. That, plus the radioactivity, is why their chemistry is described as more complicated.

**Actinoid contraction is greater than lanthanoid contraction**, because 5f electrons shield even more poorly than 4f, so effective nuclear charge rises faster and the radii shrink more sharply.

### ***The three differences to write***

|                  | <b>Lanthanoids</b>     | <b>Actinoids</b>                  |
|------------------|------------------------|-----------------------------------|
| Orbital filling  | 4f                     | 5f                                |
| Shielding        | Poor                   | Even poorer → greater contraction |
| Oxidation states | Mainly +3; some +2, +4 | +3 common, but also +4, +5, +6    |
| Radioactivity    | Only promethium        | All of them                       |

**Examiner asks:** "three points of difference" (2020) — pick orbital, contraction magnitude, and oxidation-state range. Radioactivity is the easy fourth if you need it.

## Numerical and equation patterns, collected — *the calculable parts of a mostly-descriptive chapter*

A · Spin-only magnetic moment — 2-3 marks

*Recognise it: an atomic number is given, or "calculate the magnetic moment".*

1. Write the neutral atom's configuration from  $Z$ .
2. Remove electrons for the ion — **4s first**, then 3d.
3. Fill the five d orbitals singly before pairing; count unpaired =  $n$ .
4.  $\mu = \sqrt{n(n+2)}$  BM.

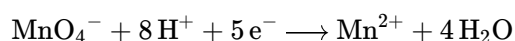
B · Balancing a redox equation — 3-5 marks

*Recognise it: "complete and balance", usually with  $\text{MnO}_4^-$  or  $\text{Cr}_2\text{O}_7^{2-}$  in acidic medium.*

1. Write reduction and oxidation halves separately.
2. Balance the metal, then O with  $\text{H}_2\text{O}$ , then H with  $\text{H}^+$ .
3. Balance charge with electrons.
4. Scale both halves to equal electrons, add, cancel.

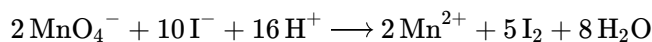
Worked · 2025 — permanganate and iodide in acid

**Reduction:**  $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ . Add 4  $\text{H}_2\text{O}$  right, 8  $\text{H}^+$  left. Charge: left  $-1 + 8 = +7$ , right  $+2$ . Add 5 electrons left:



**Oxidation:**  $2 \text{I}^- \rightarrow \text{I}_2 + 2 \text{e}^-$

Multiply reduction by 2, oxidation by 5:



C · Oxidation number from a formula — 1-2 marks

*Recognise it: "write the oxoanion in which the metal shows a state equal to its group number".*

1. Set the metal as  $x$ ; oxygen is  $-2$  each; sum to the overall charge.
2. For Mn (group 7):  $\text{MnO}_4^-$  gives  $x - 8 = -1$ , so  $x = +7$ . For Cr (group 6):  $\text{CrO}_4^{2-}$  or  $\text{Cr}_2\text{O}_7^{2-}$ , both  $+6$ .

D · Identify the compound (A, B, C chains) — 5 marks

*Recognise it: a colour-coded sequence — "black-brown solid A fused with KOH gives dark green B, which on oxidation gives dark purple C".*

1. Match colours to compounds: black-brown  $\text{MnO}_2 \rightarrow$  dark green  $\text{K}_2\text{MnO}_4 \rightarrow$  dark purple  $\text{KMnO}_4$ .
2. Write each step's equation.

## Past year questions · question types — *ranked by how often they turn up*

*From the last 10 years of boards — 25 questions, grouped by pattern. This chapter is overwhelmingly reason-based: most marks come from explaining, not calculating.*

1 · "Give reason" on configuration stability — 2-3 marks

*Recognise it: any comparison of two ions, oxidising/reducing strength, or an  $E^\circ$  anomaly.*

1. Write both configurations, before and after electron transfer.
2. Point at whichever is  $d^5$  or  $d^{10}$ .
3. State the consequence in the question's own language.

**Trap:** answering "because it's stable" without naming *which* configuration. The  $3d^5$  is the mark.

2 · Balancing redox equations — 3-5 marks

*Recognise it: "complete and balance the following".*

1. Halves separately.
2. Metal  $\rightarrow$  O with water  $\rightarrow$  H with  $H^+$   $\rightarrow$  charge with electrons.
3. Scale, add, cancel.

3 · Lanthanoid / actinoid comparison — 3 marks

*Recognise it: "lanthanoid contraction", "three differences", "why is actinoid chemistry more complicated".*

1. Name the orbital (4f vs 5f).
2. Poor shielding  $\rightarrow$  contraction; 5f worse than 4f.
3. Add the oxidation-state range and radioactivity.

4 · Magnetic and colour properties — 2-3 marks

*Recognise it: "which is colourless/paramagnetic and why", or a  $\mu$  calculation.*

1. Get the ion's d-configuration.
2. Count unpaired electrons.
3. None  $\rightarrow$  colourless and diamagnetic. Some  $\rightarrow$  coloured and paramagnetic.
4. Check for the charge-transfer exception ( $KMnO_4$ ,  $K_2Cr_2O_7$ ).

5 ·  $KMnO_4$  /  $K_2Cr_2O_7$  preparation and reactions — 3-5 marks

*Recognise it: named reagent questions, or the A/B/C identification chain.*

1. Preparation equations.
2. The pH interconversion for chromate/dichromate.
3. Oxidising action — check which medium is asked.

6 · Physical-property trends — 2-3 marks

*Recognise it: enthalpy of atomization, melting point, or a graph across the series.*

1. Count unpaired electrons.
2. More unpaired  $\rightarrow$  stronger metallic bond  $\rightarrow$  higher value.
3. For 2nd/3rd series, add the poor-shielding argument.

## Past year questions · mark slots — *what each type is worth*

| Question type                                                              | Slot                        | Time    |
|----------------------------------------------------------------------------|-----------------------------|---------|
| "Give reason" (configuration)                                              | 2–3 marker                  | 2–3 min |
| Balancing redox equations                                                  | 3-marker; both media is a 5 | 5–7 min |
| Lanthanoid/actinoid comparison                                             | 3-marker                    | 3 min   |
| Magnetic moment calculation                                                | 2–3 marker                  | 3 min   |
| KMnO <sub>4</sub> /K <sub>2</sub> Cr <sub>2</sub> O <sub>7</sub> chemistry | 3–5 marker                  | 5–7 min |
| Graph/trend interpretation                                                 | 3-marker                    | 4 min   |
| Definitions (misch metal, oxoanion)                                        | 1–2 marker                  | 2 min   |

*The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — the KMnO<sub>4</sub> chemistry and multi-part "answer the following" sets are where the 5-markers sit.*

## Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2017 Q3 · 2017 Q8 · 2019 Q2

"Transition metals show variable oxidation state" and "Zn, Cd, Hg are soft metals" — asked together, repeatedly, in the same two-part format.

2024 Q5 · lecture PYQ

Define lanthanoid contraction, then why actinoid contraction is greater. The single most reliable f-block question.

2025 Q4 · 2026 Q25(b)

Balance MnO<sub>4</sub><sup>−</sup> with I<sup>−</sup> — once in acidic medium alone, once acidic *and* alkaline in the same question.

2022 Q15 · 2022 Q16 · lecture PYQ

Graph questions on enthalpy of atomization and E° values — lowest in 3d series (Zn), why 2nd/3rd series are higher, why Mn's E° is exceptionally negative, why Cu's is exceptionally positive.

2020 Q11 · 2026 Q22

Lanthanoids vs actinoids — three differences, then why actinoid chemistry is more complicated.

2020 Q10 · 2022 Q12 · 2020 Q6

Colour and magnetism: Sc<sup>3+</sup> vs Cr<sup>3+</sup>, which ion is colourless, spin-only moment of Co<sup>2+</sup>. All answered by counting unpaired electrons.

2023 Q17 · 2017 Q13 · 2020 Q20

Mn's +7 state, transition metals as catalysts, and interstitial compound formation. Standard reason-based trio.

## Past year questions · cold practice — *answers only* — *work them before you look*

### Configuration reasoning

- 2026 Q7(a) — why is the third ionization enthalpy of Mn high?  $\text{Mn}^{2+}$  is  $3d^5$ , half-filled and stable; removing a third electron breaks it
- 2019 Q13(c) — why is  $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$  highly positive compared with  $\text{Cr}^{3+}/\text{Cr}^{2+}$ ?  $\text{Mn}^{3+}$  gains one to reach stable  $3d^5$ ;  $\text{Cr}^{3+}$  is already stable  $3d^3$
- 2017 Q9(a) — how does variability of oxidation state differ between transition and p-block? Transition differ by 1 unit; p-block by 2
- 2026 Q21(a) — why is Cu a transition element despite a filled d orbital?  $\text{Cu}^{2+}$  is  $3d^9$  — partially filled in its common oxidation state

### Magnetism and colour

- 2020 Q6 — spin-only moment of  $\text{Co}^{2+}$ ,  $Z = 27$ .  $3d^7$ ,  $n = 3$ ,  $\mu = 3.87$  BM
- 2022 Q12 — which of  $\text{Ti}^{4+}$ ,  $\text{Cr}^{3+}$ ,  $\text{V}^{3+}$  is colourless and why?  $\text{Ti}^{4+}$  —  $3d^0$ , no d-d transition possible
- 2020 Q10 —  $\text{Sc}^{3+}$  or  $\text{Cr}^{3+}$ : which shows magnetic behaviour?  $\text{Cr}^{3+}$  ( $3d^3$ , paramagnetic);  $\text{Sc}^{3+}$  is  $3d^0$ , diamagnetic
- 2026 Q21(b) —  $\text{KMnO}_4$  or  $\text{K}_2\text{MnO}_4$ : which is paramagnetic?  $\text{K}_2\text{MnO}_4$  — Mn is +6,  $3d^1$ , one unpaired electron

### Equations

- 2025 Q4 — balance  $\text{Cr}_2\text{O}_7^{2-}$  with  $\text{Fe}^{2+}$  in acid.  
$$\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \longrightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$$
- 2026 Q25(b) —  $\text{MnO}_4^-$  with  $\text{I}^-$  in alkaline medium.  
$$2\text{MnO}_4^- + \text{H}_2\text{O} + \text{I}^- \longrightarrow 2\text{MnO}_2 + 2\text{OH}^- + \text{IO}_3^-$$
- 2026 Q25(a) — identify A, B, C: black-brown solid fused with KOH in air  $\rightarrow$  dark green  $\rightarrow$  electrolytic oxidation  $\rightarrow$  dark purple. A =  $\text{MnO}_2$ , B =  $\text{K}_2\text{MnO}_4$ , C =  $\text{KMnO}_4$
- 2023 — what happens when  $\text{KMnO}_4$  is heated?  $2\text{KMnO}_4 \longrightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$  at 513 K

### f block

- 2019 Q19 — why is  $\text{Eu}^{2+}$  a strong reducing agent? Loses one electron to reach the stable +3 lanthanoid state
- 2019 Q19(b) — why is separating a lanthanoid mixture difficult? Similar radii from lanthanoid contraction  $\rightarrow$  similar chemical properties
- 2019 Q14(a) — general electronic configuration of lanthanoids.  $4f^{1-14} 5d^{0-1} 6s^2$
- 2025 Q24(a) — what is misch metal, and one use? ~95% lanthanoid + 5% Fe with traces; used in Mg alloys, bullets, shells, lighter flints

Built from Sourabh Raina's d and f Block one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 4 (Rationalised 2022–23). First-contact material: every term marked **[exposure]** is explained from scratch before use.

# Coordination Compounds

NCERT Class XII Chemistry · Chapter 5 · Coordination Compounds

*11 marks, and entirely new ground. This chapter has more vocabulary than any other in the paper, so nothing here assumes you've met a term before — every one tagged [exposure] is defined in plain words, with a worked example, before it gets used again.*

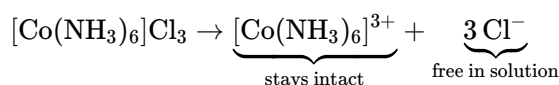
## What a coordination compound actually is — *start here — the whole chapter builds on this one picture*

**[exposure]** A **coordination compound** is a compound in which a central metal atom or ion is surrounded by, and bonded to, a fixed number of molecules or negative ions. Transition metals form these in huge numbers because they have small, highly charged ions and empty d orbitals ready to accept electron pairs. The bond isn't the usual "one electron each" kind — the surrounding species donates *both* electrons. A standard example is  $[\text{Co}(\text{NH}_3)_6]^{3+}$ : one cobalt ion at the centre, six ammonia molecules attached around it.

They matter well beyond the exam: **chlorophyll** is a coordination compound of magnesium, **haemoglobin** of iron, and **vitamin B<sub>12</sub>** of cobalt. Industrially they're used in electroplating, dyeing, medicine, metal extraction and catalysis.

### The square bracket is the whole notation

Everything **inside** the square bracket travels together as one unit and does *not* break apart in water. Everything **outside** is a separate ion that dissociates freely.



### Complex vs double salt — the distinction they examine

|                 | Double salt                                                                             | Complex                                             |
|-----------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------|
| In water        | Dissociates completely into all its ions                                                | The bracketed part keeps its identity               |
| Example         | Mohr's salt $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6 \text{H}_2\text{O}$ | $\text{K}_4[\text{Fe}(\text{CN})_6]$                |
| What you detect | $\text{Fe}^{2+}$ gives a positive test                                                  | No free $\text{Fe}^{2+}$ — the complex ion survives |

Also double salts: carnallite  $\text{KCl} \cdot \text{MgCl}_2 \cdot 6 \text{H}_2\text{O}$  and potash alum  $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$ . They exist only in the solid state.

**Examiner asks:** "difference between a complex and a double salt" (2019) — answer with the water behaviour and one example each.

### Werner's theory — the 1898 idea that made the whole subject make sense

Alfred Werner explained a puzzle: why does  $\text{CoCl}_3$  combine with different numbers of  $\text{NH}_3$  to give differently coloured compounds with different behaviour? His answer was that a metal has **two kinds of valence**.

**[exposure]** **Primary valence** is the metal's ordinary oxidation number — the charge it carries. It is **ionisable**, meaning whatever satisfies it can break away in water, and it is satisfied by **negative ions**. In  $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$  cobalt is +3, so its primary valence is 3, satisfied

by the three chloride ions sitting outside the bracket — which is exactly why those three come free in solution.

**[exposure] Secondary valence** is the number of species directly bonded to the metal — what we now call the coordination number. It is **non-ionisable**, so these stay attached in water, and it's satisfied by neutral molecules or negative ions that donate a lone pair. It is **fixed for a given metal**. In  $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$  the secondary valence is 6, satisfied by the six ammonia molecules inside the bracket. Those ligands arrange themselves in a definite shape in space — the **coordination polyhedron**.

### ***How Werner proved it — the silver nitrate experiment***

Add excess  $\text{AgNO}_3$  to each compound. Only **free** chloride ions precipitate as  $\text{AgCl}$ , so counting the moles of  $\text{AgCl}$  tells you how many chlorides are outside the bracket:

| Formula                             | Colour | Moles AgCl | Actual structure                                 |
|-------------------------------------|--------|------------|--------------------------------------------------|
| $\text{CoCl}_3 \cdot 6 \text{NH}_3$ | Yellow | 3          | $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$          |
| $\text{CoCl}_3 \cdot 5 \text{NH}_3$ | Purple | 2          | $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ |
| $\text{CoCl}_3 \cdot 4 \text{NH}_3$ | Green  | 1          | $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ |

Notice the coordination number stays **6** throughout — as ammonia leaves the bracket, chloride moves in to take its place.

### ***The vocabulary — nine terms, each with its own example***

**[exposure] Central atom or ion** — the metal at the middle, the one everything else attaches to. It acts as a **Lewis acid**: it's electron-deficient, so it *accepts* electron pairs. In  $[\text{Ni}(\text{H}_2\text{O})_4\text{Cl}_2]$  the central ion is  $\text{Ni}^{2+}$ ; in  $[\text{Fe}(\text{CN})_6]^{3-}$  it's  $\text{Fe}^{3+}$ .

**[exposure] Ligand** — a molecule or ion attached to the central metal. It must have at least one **lone pair of electrons** to donate, which is what lets it act as a **Lewis base**. Ligands range from simple ions ( $\text{Cl}^-$ ,  $\text{CN}^-$ ) through small molecules ( $\text{H}_2\text{O}$ ,  $\text{NH}_3$ ) to whole proteins. The specific atom in the ligand that donates the pair is the **donor atom** — in  $\text{NH}_3$  it's the nitrogen.

**[exposure] Coordinate bond** — the bond formed when *one* species supplies *both* electrons of the shared pair. It exists as a separate name because an ordinary covalent bond takes one electron from each partner. Once formed it's indistinguishable from a normal covalent bond; only its origin differs. Every metal-ligand bond in this chapter is one.

**[exposure] Coordination number** — the number of donor atoms directly bonded to the metal. Count **donor atoms, not ligands**, and count only sigma bonds. In  $[\text{Co}(\text{NH}_3)_6]^{3+}$  six ammonias give six donor atoms, so  $\text{CN} = 6$ . But in  $[\text{Co}(\text{en})_3]^{3+}$  there are only three ligands — each supplies *two* nitrogen donors, so  $\text{CN} = 3 \times 2 = 6$  again.

**[exposure] Coordination sphere** — the central metal plus its ligands, written inside the square bracket together with the overall charge. Whatever sits outside is the **counter ion**, which is the ionisable part. In  $\text{K}_4[\text{Fe}(\text{CN})_6]$  the coordination sphere is  $[\text{Fe}(\text{CN})_6]^{4-}$  and the counter ions are four  $\text{K}^+$ .

**[exposure] Coordination polyhedron** — the shape the ligands make in space around the metal. It exists as a term because the geometry, not just the count, determines the

compound's properties and isomers. Six ligands give an **octahedron** (90° angles), four give either a **tetrahedron** (109.5°) or a **square planar** arrangement (90°, all in one plane).

### ***Oxidation number of the central atom***

The charge the metal would carry if every ligand were removed along with its donated electron pair. Write it as a Roman numeral in brackets after the metal's name.

**Method:** set the metal as  $x$ , add each ligand's charge, set the sum equal to the overall charge on the bracket. For  $[\text{Cu}(\text{CN})_4]^{3-}$ :  $x + 4(-1) = -3$ , so  $x = +1 \rightarrow \text{copper(I)}$ .

### ***Homoleptic vs heteroleptic***

**Homoleptic** — one type of ligand only, e.g.  $[\text{Co}(\text{NH}_3)_6]^{3+}$ . **Heteroleptic** — more than one type, e.g.  $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ .

### ***Types of ligand — denticity, ambidentate, and the chelate effect***

**[exposure] Denticity** is simply how many donor atoms a single ligand uses to grip the metal — from Latin *dens*, tooth. The word exists because a ligand with several donor atoms behaves very differently from one with a single donor: it can clamp onto the metal at multiple points at once. A ligand with one donor site is **unidentate**, two is **didentate**, three or more is **polydentate**.

| Type        | Donor sites | Examples                                                                             |
|-------------|-------------|--------------------------------------------------------------------------------------|
| Unidentate  | 1           | $\text{Cl}^-$ , $\text{NH}_3$ , $\text{H}_2\text{O}$ , $\text{OH}^-$ , $\text{CN}^-$ |
| Didentate   | 2           | ethane-1,2-diamine (en); oxalate $\text{C}_2\text{O}_4^{2-}$ (ox)                    |
| Polydentate | 3+          | $\text{EDTA}^{4-}$ — <b>hexadentate</b>                                              |

**$\text{EDTA}^{4-}$**  is worth knowing precisely: it binds through **two nitrogen atoms and four oxygen atoms**, six donor sites in total.

**[exposure] An ambidentate ligand** is a unidentate ligand that has *more than one kind* of donor atom available, and can attach through either one — but only one at a time. It's a distinct category because it creates isomers that differ purely in which atom did the bonding.  $\text{NO}_2^-$  can bind through nitrogen (called **nitrito-N**) or through oxygen (**nitrito-O**).  $\text{SCN}^-$  can bind through sulphur (**thiocyanato**) or nitrogen (**isothiocyanato**).

**[exposure] A chelate ligand** is a didentate or polydentate ligand that grips one metal ion at two or more points, forming a **ring** that includes the metal — the word comes from the Greek for a crab's claw. It matters because chelate complexes are markedly **more stable** than equivalent complexes made from separate unidentate ligands. In  $[\text{Pt}(\text{en})\text{Cl}_2]$ , the "en" ligand bonds through both its nitrogens, closing a five-membered ring around the platinum.

**Why chelates are more stable:** replacing six separate unidentate ligands with three didentate ones releases more free particles into solution, so entropy rises; combined with a favourable enthalpy this makes  $\Delta G$  more negative. More rings means more stability.

### ***IUPAC nomenclature — the rules in order, then worked examples***

#### ***Naming a complex — the rules***

1. **Cation first, then anion** — as in any ionic compound.

2. **Ligands before the metal**, in **alphabetical order** regardless of charge.
3. **Anionic ligands end in -o**: chlorido, bromido, cyanido, hydroxido, sulphato, oxalato.
4. **Neutral ligands** keep special names:  $\text{H}_2\text{O}$  = **aqua**,  $\text{NH}_3$  = **ammine** (two m's),  $\text{CO}$  = **carbonyl**,  $\text{NO}$  = nitrosyl.
5. **Prefixes** di, tri, tetra for simple ligands; **bis, tris, tetrakis** when the ligand's own name already contains a prefix (then bracket the ligand).
6. Prefixes are **ignored** when alphabetising.
7. **Oxidation number** in Roman numerals, in brackets, right after the metal.
8. If the complex ion is an **anion**, the metal name ends in **-ate** (ferrate, cuprate, argentate, zincate, cobaltate).

Worked — three names built step by step

$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$  : alphabetise ammine, aqua, chlorido. Four ammines  $\rightarrow$  tetraammine.

Charge:  $x + 0 + 0 - 1 = +2$ , so  $x = +3$ .

$\rightarrow$  **tetraamminaquachloridocobalt(III) chloride**

$[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$  : neutral overall, so  $x - 3 = 0$ ,  $x = +3$ .

$\rightarrow$  **triamminetrichloridochromium(III)**

$[\text{PtCl}_2(\text{en})_2]^{2+}$  : "c" before "e", so chlorido first. "en" already contains "di", so use **bis** and bracket it.  $x + 0 - 2 = +2$ ,  $x = +4$ .

$\rightarrow$  **dichloridobis(ethane-1,2-diamine)platinum(IV)**

$\text{K}_4[\text{Fe}(\text{CN})_6]$  : anionic complex, so iron becomes **ferrate**.  $x - 6 = -4$ ,  $x = +2$ .

$\rightarrow$  **potassium hexacyanidoferrate(II)**

### **Writing the formula from a name**

1. Metal first, then ligands in **alphabetical order**.
2. Polydentate and abbreviated ligands go in **parentheses**; alphabetise abbreviations by their first letter (en  $\rightarrow$  e, ox  $\rightarrow$  o).
3. Whole coordination entity in **square brackets**, no spaces inside.
4. Charge as a right superscript outside the bracket, number before sign.

**Trap:** "ammine" with two m's is  $\text{NH}_3$ ; "amine" with one m is an organic amine like ethane-1,2-diamine. NCERT uses both, in these exact senses. Also: current IUPAC is **chlorido**, not chloro.

## **Isomerism — two families, seven types — the biggest mark-earner in the chapter**

**[exposure] Isomers** are compounds with the same formula but a different arrangement of atoms. In coordination chemistry they split into two families: **stereoisomers**, where the connections are the same but the spatial arrangement differs, and **structural isomers**, where what is bonded to what actually differs. The distinction matters because the two families are detected in completely different ways — stereoisomers by geometry and optical activity, structural ones by simple chemical tests.

## Stereoisomerism 1 • Geometrical (cis-trans)

Same ligands, different positions. **Cis** = identical ligands adjacent; **trans** = opposite.

- **Square planar**  $\text{MX}_2\text{L}_2$  — shows cis and trans. Example:  $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ .
- **Octahedral**  $\text{MX}_2\text{L}_4$  — shows cis and trans. Example:  $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ .
- **Octahedral**  $[\text{CoCl}_2(\text{en})_2]^+$  — cis and trans.
- **Octahedral**  $\text{MA}_3\text{B}_3$  — gives **facial (fac)** where the three identical ligands occupy one triangular face, and **meridional (mer)** where they lie around a meridian. Example:  $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$ .

**Trap: tetrahedral complexes do not show geometrical isomerism.** In a tetrahedron every position is adjacent to every other — all angles stay  $109.5^\circ$  however you swap ligands, so there's no "opposite" to be trans to. This is a standing exam question.

## Stereoisomerism 2 • Optical

**[exposure]** A complex is **optically active** when it has no plane of symmetry, so its mirror image cannot be superimposed on it — like your left and right hands. Such a molecule is **chiral**, and the two mirror-image forms are **enantiomers**: one rotates plane-polarised light to the right (**dextro, d**), the other equally to the left (**laevo, l**). It matters because two compounds identical in every other property can be told apart this way.

Most common in octahedral complexes with didentate ligands:

- $[\text{Cr}(\text{ox})_3]^{3-}$  — type  $\text{M}(\text{AA})_3$ , optically active.
- $[\text{Co}(\text{en})_2\text{Cl}_2]^+$  — type  $\text{M}(\text{AA})_2\text{X}_2$ : the **cis** form is optically active, the **trans** form is not (its mirror image is superimposable).

**Examiner asks:** "draw the geometrical isomer of  $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$  which is optically active" — draw the **cis** form. Trans is always the optically inactive one in these.

## Structural isomerism — four types

| Type                | What differs                                        | Example pair                                                                                                                                |
|---------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Linkage</b>      | Which atom of an ambidentate ligand bonds           | $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ vs $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$                                             |
| <b>Ionisation</b>   | Ligand and counter ion swap places                  | $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ vs $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$                                        |
| <b>Hydrate</b>      | Water inside vs outside the sphere                  | $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (violet) vs $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ |
| <b>Coordination</b> | Ligands swap between a cationic and anionic complex | $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ vs $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$                                |

**The chemical test for ionisation isomers** (asked directly): add  $\text{BaCl}_2$  — the isomer with free sulphate gives a white  $\text{BaSO}_4$  precipitate. Add  $\text{AgNO}_3$  — the isomer with free chloride gives white  $\text{AgCl}$ . Whichever ion is *outside* the bracket is the one that reacts.

## Valence bond theory — the hybridisation method — work every NCERT example the same way

VBT says the metal makes empty **hybrid orbitals** of equal energy, and each ligand donates a lone pair into one of them.

| CN | Hybridisation | Geometry                   |
|----|---------------|----------------------------|
| 4  | $sp^3$        | Tetrahedral                |
| 4  | $dsp^2$       | Square planar              |
| 6  | $d^2sp^3$     | Octahedral (inner orbital) |
| 6  | $sp^3d^2$     | Octahedral (outer orbital) |

**[exposure]** A **strong field ligand** pushes hard enough on the metal's d electrons to force them to **pair up**, freeing inner d orbitals for bonding. A **weak field ligand** doesn't, so the electrons stay unpaired. This single distinction decides the hybridisation, the geometry, and the magnetism of every complex below. **Strong:**  $\text{CN}^-$ , CO, and  $\text{NH}_3$  when the metal is highly charged ( $\text{Co}^{3+}$ ,  $\text{Cr}^{3+}$ ). **Weak:**  $\text{F}^-$ ,  $\text{Cl}^-$ ,  $\text{H}_2\text{O}$ .

### The method — five steps, every time

1. Find the metal's oxidation number, then its d-electron count.
2. Draw the d orbitals and fill them.
3. Ask: strong ligand (pair up) or weak (leave unpaired)?
4. Count how many orbitals the ligands need, and take that many empty ones — **inner** 3d if pairing freed some, otherwise **outer** 4d.
5. Name the hybridisation, the geometry, and the magnetism.

| Complex                           | Metal                     | Ligand | Hybrid    | Geometry      | Magnetism    |
|-----------------------------------|---------------------------|--------|-----------|---------------|--------------|
| $[\text{Co}(\text{NH}_3)_6]^{3+}$ | $\text{Co}^{3+}$ , $3d^6$ | Strong | $d^2sp^3$ | Octahedral    | Diamagnetic  |
| $[\text{CoF}_6]^{3-}$             | $\text{Co}^{3+}$ , $3d^6$ | Weak   | $sp^3d^2$ | Octahedral    | Paramagnetic |
| $[\text{NiCl}_4]^{2-}$            | $\text{Ni}^{2+}$ , $3d^8$ | Weak   | $sp^3$    | Tetrahedral   | Paramagnetic |
| $[\text{Ni}(\text{CN})_4]^{2-}$   | $\text{Ni}^{2+}$ , $3d^8$ | Strong | $dsp^2$   | Square planar | Diamagnetic  |
| $[\text{Ni}(\text{CO})_4]$        | Ni, $3d^8 4s^2$           | Strong | $sp^3$    | Tetrahedral   | Diamagnetic  |

Worked —  $[\text{Ni}(\text{CO})_4]$ , the one that catches people out

CO is neutral, so nickel's oxidation state is **0** — configuration  $3d^8 4s^2$ .

CO is a strong ligand: it forces pairing. The two 4s electrons move into 3d, filling it to  $3d^{10}$  and leaving 4s empty.

Four CO ligands need four orbitals: one 4s + three 4p  $\rightarrow sp^3$ , tetrahedral. All electrons paired  $\rightarrow$  **diamagnetic**.

**Trap:**  $[\text{NiCl}_4]^{2-}$  and  $[\text{Ni}(\text{CN})_4]^{2-}$  have the same metal in the same oxidation state but *different* geometry and magnetism — purely because  $\text{Cl}^-$  is weak and  $\text{CN}^-$  is strong. This exact pair is a favourite comparison.

**[exposure]** **Inner orbital** (or **low spin**, or **spin-paired**) complexes use the metal's *inner* 3d orbitals, which only become free when a strong ligand forces pairing — so they have few or no unpaired electrons. **Outer orbital** (**high spin**, **spin-free**) complexes use the outer 4d orbitals because a weak ligand left the 3d electrons unpaired and in the way. The names are just describing which shell supplied the orbitals:  $[\text{Co}(\text{NH}_3)_6]^{3+}$  is inner/low-spin,  $[\text{CoF}_6]^{3-}$  is outer/high-spin.

## Magnetic moment

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

Measuring  $\mu$  tells you  $n$ , which tells you whether the ligand was strong or weak — this is how VBT is tested experimentally.

### Limitations of VBT — asked as a list

1. Involves many assumptions.
2. Gives no quantitative account of magnetic data.
3. Does not explain the **colour** of complexes.
4. Gives no quantitative account of stability.
5. Cannot predict whether a 4-coordinate complex will be tetrahedral or square planar.
6. Does not distinguish weak from strong ligands — it just assumes the answer.

## Crystal field theory — the model that finally explains colour

**[exposure] Crystal field theory** treats the metal-ligand bond as purely **electrostatic attraction** — the positive metal ion pulling on negative ions or on the negative end of polar molecules. It replaced VBT's covalent picture because it can explain what VBT could not: why complexes are coloured, and why some are high spin and others low. Anionic ligands like  $\text{Cl}^-$  are treated as point negative charges; neutral ones like  $\text{H}_2\text{O}$  as dipoles pointing their negative end at the metal.

**[exposure] Crystal field splitting** is the central idea. In a free metal ion all five d orbitals have identical energy — they are **degenerate**. If you surrounded the ion with an evenly spread sphere of negative charge, all five would rise in energy equally and stay degenerate. But real ligands arrive from *specific directions*, so some d orbitals get pushed much harder than others. The degeneracy breaks and the five orbitals split into two groups at different energies. That energy gap is what visible light interacts with — which is where colour comes from.

### Octahedral field

Ligands approach **along the x, y and z axes**.

- $d_{x^2-y^2}$  and  $d_{z^2}$  point **along** the axes → head-on repulsion → energy **raised**. These two are the  $e_g$  set.
- $d_{xy}$ ,  $d_{yz}$ ,  $d_{zx}$  point **between** the axes → less repulsion → energy **lowered**. These three are the  $t_{2g}$  set.

The gap is the **crystal field splitting energy**,  $\Delta_o$ .

### High spin or low spin — the $\Delta_o$ vs P contest

**[exposure] Pairing energy (P)** is the energy cost of forcing a second electron into an orbital that already has one, against their mutual repulsion. It matters because the fourth d electron faces a genuine choice: pay P and pair up in the lower  $t_{2g}$  set, or pay  $\Delta_o$  and go alone into the upper  $e_g$  set. Whichever costs less, wins.

- $\Delta_o < P$  (weak field ligand): cheaper to jump up. Fourth electron enters  $e_g \rightarrow t_{2g}^3 e_g^1 \rightarrow$  more unpaired electrons → **high spin**.

- $\Delta_o > P$  (strong field ligand): cheaper to pair. Fourth electron pairs in  $t_{2g} \rightarrow t_{2g}^4 \rightarrow$  fewer unpaired  $\rightarrow$  **low spin**.

### ***Tetrahedral field***

The geometry inverts: now  $d_{xy}$ ,  $d_{yz}$ ,  $d_{zx}$  lie closer to the incoming ligands and are **raised** ( $t_2$ ), while  $d_{x^2-y^2}$  and  $d_{z^2}$  are **lowered** ( $e$ ).

The splitting is much smaller, because there are only four ligands instead of six and they don't point directly at any orbital:

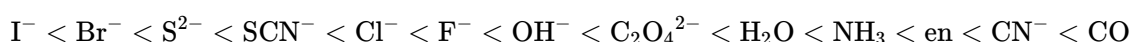
$$\Delta_t = \frac{4}{9}\Delta_o$$

Since  $\Delta_t$  is always smaller than  $P$ , **tetrahedral complexes are essentially always high spin**.

**Examiner asks:** "how is  $\Delta_o$  related to  $\Delta_t$ ?" (2024) — quote  $\Delta_t = \frac{4}{9}\Delta_o$  and say why: fewer ligands, none pointing directly at an orbital.

### ***Spectrochemical series***

Ligands arranged by the size of  $\Delta$  they produce, determined experimentally from light absorption. Weak field on the left, strong on the right:



### ***Colour, explained at last***

An electron absorbs light of exactly the energy  $\Delta_o$  and jumps from  $t_{2g}$  to  $e_g$  — a **d-d transition**. The compound transmits what it didn't absorb, so we see the **complementary** colour. A complex with  $d^0$  or  $d^{10}$  has no possible jump and is colourless.

**Examiner asks:** why does a complex lose its colour when the ligands are removed? Without ligands there is no crystal field, so no splitting, so no d-d transition — it turns colourless.

## **Stability and applications — the short sections that still carry marks**

### ***Stability constant***

For  $\text{M} + 4\text{L} \rightleftharpoons \text{ML}_4$ , the stability (formation) constant is  $K = \frac{[\text{ML}_4]}{[\text{M}][\text{L}]^4}$ . Higher  $K$  means a more stable complex. Stability rises with the metal's charge, with smaller metal size, and — strongly — with chelation.

### ***Applications worth naming***

- **Wilkinson's catalyst**,  $[(\text{Ph}_3\text{P})_3\text{RhCl}]$  — catalyses the **hydrogenation of alkenes**.
- **Cisplatin**,  $\text{cis-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$  — anti-cancer drug. Only the *cis* isomer works.
- **EDTA** — estimation of water hardness; treatment of lead poisoning.
- **Extraction of metals** — gold and silver via cyanide complexes.
- **Biological** — chlorophyll (Mg), haemoglobin (Fe), vitamin B<sub>12</sub> (Co).
- **Electroplating** and photography (silver thiosulphate complex).

## Patterns, collected — the four things you'll be asked to actually do

A · IUPAC name ↔ formula — 2-3 marks

*Recognise it: "write the IUPAC name" or "write the formula".*

1. Get the oxidation number first — it's needed either way.
2. Naming: ligands alphabetically, anionic ones in -o, prefixes ignored for alphabetising, Roman numeral, -ate if the complex is an anion.
3. Formula: metal first, then ligands alphabetically, abbreviations in parentheses.

B · Hybridisation, geometry, magnetism — 3 marks

*Recognise it: an atomic number is given alongside a complex.*

1. Oxidation number → d-electron count.
2. Strong or weak ligand? Pair or don't.
3. Count the orbitals needed; inner 3d if free, else outer 4d.
4. State hybridisation, geometry, and paramagnetic/diamagnetic.

C · Identify the isomerism — 2-3 marks

*Recognise it: "what type of isomerism is shown by..."*

1. Ambidentate ligand ( $\text{NO}_2^-$ ,  $\text{SCN}^-$ ) present → **linkage**.
2. An ion could swap in/out of the bracket → **ionisation**.
3. Water present both inside and out → **hydrate**.
4. Two complex ions in one compound → **coordination**.
5. Otherwise check geometry: square planar or octahedral → **geometrical**; didentate ligands → check **optical** too.

D · Magnetic moment — 2 marks

*Recognise it: "calculate the magnetic moment" or "predict the magnetic behaviour".*

1. d-count, then apply the ligand's field strength.
2. Count unpaired  $n$ .
3.  $\mu = \sqrt{n(n+2)}$  BM.

## Past year questions · question types — ranked by how often they turn up

*From the last 10 years of boards — 25 questions, grouped by pattern.*

1 · IUPAC naming / formula writing — 2-3 marks

*Recognise it: a complex given with "name this" or a name given with "write the formula".*

1. Oxidation number.
2. Alphabetical ligands, correct endings.
3. -ate if anionic.

**Trap:** "ammine" vs "amine", and forgetting **bis/tris** when the ligand name already has a prefix.

2 · Isomerism identification and drawing — 2-3 marks

*Recognise it: "what type of isomerism", or "draw the isomer which is optically active".*

1. Scan for ambidentate ligands, swappable counter ions, or two complex ions.
2. If none, it's stereoisomerism — check cis/trans, then optical.
3. Optically active = **cis** with didentate ligands.

### 3 · Hybridisation and magnetic behaviour — 3 marks

*Recognise it: atomic number supplied.*

1. Oxidation number → d count.
2. Ligand strength → pairing.
3. Hybridisation → geometry → magnetism.

**Trap:** in  $[\text{Ni}(\text{CO})_4]$  nickel is in oxidation state **zero**, not +2.

### 4 · Crystal field theory reasoning — 3 marks

*Recognise it:  $\Delta_o$ ,  $\Delta_t$ , high/low spin, or "why is it coloured".*

1. Compare  $\Delta_o$  with P.
2.  $\Delta_o < P \rightarrow$  high spin;  $\Delta_o > P \rightarrow$  low spin.
3. For colour, name the d-d transition.

### 5 · Definitions and comparisons — 2 marks

*Recognise it: "define", "difference between", "what is denticity/chelate/ambidentate".*

1. Give the definition in NCERT's own words.
2. Add one concrete example — it's usually worth half the mark.

### 6 · Chemical tests and applications — 2-3 marks

*Recognise it: "give a chemical test", or "write the formula and use of..."*

1. Ionisation isomers →  $\text{BaCl}_2$  for sulphate,  $\text{AgNO}_3$  for chloride.
2. Named catalysts: Wilkinson's for alkene hydrogenation.

## Past year questions · mark slots — *what each type is worth*

| Question type                     | Slot                                    | Time    |
|-----------------------------------|-----------------------------------------|---------|
| IUPAC naming / formula            | 2-3 marker                              | 3 min   |
| Isomerism ID + drawing            | 2-3 marker                              | 3-4 min |
| Hybridisation + magnetism         | 3-marker                                | 4 min   |
| CFT reasoning                     | 3-marker; full splitting diagram is a 5 | 4-6 min |
| Definitions / differences         | 2-marker                                | 2 min   |
| Chemical test / application       | 2-marker                                | 2 min   |
| Multi-part "answer the following" | 5-marker                                | 7 min   |

*The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — the multi-part sets combining naming, isomerism and CFT are where the 5-markers sit.*

**Past year questions · repeat offenders — *appeared more than once* — *highest probability in the chapter***

2026 Q1 · 2019 Q4 · 2023 Q2

IUPAC naming — including naming the **linkage isomer** of a red complex, and writing a formula from a given name. Naming appears in essentially every paper.

2017 Q14 · 2022 Q19 · 2026 Q5

"What type of isomerism is shown by..." — run across linkage, ionisation and coordination isomers in the same question format.

2025 Q13 · 2022 Q16

Hybridisation and magnetic behaviour of  $[\text{Ni}(\text{CO})_4]$  with  $Z = 28$  supplied. The oxidation-state-zero trap is the point of it.

2024 Q24(a) · 2026 Q17

$\Delta_t = \frac{4}{9}\Delta_o$ , and the  $\Delta_o$  vs pairing-energy comparison deciding high vs low spin.

2019 Q25(a) · lecture PYQ

Difference between a complex and a double salt, with examples. Straight recall, always available.

2025 Q18 · 2022 Q23

Geometrical isomerism: which complexes show it, fac vs mer for  $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ , and why tetrahedral complexes don't.

2025 Q22 · 2026 Q20

Wilkinson's catalyst formula and use, and why a complex becomes colourless when ligands are absent.

**Past year questions · cold practice — *answers only* — *work them before you look***

***Naming and formulae***

- 2019 — write the formula for pentaamminenitrito-O-cobalt(III).  $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$
- Lecture PYQ — IUPAC name of  $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ . triamminetrichloridochromium(III)
- Lecture PYQ — IUPAC name of  $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$ . pentaamminecarbonatocobalt(III) chloride
- Lecture PYQ — IUPAC name of  $[\text{Zn}(\text{OH})_4]^{2-}$ . tetrahydroxidozincate(II) ion

***Isomerism***

- 2026 Q5 — isomerism in  $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ . Linkage —  $\text{NO}_2^-$  is ambidentate
- 2017 Q14 — isomerism in  $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ . Ionisation — sulphate and chloride can swap
- 2017 Q14 — isomerism in  $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ . Coordination — ligands exchange between the two spheres
- 2025 Q18 — draw the optically active geometrical isomer of  $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$ . The **cis** form; trans is optically inactive

- Lecture PYQ — which shows geometrical isomerism:  $[\text{Co}(\text{NH}_3)_6]^{3+}$  or  $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ ? The second — fac and mer. The first has all identical ligands

### **Bonding**

- 2025 Q13 — hybridisation and magnetic behaviour of  $[\text{Ni}(\text{CO})_4]$ ,  $Z = 28$ .  $sp^3$ , tetrahedral, diamagnetic — Ni is in oxidation state 0
- Lecture PYQ — compare  $[\text{NiCl}_4]^{2-}$  and  $[\text{Ni}(\text{CN})_4]^{2-}$ .  $sp^3$  tetrahedral paramagnetic vs  $dsp^2$  square planar diamagnetic
- Lecture PYQ — why is  $[\text{CoF}_6]^{3-}$  paramagnetic but  $[\text{Co}(\text{NH}_3)_6]^{3+}$  diamagnetic?  $\text{F}^-$  weak (no pairing,  $sp^3d^2$ );  $\text{NH}_3$  strong with  $\text{Co}^{3+}$  (pairs,  $d^2sp^3$ )

### **CFT and applications**

- 2024 Q24(a) — relate  $\Delta_o$  and  $\Delta_t$ .  $\Delta_t = \frac{4}{9}\Delta_o$  — fewer ligands, none pointing directly at an orbital
- 2026 Q17 — what does  $\Delta_o < P$  mean for the complex? Weak field, high spin — the fourth electron goes to  $e_g$  rather than pairing
- 2025 Q22 — formula and use of Wilkinson's catalyst.  $[(\text{Ph}_3\text{P})_3\text{RhCl}]$ ; hydrogenation of alkenes
- 2026 Q6 — chemical test distinguishing  $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$  from  $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ .  $\text{AgNO}_3 \rightarrow$  white  $\text{AgCl}$  for the first;  $\text{BaCl}_2 \rightarrow$  white  $\text{BaSO}_4$  for the second

Built from Sourabh Raina's Coordination Compounds one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 5 (Rationalised 2022-23). Nomenclature verified against NCERT: **ammine** (two m's) for  $\text{NH}_3$  versus **amine** for organic amines, and **chlorido** rather than chloro. First-contact material: every term marked **[exposure]** is explained from scratch before use.

# Haloalkanes and Haloarenes

NCERT Class XII Chemistry · Chapter 6 · Haloalkanes and Haloarenes

*6 marks, whole chapter 6.1–6.8, and your first organic chapter of the paper. Organic chemistry runs on a small set of ideas repeated everywhere — nucleophile, leaving group, carbocation, steric hindrance. Every one of them is tagged [exposure] and explained from scratch here, because once you have those four, most of this chapter answers itself.*

## What these compounds are — *start here* — *one sentence defines the whole chapter*

**[exposure]** A **haloalkane** (also called an **alkyl halide**, written R–X) is what you get when you take a hydrocarbon chain and replace one or more hydrogen atoms with a halogen — fluorine, chlorine, bromine or iodine. That's the entire definition. Ethane  $\text{CH}_3\text{CH}_3$  with one hydrogen swapped for chlorine becomes chloroethane  $\text{CH}_3\text{CH}_2\text{Cl}$ . The defining structural feature: the carbon carrying the halogen is  $sp^3$  **hybridised** — it has four single bonds, no double bond anywhere on it.

**[exposure]** A **haloarene** (or **aryl halide**, Ar–X) is the same swap performed on an *aromatic* ring instead. Benzene with one hydrogen replaced by chlorine is chlorobenzene. The defining feature here: the carbon carrying the halogen is  $sp^2$  **hybridised**, because it is part of the ring's double-bond system. This single difference —  $sp^3$  versus  $sp^2$  — is why haloalkanes and haloarenes behave almost oppositely, and it comes up in the exam again and again.

Throughout the chapter, X means any halogen and R means any alkyl group.

## Why the chapter exists at all

Halogen-containing organic compounds are everywhere, natural and synthetic:

- **Chloramphenicol** — a chlorine-containing antibiotic made by soil microorganisms, used against typhoid.
- **Thyroxine** — the iodine-containing hormone your own thyroid produces. Too little iodine in the diet and you get **goitre**.
- **Chloroquine** — synthetic, used against malaria.
- **Halothane** — a synthetic anaesthetic used during surgery.
- **Fully fluorinated compounds** — every hydrogen replaced by fluorine. They dissolve and deliver oxygen to tissue, so they are being explored as **blood substitutes** in surgery.

**Examiner asks:** one-mark recall of a named compound and its use. Chloramphenicol/typhoid and thyroxine/goitre are the two that appear.

## 6.1 Classification — *three separate cuts* — *count, hybridisation, and where the halogens sit*

### By number of halogen atoms

**Mono, di or poly** halogen compounds, depending on whether they carry one, two, or more halogen atoms.  $\text{CH}_3\text{Cl}$  is monohalo;  $\text{CH}_2\text{Cl}_2$  is dihalo;  $\text{CHCl}_3$  is trihalo. The same labelling works on rings: one chlorine on benzene is a monohaloarene, two is a dihaloarene.

### By hybridisation of the carbon holding the halogen

This is the classification that actually predicts behaviour, so learn these five names properly.

#### Compounds with an $sp^3$ C-X bond:

**[exposure] Alkyl halides** — the halogen sits on an ordinary  $sp^3$  carbon of a chain. They are sub-classified by *what else* that carbon is attached to: if the carbon carries one other carbon it is **primary (1°)**, two others **secondary (2°)**, three others **tertiary (3°)**. This count matters more than anything else in the chapter — it decides which mechanism the compound follows and how fast.  $\text{CH}_3\text{CH}_2\text{Cl}$  is primary;  $(\text{CH}_3)_2\text{CHCl}$  is secondary;  $(\text{CH}_3)_3\text{CCl}$  is tertiary.

**[exposure] An allylic halide** has the halogen on an  $sp^3$  carbon that sits *next to* a carbon-carbon double bond. The word exists because that neighbouring double bond changes everything: when the halogen leaves, the positive charge left behind can spread into the double bond by resonance, so allylic halides react far faster than ordinary ones. Example:  $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$  — the chlorine-bearing carbon has four single bonds, the carbon beside it has the double bond.

**[exposure] A benzylic halide** is the same idea with a benzene ring instead of a double bond: the halogen is on an  $sp^3$  carbon directly attached to an aromatic ring.  $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$  (benzyl chloride) is the standard one. Benzylic halides are the *most* reactive of all in this respect, because the ring gives the intermediate the largest resonance stabilisation available. These too can be 1°, 2° or 3° by the same counting rule.

#### Compounds with an $sp^2$ C-X bond:

**[exposure] A vinylic halide** has the halogen bonded *directly to* a carbon of a carbon-carbon double bond. Not next to it — on it.  $\text{CH}_2=\text{CH}-\text{Cl}$  is vinyl chloride. Because that carbon is  $sp^2$ , the C-X bond is shorter and stronger, and these are very unreactive towards substitution.

**[exposure] An aryl halide** has the halogen bonded directly to an  $sp^2$  carbon of an aromatic ring — chlorobenzene. Same story as vinylic, for the same reason plus resonance.

**Trap:** allylic/benzylic (fast) versus vinylic/aryl (slow) is decided by *one* question — is the halogen **on** the unsaturated carbon, or **next to** it? On it  $\rightarrow sp^2 \rightarrow$  unreactive. Next to it  $\rightarrow sp^3 \rightarrow$  very reactive. This exact discrimination appeared as 2017 Q5(b).

### Dihaloalkanes: where the two halogens sit

**[exposure] Geminal (gem) dihalides** carry both halogens on the *same* carbon — from Latin *geminus*, twin.  $\text{CH}_3\text{CHCl}_2$  is one. Their common name is built from the alkylidene group:  $\text{CH}_3\text{CHCl}_2$  is **ethylidene chloride**.

**[exposure] Vicinal (vic) dihalides** carry the halogens on *adjacent* carbons — from Latin *vicinus*, neighbour.  $\text{CH}_2\text{ClCH}_2\text{Cl}$  is one. Their common name uses the alkylene group: **ethylene dichloride**. Vicinal dihalides matter because that's what you get when bromine adds across a double bond.

**Examiner asks:** "write the structures of all dihalogen derivatives of propane" (NCERT intext 6.3) — there are four: 1,1- and 2,2- (both geminal), 1,2- (vicinal), and 1,3-.

## 6.2 Nomenclature — IUPAC and common names — both are examinable

### IUPAC rules, in the order you apply them

1. Find the **longest carbon chain** containing the principal functional group. Count carbons only — hydrogens don't matter.
2. The halogen is a **substituent**, named as a prefix: **fluoro-, chloro-, bromo-, iodo-**.
3. If there is a **double or triple bond**, it takes **priority** for the lowest number, not the halogen.
4. With no multiple bond, number so the **substituent** gets the lowest locant.
5. If numbering from both ends gives the same set of numbers, give the lowest number to whichever substituent comes **first alphabetically**. Bromo beats methyl; chloro beats methyl.
6. Write substituents alphabetically in the name.

| Structure                                               | IUPAC name              | Common name                |
|---------------------------------------------------------|-------------------------|----------------------------|
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$            | 1-bromopropane          | <i>n</i> -propyl bromide   |
| $(\text{CH}_3)_2\text{CHCl}$                            | 2-chloropropane         | isopropyl chloride         |
| $\text{CH}_3\text{CH}_2\text{CHClCH}_3$                 | 2-chlorobutane          | <i>sec</i> -butyl chloride |
| $(\text{CH}_3)_3\text{CBr}$                             | 2-bromo-2-methylpropane | <i>tert</i> -butyl bromide |
| $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$                 | 1-bromo-2-methylpropane | isobutyl bromide           |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ | 1-chlorobutane          | <i>n</i> -butyl chloride   |

The straight-chain compound always takes ***n*-** in the common name. A three-carbon group halogenated at the middle carbon is **isopropyl**. Four carbons branched at the end is **isobutyl**; branched at the halogen-bearing carbon is *sec*- or *tert*-butyl depending on the count.

### Common names you're expected to know cold

| Formula                  | IUPAC              | Common               |
|--------------------------|--------------------|----------------------|
| $\text{CHCl}_3$          | trichloromethane   | chloroform           |
| $\text{CHBr}_3$          | tribromomethane    | bromoform            |
| $\text{CHI}_3$           | triiodomethane     | iodoform             |
| $\text{CCl}_4$           | tetrachloromethane | carbon tetrachloride |
| $\text{CH}_2\text{Cl}_2$ | dichloromethane    | methylene chloride   |

### Aromatic compounds

For haloarenes the common name and the IUPAC name are the same word — chlorobenzene is both. Three side-chain names you must recognise:

- **Benzyl** — benzene ring plus  $\text{CH}_2$ . So  $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$  is benzyl chloride.
- **Benzal** — ring plus CH carrying two halogens.  $\text{C}_6\text{H}_5\text{CHCl}_2$  is benzal dichloride.
- **Benzo-** — ring plus a carbon with no hydrogen left.  $\text{C}_6\text{H}_5\text{CCl}_3$  is benzo-trichloride.

For two substituents on a ring, positions 1,2 are **ortho (o-)**, 1,3 are **meta (m-)**, 1,4 are **para (p-)**. So 1,4-dibromobenzene is *p*-dibromobenzene. Three substituents at 1,3,5 is **sym-**

(symmetrical).

**Examiner asks:** "give the IUPAC name of the following" is a standing 3-marker (2017 Q14, 2024 Q11a). Always check for a double bond before you number.

### 6.3 Nature of the C-X bond — *short section, but it explains every reaction that follows*

Halogens are more electronegative than carbon, so the shared electron pair sits closer to the halogen. The result is a **polar bond**: the carbon carries a partial positive charge  $\delta+$  and the halogen a partial negative charge  $\delta-$ .



That  $\delta+$  carbon is the target of essentially every reaction in this chapter. Anything electron-rich will go for it.

Down the group  $\text{F} \rightarrow \text{Cl} \rightarrow \text{Br} \rightarrow \text{I}$ , the halogen gets bigger, so the **bond length increases** and the **bond enthalpy decreases**. A longer, weaker bond breaks more easily — which is why  $\text{R-I}$  is the most reactive alkyl halide and  $\text{R-F}$  the least in substitution reactions. Under ordinary conditions  $\text{R-F}$  barely reacts at all.

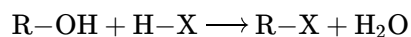
**Trap:** the C-Cl bond in a **haloalkane** is **177 pm**, but in a **haloarene** only **169 pm** — shorter, therefore stronger, therefore harder to break. That number is the physical reason haloarenes resist nucleophilic substitution, and it is worth quoting.

### 6.4 Preparation of haloalkanes — *four routes — from alcohols, from alkanes, from alkenes, by exchange*

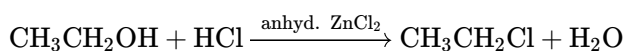
#### **A · From alcohols**

The general move: replace the  $-\text{OH}$  group with a halogen.

##### **(i) With halogen acids (HX)**

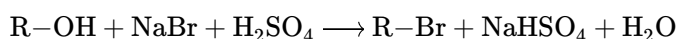


For **chloroalkanes**, use  $\text{HCl}$  with **anhydrous**  $\text{ZnCl}_2$  as catalyst:



The  $\text{ZnCl}_2$  works as a **dehydrating agent**: it soaks up the water product. By Le Chatelier's principle, removing a product drives the equilibrium forward, so you get more chloroethane.

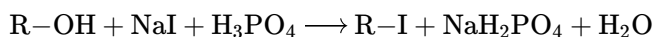
For **bromoalkanes**, heat the alcohol with  $\text{NaBr}$  (or  $\text{KBr}$ ) and concentrated  $\text{H}_2\text{SO}_4$ :



**[exposure]** The  $\text{HBr}$  is generated **in situ** — Latin for "in place", meaning the reagent is made inside the reaction vessel at the moment it's needed rather than being bought and stored. Chemists do this whenever a reagent is nasty or unstable to keep around.  $\text{HBr}$  is a corrosive

acid that gives off heavy fumes, so making it in the flask from  $\text{NaBr} + \text{H}_2\text{SO}_4$  is simply safer. You can also just use 48% concentrated  $\text{HBr}$  directly.

For **iodoalkanes**, use  $\text{NaI}$  (or  $\text{KI}$ ) with **95% phosphoric acid**,  $\text{H}_3\text{PO}_4$  — *not* sulphuric acid:

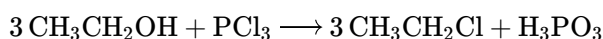


**Trap — this is NCERT intext 6.2 and it appears in boards:** why not  $\text{H}_2\text{SO}_4$  with  $\text{KI}$ ?

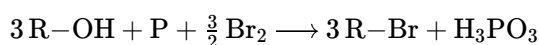
Concentrated  $\text{H}_2\text{SO}_4$  *would* make  $\text{HI}$ , but it is also a **strong oxidising agent** and immediately oxidises that  $\text{HI}$  to  $\text{I}_2$ . The  $\text{HI}$  never survives to reach the alcohol.  $\text{H}_3\text{PO}_4$  is not an oxidising agent, so it works.

**Reactivity orders to memorise** — halogen acid:  $\text{HI} > \text{HBr} > \text{HCl}$  (because  $\text{HI}$  has the longest, weakest bond, so it ionises most easily). Alcohol: **tertiary** > **secondary** > **primary** (because these reactions go through a carbocation, and a tertiary carbocation is the most stable).

**(ii) With phosphorus halides**



For bromides and iodides, use **red phosphorus with**  $\text{Br}_2$  **or**  $\text{I}_2$ . The  $\text{PBr}_3$  or  $\text{PI}_3$  is generated in situ, again because these are too unstable to store:



**(iii) With thionyl chloride — the preferred method**

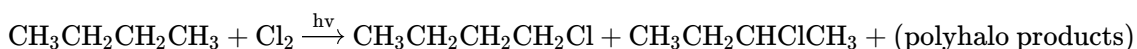


**Examiner asks:** "why is thionyl chloride preferred for preparing alkyl chlorides from alcohols?" (2024 Q11b) — because both by-products,  $\text{SO}_2$  and  $\text{HCl}$ , are **gases that escape**, leaving pure alkyl chloride behind with no separation step. Say "escapable gases, hence pure product" and you have the mark.

**Trap: none of these routes work on phenol.** In phenol, the oxygen's lone pair delocalises into the ring, giving the C-O bond **partial double-bond character**. It's too strong to break, so you cannot make an aryl halide from phenol this way. This is a stock 2-marker and it also shows up embedded in "major monohalo product" questions where a molecule has two  $-\text{OH}$  groups: only the non-phenolic one reacts.

**B • From hydrocarbons — free radical halogenation**

Treat an alkane with  $\text{Cl}_2$  or  $\text{Br}_2$  in the presence of UV light or heat, and hydrogens get replaced one at a time:



The problem: you get a **complex mixture** of isomeric mono- and polyhalogenated products whose boiling points are so close they are almost impossible to separate. So this is **not a preferred preparative method** — but it is heavily examined for *predicting* products.

When asked for the **major** free-radical product, use the free-radical stability order: **benzylic**  $\approx$  **allylic** > **tertiary** > **secondary** > **primary**. Benzylic and allylic radicals win because the ring or double bond stabilises them by resonance.

### C • From alkenes

#### (i) Addition of a halogen acid

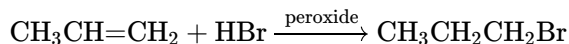
With a **symmetrical** alkene, there's no choice to make:



**[exposure]** With an **unsymmetrical** alkene the two carbons of the double bond are different, so you need a rule to say which end the halogen goes to. **Markovnikov's rule** says the **negative part of the adding molecule attaches to the carbon that carries fewer hydrogens**. The reason underneath is carbocation stability — the proton adds first, wherever it produces the more stable carbocation, and the halide then goes to that carbon. So propene plus HBr gives 2-bromopropane, not 1-bromopropane.

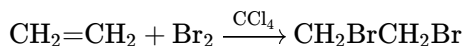


**Anti-Markovnikov (peroxide / Kharasch effect):** add a peroxide such as benzoyl peroxide and the rule flips — bromine goes to the carbon with *more* hydrogens.



**Trap:** the peroxide effect works with HBr **only**. It does not happen with HCl or HI. If a question gives you peroxide with HCl, the answer is still Markovnikov.

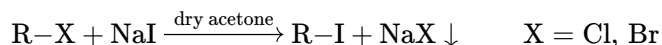
#### (ii) Addition of a halogen



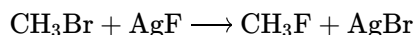
This gives a **vicinal dibromide**. The reddish-brown colour of the bromine solution **discharges** as the reaction proceeds — which is exactly the standard laboratory **test for unsaturation**.

### D • Halogen exchange

**[exposure]** The **Finkelstein reaction** makes alkyl *iodides* by swapping out a chloride or bromide, using NaI in **dry acetone**. It needs a named trick because the swap is an equilibrium that wouldn't otherwise favour the product. The trick: NaCl and NaBr are **insoluble in dry acetone and precipitate out**, while NaI is soluble. Removing a product drags the equilibrium forward (Le Chatelier again).



**[exposure]** The **Swarts reaction** makes alkyl *fluorides*, which the ordinary routes can't touch. Heat an alkyl chloride or bromide with a **metallic fluoride** — AgF, Hg<sub>2</sub>F<sub>2</sub>, CoF<sub>2</sub> or SbF<sub>3</sub>:

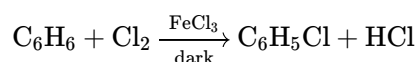


**Examiner asks:** "what happens when methyl bromide is treated with silver fluoride?" (2025 Q20a) — name it as Swarts and give the equation.

## 6.5 Preparation of haloarenes — *two routes — and one of them is a named reaction you must know*

### A • From arenes, by electrophilic substitution

Treat benzene with Cl<sub>2</sub> or Br<sub>2</sub> in the presence of a **Lewis acid catalyst** — FeCl<sub>3</sub>, FeBr<sub>3</sub> or iron filings — in the dark:



Mechanically: the Lewis acid is electron-deficient, so it pulls Cl<sup>−</sup> off Cl<sub>2</sub>, leaving Cl<sup>+</sup>. That Cl<sup>+</sup> is the **electrophile** that attacks the electron-rich ring.

With **toluene**, the −CH<sub>3</sub> group is ortho/para directing, so you get a mixture of *o*- and *p*-chlorotoluene. Conveniently, these two are **easy to separate** because their melting points differ widely.

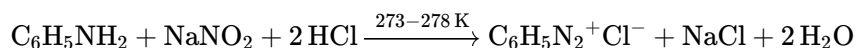
Two limits on this method:

- **Fluoro compounds can't be made this way** — fluorine is too reactive and the reaction is uncontrollable.
- **Iodination is reversible** and gives poor yield. Fix it by adding an **oxidising agent** (HNO<sub>3</sub> or HIO<sub>4</sub>), which oxidises the HI by-product to I<sub>2</sub> and so pulls the equilibrium forward.

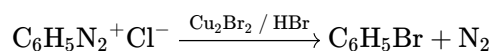
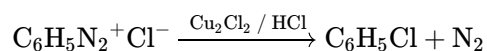
### B • From amines — the Sandmeyer reaction

**[exposure]** A **diazonium salt** is an aromatic compound carrying the group −N<sub>2</sub><sup>+</sup> — a nitrogen pair on its way out. It exists as an intermediate because that group is an *outstanding* leaving group: it departs as harmless N<sub>2</sub> gas, dragging the reaction forward. That makes it the standard way to put almost anything onto a benzene ring. Diazonium salts are only stable cold, which is why the next step specifies 0–5 °C.

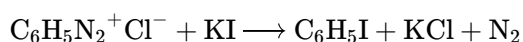
**Step 1 — diazotisation.** Treat a primary aromatic amine (aniline) with nitrous acid, generated in situ from NaNO<sub>2</sub> + HCl, at **273–278 K (0–5 °C)**:



**Step 2 — Sandmeyer.** Treat the diazonium salt with a **cuprous halide**:



**Trap:** for the **iodo** compound you need **no copper salt at all** — simply shaking the diazonium salt with KI does it. This exception is asked directly.



## 6.6 Physical properties — *every point here is a "give the reason" question*

### **Physical state and colour**

$\text{CH}_3\text{Cl}$ ,  $\text{CH}_3\text{Br}$ ,  $\text{CH}_3\text{CH}_2\text{Cl}$  and some chlorofluoromethanes are **gases** at room temperature; higher members are liquids or solids. Pure haloalkanes are **colourless**, but bromo- and iodo-compounds develop colour on standing in light.

### **Boiling point — four separate rules**

1.  $\text{RF} < \text{RCl} < \text{RBr} < \text{RI}$ . Down the group the halogen's mass and size increase, so the **van der Waals forces** between molecules get stronger and more energy is needed to separate them.
2. **A haloalkane always boils higher than the hydrocarbon of comparable mass.** Two reasons together: the C-X bond is polar, so molecules attract each other by **dipole-dipole interaction**; and the heavy halogen raises the molecular mass, strengthening van der Waals forces.
3. **Among isomers, boiling point falls as branching rises.** Branching makes a molecule more compact and spherical, cutting the **surface area of contact** between molecules, which weakens van der Waals forces.
4. **More halogen atoms → higher boiling point**, again through mass.  
 $\text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_2\text{Br}_2 < \text{CHBr}_3$ .

**Worked:** Arrange in increasing boiling point: 1-chloropropane, 2-chloropropane, 1-chlorobutane. 1-chlorobutane has four carbons — highest mass, so it boils highest. The other two are isomers, so compare branching: 2-chloropropane is branched, so it boils lowest.  
Answer: **2-chloropropane < 1-chloropropane < 1-chlorobutane.**

### **Melting point — the para exception**

The boiling points of *o*-, *m*- and *p*-dihalobenzenes are nearly identical. But the **para isomer melts much higher** than the other two, because it is more **symmetrical** and so packs more tightly into the crystal lattice. More energy is needed to break a well-packed lattice apart.

### **Density**

Bromo, iodo and polychloro derivatives are **heavier than water**. Density increases with the number of carbon atoms, the number of halogen atoms, and the atomic mass of the halogen.

### **Solubility — a favourite "give the reason"**

Haloalkanes are polar, yet they are **insoluble in water**. Dissolving anything in water requires breaking the strong **hydrogen bonds between water molecules** and also the dipole-dipole attractions between haloalkane molecules. The energy released when new haloalkane-water attractions form is **less** than the energy needed to break the old ones — so it doesn't happen.

They *are* soluble in organic solvents (alcohol, ether, benzene), because there the new attractions formed are about as strong as the ones broken, so the books balance.

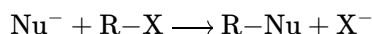
**Examiner asks:** "why are alkyl halides, though polar, immiscible with water?" (2017 Q5a) — the two energies, in one sentence each. It is a repeat question.

## 6.7 Reactions of haloalkanes — nucleophilic substitution — *the core of the chapter; get the two words first*

**[exposure]** A **nucleophile** is a species that is **electron-rich** — it has a lone pair or a negative charge — and goes looking for a positive centre to donate that pair to. The name means "nucleus-loving".  $\text{OH}^-$ ,  $\text{CN}^-$ ,  $\text{NH}_3$  and  $\text{RO}^-$  are all nucleophiles. In this chapter the positive centre it attacks is always the  $\delta+$  carbon of the C-X bond.

**[exposure]** A **leaving group** is the species that departs, taking the bonding electron pair with it. Here it's the halide ion  $\text{X}^-$ . A group leaves easily when the resulting ion is stable and the bond was weak — so  $\text{I}^-$  **is the best leaving group** and  $\text{F}^-$  **the worst**, following bond strength. That single fact explains the reactivity order  $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$ .

Put those together and the reaction reads: a nucleophile attacks the  $\delta+$  carbon, the halide leaves, and you have a new compound. The substitution only happens if the incoming nucleophile is **stronger than the halide it displaces**.



### *The substitution products you must be able to write*

| Reagent                                      | Product                | Class of product           |
|----------------------------------------------|------------------------|----------------------------|
| aqueous KOH / NaOH                           | $\text{R-OH}$          | alcohol                    |
| NaOR (sodium alkoxide)                       | $\text{R-O-R}$         | ether                      |
| KCN                                          | $\text{R-CN}$          | alkyl cyanide (nitrile)    |
| AgCN                                         | $\text{R-NC}$          | alkyl <b>isocyanide</b>    |
| $\text{NH}_3$                                | $\text{R-NH}_2$        | primary amine              |
| $\text{KNO}_2$                               | $\text{R-O-N=O}$       | alkyl nitrite              |
| $\text{AgNO}_2$                              | $\text{R-NO}_2$        | <b>nitroalkane</b>         |
| $\text{R}'\text{COOAg}$ (silver carboxylate) | $\text{R}'\text{COOR}$ | ester                      |
| $\text{LiAlH}_4$                             | $\text{R-H}$           | alkane (same carbon count) |
| $\text{R}'\text{-Li}$ (or Na)                | $\text{R-R}'$          | alkane (carbons added)     |

**[exposure]** Look at the four pairs above that come in twos. An **ambident nucleophile** is a nucleophile with **two different donor atoms**, either of which can do the attacking — though only one at a time. The category exists because *which* atom attacks decides which product you get, and that in turn depends on whether the reagent is ionic or covalent. The two examples are  $\text{CN}^-$  (attacks through C or N) and  $\text{NO}_2^-$  (attacks through N or O).

**Trap — and one of the most-asked questions in the chapter:** KCN is **ionic**, so it dissociates and gives a free  $\text{CN}^-$  ion which attacks through its **carbon** → alkyl **cyanide**.

AgCN is **covalent**, so it does not dissociate; only the **nitrogen** lone pair is available → alkyl **isocyanide**. The same logic runs the KNO<sub>2</sub> / AgNO<sub>2</sub> pair: ionic gives attack through oxygen (nitrite ester), covalent gives attack through nitrogen (nitroalkane). Remember it as "**silver goes through nitrogen**".

### **Mechanism 1 · S<sub>N</sub>2 — bimolecular**

**[exposure]** S<sub>N</sub>2 stands for **Substitution, Nucleophilic, bimolecular**. "Bimolecular" means **two** species are involved in the slow, rate-deciding step, so the rate depends on the concentration of *both*:

$$\text{Rate} = k [\text{R-X}][\text{Nu}^-]$$

Everything happens in **one single step**. The nucleophile attacks the carbon from the side **opposite** the halogen — it has to, because the halogen's lone pairs would repel it if it came from the front. As the new C-Nu bond forms, the old C-X bond breaks, **simultaneously**.

**[exposure]** Halfway through, the carbon is momentarily bonded to **five** things at once — the three groups it keeps, the incoming nucleophile, and the leaving halogen. That arrangement is the **transition state**. It is not an intermediate: it is a fleeting, very high-energy structure that **cannot be isolated**, existing only at the top of the energy hill. In S<sub>N</sub>2 there is **no intermediate at all** — no carbocation, no free radical.

Because the nucleophile comes in from the back and the halogen leaves the front, the other three groups get pushed through, like an umbrella flipping in the wind. The result is **inversion of configuration**, called Walden inversion.

**Worked:** (–)-2-bromooctane treated with OH<sup>–</sup> gives (+)-octan-2-ol. The OH ends up in the position the Br did not occupy — the configuration is inverted, and the sign of optical rotation flips.

**[exposure]** **Steric hindrance** is physical crowding: bulky groups around the reaction centre get in the way of an incoming reagent and slow the reaction down. It matters here because the S<sub>N</sub>2 nucleophile must physically reach the carbon. A methyl halide has only three small hydrogens in the way, so it reacts fastest. A tertiary halide has three fat CH<sub>3</sub> groups blocking the approach, so it reacts slowest.

S<sub>N</sub>2 **reactivity:** CH<sub>3</sub>X > **primary** > **secondary** > **tertiary**.

**Trap:** when both compounds in a pair are primary, compare **how close the branching is to the halogen**. A methyl group nearer the C-X carbon crowds it more and slows S<sub>N</sub>2 further. That comparison is NCERT intext 6.7 and appears in boards.

### **Mechanism 2 · S<sub>N</sub>1 — unimolecular**

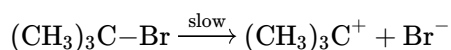
**[exposure]** S<sub>N</sub>1 is **Substitution, Nucleophilic, unimolecular**. "Unimolecular" means only **one** species appears in the slow step — the alkyl halide. So the rate depends only on *its* concentration; changing the nucleophile's concentration does nothing at all:

$$\text{Rate} = k [\text{R-X}]$$

It runs in **two steps**, and it requires a **polar protic solvent** — polar (it has  $\delta + / \delta -$  ends) and protic (it contains an O-H or N-H that can hand out protons). Water, alcohols and acetic acid all qualify.

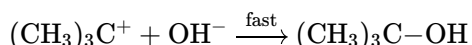
**[exposure]** A **carbocation** is a carbon atom bearing a **full positive charge**, left behind when a leaving group departs with both bonding electrons. It has only three bonds, so it is  $sp^2$  hybridised and **flat (planar)**. Being electron-deficient it is very unstable and short-lived — but its stability varies enormously, and that variation drives this whole mechanism. **Tertiary > secondary > primary > methyl**, because the surrounding alkyl groups push electron density in and cushion the positive charge. Benzylic and allylic carbocations beat all of these, because resonance spreads the charge over several atoms.

**Step 1 (slow, reversible — the rate-determining step):**



Breaking that C-Br bond costs energy. The energy comes from **solvation**: the protic solvent's  $\delta +$  hydrogens crowd around the departing  $\text{Br}^-$ , stabilise it, and release energy in doing so. That is exactly why a polar protic solvent is required.

**Step 2 (fast):** the nucleophile attacks the carbocation.



**$S_N1$  reactivity: tertiary > secondary > primary** — the exact opposite of  $S_N2$ , and for a completely different reason (carbocation stability, not crowding).

**Trap:** for allylic and benzylic halides, **both** mechanisms are fast —  $S_N1$  because their carbocations are resonance-stabilised,  $S_N2$  because the carbon is usually primary and uncrowded. Don't let "it's primary" push you to the wrong answer on an  $S_N1$  question about benzyl chloride: benzyl chloride is the *faster*  $S_N1$  substrate against cyclohexyl chloride, because the benzyl carbocation is resonance-stabilised.

**$S_N1$  vs  $S_N2$  — the comparison table**

|                   | $S_N1$                        | $S_N2$                            |
|-------------------|-------------------------------|-----------------------------------|
| Order of reaction | first order                   | second order                      |
| Rate depends on   | [RX] only                     | [RX] and [Nu]                     |
| Steps             | two                           | one                               |
| Goes through      | carbocation intermediate      | transition state, no intermediate |
| Stereochemistry   | <b>racemisation</b>           | <b>inversion</b>                  |
| Favoured by       | $3^\circ > 2^\circ > 1^\circ$ | $1^\circ > 2^\circ > 3^\circ$     |
| Solvent           | polar protic                  | polar aprotic                     |

**Examiner asks:** "write two differences between  $S_N1$  and  $S_N2$ " (2021 Q21) — pick order + steps, or intermediate + stereochemistry. Two rows, two marks.

## Optical isomerism — the stereochemistry vocabulary — six terms, each needed for the $S_N$ answers

**[exposure]** **Optical activity** is a property some substances have of **rotating the plane of plane-polarised light**. Ordinary light vibrates in every direction; pass it through a polariser and it vibrates in one plane only. Send that through certain liquids or solutions and the plane comes out twisted by some angle. Substances that do this are **optically active**; those that don't are optically inactive. The angle is measured with an instrument called a **polarimeter**.

A substance rotating the plane **clockwise (right)** is **dextrorotatory**, written **d** or **(+)**. One rotating it **anticlockwise (left)** is **laevorotatory**, written **l** or **(-)**.

**[exposure]** A **chiral carbon** (also called an asymmetric carbon or stereocentre) is an  $sp^3$  carbon attached to **four different atoms or groups**. It is marked with an asterisk. It matters because it is the structural cause of optical activity — with four different groups the molecule has no plane of symmetry, and that asymmetry is what twists the light. In  $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$  the second carbon holds methyl, ethyl, bromine and hydrogen: four different things, so it is chiral.

**[exposure]** A **chiral molecule** has no plane of symmetry, and so is **non-superimposable on its mirror image** — exactly like your left and right hands, which is where the name comes from (Greek *cheir*, hand). Lay one over the other and they never match. A molecule with a chiral carbon is normally chiral, and chiral molecules are optically active.

**[exposure]** **Enantiomers** are the two mirror-image forms of a chiral molecule. They are stereoisomers — same connections, different spatial arrangement. They rotate plane-polarised light by the **same angle in opposite directions**: if one is dextro, the other is laevo. Every other physical property — melting point, boiling point, solubility, refractive index — is **identical**.

**Examiner asks:** "why are the d- and l-isomers of butan-2-ol difficult to separate by fractional distillation?" (2024 Q22) — because fractional distillation separates on boiling point, and enantiomers have **identical** boiling points. One line, full marks.

**[exposure]** A **racemic mixture** is a **1:1 (equimolar)** mixture of the two enantiomers. It is **optically inactive**, because the rotation produced by one is exactly cancelled by the other. It's written **dl** or **(±)**, as in (±)-butan-2-ol. The process of converting a single enantiomer into such a mixture is **racemisation**.

**[exposure]** Three words for what a substitution does to a chiral centre. **Retention** — the new group takes the same spatial position the old one had; the configuration is unchanged.

**Inversion** — the new group takes the position *opposite*; the configuration is flipped.

**Racemisation** — you get roughly 50% of each, so the product is optically inactive.

### Which mechanism gives which:

- $S_N2 \rightarrow$  **inversion**. The nucleophile can only attack from the back, so there is only one possible outcome.
- $S_N1 \rightarrow$  **racemisation**. The carbocation intermediate is **planar**, so the nucleophile can attack from either face with equal ease. Two products form in equal amounts, they are enantiomers, and the mixture is racemic.

**Examiner asks:** "out of  $S_N1$  and  $S_N2$ , which occurs with inversion and which with racemisation?" (2026 Q23c) — and expect to have to *explain* it: back-side attack for one,

planar carbocation for the other.

## Elimination and reactions with metals — *the other two things a haloalkane does*

### Elimination — dehydrohalogenation

**[exposure] Dehydrohalogenation** means removing H and X from adjacent carbons to make a double bond. It is also called  **$\beta$ -elimination**, because of where the two pieces come from: the carbon bearing the halogen is the  **$\alpha$ -carbon**, the one next to it is the  **$\beta$ -carbon**, and the reaction takes the halogen from  $\alpha$  and a hydrogen from  $\beta$ . The reagent is **alcoholic KOH** — and the "alcoholic" is the whole point, because aqueous KOH gives substitution instead.



**Trap — the single most reliable trick in the chapter: aqueous KOH  $\rightarrow$  alcohol (substitution). Alcoholic KOH  $\rightarrow$  alkene (elimination).** One word in the question decides the whole answer, and it is tested constantly.

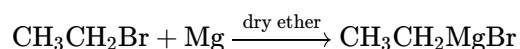
**[exposure]** When a molecule has **more than one  $\beta$ -hydrogen**, more than one alkene can form, and **Saytzeff's rule** (NCERT spells it **Zaitsev**, "also pronounced as Saytzeff") tells you which one dominates: the **more substituted alkene** — the one whose double-bonded carbons carry the most alkyl groups — is the major product. It exists as a rule because more-substituted alkenes are more stable, and the reaction settles into the more stable product.

**Worked:** 2-bromopentane with alcoholic KOH. There are  $\beta$ -hydrogens on both sides, so two alkenes are possible. Removing one gives pent-1-ene, the other gives pent-2-ene. Pent-2-ene has an alkyl group on *each* double-bonded carbon, pent-1-ene has only one — so **pent-2-ene is major, 81% against 19%**.

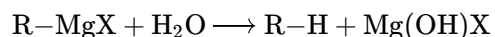
### Reaction with metals

**[exposure]** An **organometallic compound** is any compound containing a direct **carbon-metal bond**. They matter because the metal, being electropositive, pushes electron density onto the carbon, giving that carbon a  $\delta^-$  charge — the exact reverse of its usual state. A carbon that is normally attacked becomes a carbon that attacks.

**[exposure]** The **Grignard reagent**,  $\text{R-Mg-X}$  (alkyl magnesium halide), is the most important organometallic compound in the syllabus. Make it by reacting a haloalkane with **magnesium metal in dry ether**:

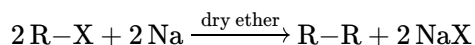


The C-Mg bond is covalent but **highly polar**: carbon  $\delta^-$ , magnesium  $\delta^+$ . That  $\delta^-$  carbon is so electron-rich that it grabs a proton from **any** source — water, alcohol, an amine — and turns into an alkane:



**Examiner asks:** "why should Grignard reagents be prepared under anhydrous conditions?" (2024 Q13c) — because they react instantly with any water present to give the alkane, destroying the reagent before it can be used. Repeat question; know it verbatim.

**[exposure]** The **Wurtz reaction** joins two alkyl halides end to end using **sodium in dry ether**, producing an alkane with **double the carbon count**. It exists as a named reaction because it's the standard way to build a longer chain, and because it's the only route that *doubles* — which is what lets you work backwards from the product in exam questions.



**Trap:** Wurtz only gives a clean product with a **symmetrical** alkane. So when a question gives you the product and asks for the starting halide, **cut the product exactly in half** and put X on the cut. 2,5-dimethylhexane cuts into two 1-bromo-2-methylpropane molecules. That reverse-engineering step is asked repeatedly (2026 Q10b).

## Reactions of haloarenes — *why they behave backwards, and the two named ring reactions*

### Why haloarenes resist nucleophilic substitution

This is a guaranteed question. There are four reasons; two or three earn full marks.

1. **Resonance.** The halogen's lone pair delocalises into the ring, so the C-X bond acquires **partial double-bond character**. A partial double bond is far harder to break than a single one.
2.  **$sp^2$  hybridisation.** The carbon holding the halogen is  $sp^2$ , so it has more **s-character** and is more electronegative than an  $sp^3$  carbon. It holds the bonding pair more tightly. The bond is also shorter — 169 pm against 177 pm in a haloalkane.
3. **Instability of the phenyl cation.** An  $S_N1$  route would need a phenyl carbocation, which gets no resonance stabilisation and is therefore not formed.
4. **Electronic repulsion.** The ring's  $\pi$ -electron cloud repels an approaching nucleophile.

### The nitro group changes everything

Put a **strongly electron-withdrawing group** such as  $-\text{NO}_2$  at the **ortho or para** position and the picture flips. The nitro group pulls electron density away, weakening the C-X bond and activating the ring towards nucleophilic attack. Each extra nitro group makes it easier:

| Compound                     | Conditions needed                     |
|------------------------------|---------------------------------------|
| chlorobenzene                | NaOH, 623 K, 300 atm                  |
| <i>p</i> -nitrochlorobenzene | NaOH, 443 K, then acidify             |
| 2,4-dinitrochlorobenzene     | NaOH, 368 K, then acidify             |
| 2,4,6-trinitrochlorobenzene  | warm water alone → <b>picric acid</b> |

**Trap:** a nitro group at the **meta** position has **no effect at all**. Resonance only pushes and pulls electron density at the ortho and para positions — meta is never in the delocalisation path. Stating this earns a mark on its own.

## Electrophilic substitution — the halogen's split personality

Halogens are **ortho/para directing but deactivating**. Both halves of that need explaining:

- **Ortho/para directing** because the halogen's lone pair delocalises *into* the ring (**+R effect**), and resonance raises the electron density specifically at the ortho and para positions. An incoming electrophile goes where the electrons are.
- **Deactivating** because the halogen is also strongly electronegative and pulls electron density *out* along the sigma bond (**-I effect**). Overall the ring ends up poorer in electrons than plain benzene, so the reaction is slower and needs harsher conditions.

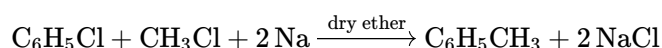
**Examiner asks:** "although chlorine shows a strong -I effect, why is it ortho/para directing?" (2026 Q25a) — answer with the two effects working in opposite directions: -I decides *how fast*, +R decides *where*.

The four standard electrophilic substitutions on chlorobenzene, all giving ortho + para with **para as the major product** (it's more symmetrical, and less crowded than ortho):

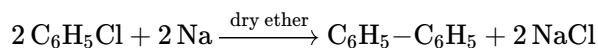
- **Halogenation** —  $\text{Cl}_2$ , anhydrous  $\text{FeCl}_3 \rightarrow$  1,2- and 1,4-dichlorobenzene.
- **Nitration** — conc.  $\text{HNO}_3$  + conc.  $\text{H}_2\text{SO}_4 \rightarrow$  1-chloro-2-nitrobenzene and 1-chloro-4-nitrobenzene.
- **Sulphonation** — conc.  $\text{H}_2\text{SO}_4$ , heat  $\rightarrow$  2- and 4-chlorobenzenesulphonic acid.
- **Friedel-Crafts**, anhydrous  $\text{AlCl}_3$  — **alkylation** with  $\text{CH}_3\text{Cl}$  gives 1-chloro-2/4-methylbenzene; **acylation** with  $\text{CH}_3\text{COCl}$  gives 2- and 4-chloroacetophenone.

## Reactions with metals

**[exposure]** The **Wurtz-Fittig reaction** is the Wurtz reaction run with a **mixture** of an alkyl halide and an aryl halide, using sodium in dry ether. The alkyl group ends up attached to the ring, giving an **alkylarene**. It's the standard way to put a side chain on benzene: chlorobenzene plus  $\text{CH}_3\text{Cl}$  gives **toluene**.



**[exposure]** The **Fittig reaction** is the same setup with **aryl halide alone** — so two rings couple to each other, giving **diphenyl (biphenyl)**.



**Trap:** one reagent tells you which. **Both** halides present  $\rightarrow$  Wurtz-Fittig  $\rightarrow$  alkylarene. **Only** the aryl halide  $\rightarrow$  Fittig  $\rightarrow$  biphenyl. "How do you convert chlorobenzene to biphenyl?" (2025 Q12a) is Fittig, not Wurtz-Fittig.

## 6.8 Polyhalogen compounds — pure recall — uses and harmful effects, and it's free marks

**[exposure]** A **polyhalogen compound** is simply a carbon compound carrying **more than one halogen atom**. The section exists in the syllabus not for the chemistry but for the applications and the environmental damage — so the marks here come from recall, not reasoning.

### **Dichloromethane, CH<sub>2</sub>Cl<sub>2</sub> (methylene chloride)**

**Uses:** solvent, paint remover, propellant in aerosols, degreasing agent.

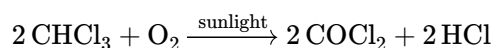
**Harm:** affects the **central nervous system**. At low levels in air, impairs hearing and vision. At high levels, causes dizziness, nausea, and tingling or numbness in the fingers and toes. Direct contact burns the skin; contact with the eye burns the cornea.

### **Trichloromethane, CHCl<sub>3</sub> (chloroform)**

**Uses:** solvent for fats, alkaloids and iodine; feedstock for making **Freon R-22**. It was once used as a general anaesthetic in surgery but has been replaced by ether, which is less harmful.

**Harm:** inhaling it **depresses the central nervous system**. Breathing about 900 ppm briefly causes dizziness, fatigue and headache. Chronic exposure damages the **liver and kidneys**.

**Examiner asks:** "why is chloroform stored in dark coloured bottles?" (2026 Q24a) — because in **light and air it is slowly oxidised to phosgene**, COCl<sub>2</sub>, an extremely poisonous gas. The bottles are dark **and filled to the brim**, so no air is left inside. Repeat question — memorise the equation:



### **Triiodomethane, CHI<sub>3</sub> (iodoform)**

Formerly used as an **antiseptic**, working through the free iodine it liberates. Now replaced by other iodine formulations because of its **objectionable smell**.

### **Tetrachloromethane, CCl<sub>4</sub> (carbon tetrachloride)**

**Uses:** refrigerant manufacture, aerosol propellant, feedstock for chlorofluorocarbons, solvent. Until the 1960s it was a cleaning fluid — an industrial **degreasing agent** and a household spot remover.

**Harm:** long exposure causes **liver cancer**. Inhalation causes dizziness, light-headedness, nausea and vomiting, and can permanently damage nerve cells. Severe cases lead to stupor, coma, unconsciousness, irregular heartbeat or death. It also **depletes the ozone layer**.

### **Freons**

**Freons** are the **chlorofluorocarbons (CFCs)** of methane and ethane. They are extremely stable, unreactive, non-toxic, non-corrosive and easily liquefiable — which is exactly what made them so useful and so damaging. The commonest is **Freon-12**, CCl<sub>2</sub>F<sub>2</sub>, made from CCl<sub>4</sub> by the **Swarts reaction**.

**Uses:** aerosol propellant, refrigeration, air conditioning.

**Harm:** because they are unreactive, they survive the journey up to the **stratosphere** unchanged. There they start a **chain reaction with ozone** and destroy the ozone layer.

### **DDT**

**p,p'-Dichlorodiphenyltrichloroethane** — two benzene rings each carrying a chlorine at the para position, joined by a carbon bearing a CCl<sub>3</sub> group. Prepared by the **condensation of chloral with chlorobenzene in sulphuric acid**.

**Use:** the first modern insecticide — effective against the mosquitoes that spread malaria and the lice that spread typhus.

**Harm:** many insect species **developed resistance** to it. It is highly toxic to fish. And it is chemically very stable and **fat-soluble**, so animals cannot metabolise it — it accumulates in fatty tissue and builds up along the food chain.

## Patterns, collected — *the six things you'll actually be asked to do*

A · Predict the product — 2-3 marks

*Recognise it: a reaction arrow with reagents written over it, and "write the major product" or "identify A and B".*

1. **Read the reagent first** — it decides everything. Aqueous KOH → alcohol. Alcoholic KOH → alkene. Mg/dry ether → Grignard. Na/dry ether → Wurtz. NaI/acetone → Finkelstein. AgF → Swarts.  $\text{SOCl}_2$  → alkyl chloride.
2. If a double bond is being attacked: Markovnikov unless "peroxide" appears, and peroxide only counts with HBr.
3. If an alkene is being formed: apply Saytzeff — more substituted is major.
4. If a ring is being attacked: ortho and para, para major.

B · Which reacts faster, and why — 2-3 marks

*Recognise it: a pair of compounds and "which undergoes faster  $S_N1/S_N2$ ".*

1. Identify which mechanism the question names. They run in opposite directions.
2.  $S_N2$  → think **steric hindrance**. Fewer/smaller groups around the carbon wins.  $1^\circ > 2^\circ > 3^\circ$ .
3.  $S_N1$  → think **carbocation stability**. Tertiary, benzylic or allylic wins.
4. If the halogens differ, the **leaving group** decides:  $\text{I} > \text{Br} > \text{Cl} > \text{F}$ .
5. Always give the *reason* word — "less steric hindrance" or "resonance-stabilised carbocation". The reason carries the mark, not the choice.

C · Give the reason — 1-2 marks

*Recognise it: "account for the following" or "why is...".*

Almost every one of these reduces to one of five stock answers: **resonance / partial double bond** (haloarene unreactivity, phenol won't react, allyl beats propyl),  **$sp^2$  vs  $sp^3$  and s-character** (dipole moment, bond length), **energy of bonds broken vs formed** (water insolubility), **van der Waals and surface area** (boiling points), **escaping gaseous by-products** (thionyl chloride).

D · Arrange in order — 1-2 marks

*Recognise it: three or four compounds, "increasing order of...".*

1. **Boiling point** → molecular mass first, branching second.
2. **Reactivity to  $S_N1$**  → carbocation stability.
3. **Reactivity to  $S_N2$**  → steric hindrance.
4. **Reactivity of RX generally** → C-X bond strength:  $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$ .

E · Conversions — 3 marks

*Recognise it: "how would you convert X into Y".*

1. Count the carbons on both sides. Same count → substitution or addition. Doubled → Wurtz.
2. Alkene → halide: HX, checking Markovnikov versus peroxide.
3. Need an iodide → make the bromide first, then Finkelstein.
4. Halide → alkene: alcoholic KOH, then check Saytzeff.
5. Halide → alkane: Mg/dry ether then water, or  $\text{LiAlH}_4$  directly.

F · Work backwards from the product — 2-3 marks

*Recognise it: the product is given and the starting material is A, or "identify the hydrocarbon".*

1. **Wurtz product** → cut the alkane in half at the middle and add X.
2. **Grignard then water** → the alkane's carbon skeleton is the halide's; put MgX back where the new hydrogen went.
3. **"Only one monochlorination product"** → all the hydrogens must be equivalent → **2,2-dimethylpropane (neopentane)**.
4. **"Three isomeric products"** → pentane. **"Four"** → 2-methylbutane.

### **Past year questions · question types — ranked by how often they turn up**

*From the last 10 years of boards — 25 questions, grouped by pattern.*

1 · Predict the product / identify A, B, C — 3-5 marks

*Recognise it: a reaction scheme, sometimes a chain of three or four arrows.*

1. Work one arrow at a time; never skip to the end.
2. Name the reaction under each arrow before you draw anything — Finkelstein, Wurtz, Friedel-Crafts, Sandmeyer.
3. Where two products are possible, say which is major and give the rule by name (Markovnikov, Saytzeff, para-major).

**Trap:** a molecule with two  $\text{-OH}$  groups, one on the ring and one on a side chain, treated with  $\text{SOCl}_2$  or  $\text{HCl}$ . Only the **side-chain**  $\text{-OH}$  reacts; the phenolic one has partial double-bond character and stays put.

2 ·  $S_N1$  /  $S_N2$  comparison — 2-3 marks

*Recognise it: a pair of halides, or "write the mechanism of the following".*

1. For a mechanism question, draw **both steps** of  $S_N1$  and label the slow step as rate-determining, or draw the  $S_N2$  transition state in square brackets.
2. For a comparison, name the deciding factor before you name the winner.

**Trap:**  $S_N2$  ordering is  $1^\circ > 2^\circ > 3^\circ$ ;  $S_N1$  is the exact reverse. Writing the reactivity order without checking which mechanism the question asked about is the commonest way to lose these marks.

3 · "Give the reason" / "account for the following" — 1-2 marks

*Recognise it: a statement of fact, then "why?".*

1. Identify which of the five stock explanations applies (see Patterns, C).
2. Write it as cause → effect in one or two sentences. No padding.

#### 4 · IUPAC naming and structure drawing — 1-3 marks

*Recognise it: "give the IUPAC name" or "draw the structure of..."*

1. Longest chain, then check for a double bond — **it takes numbering priority over the halogen**.
2. Substituents alphabetically in the name; lowest locants where there is a tie.

**Trap:** numbering from the halogen out of habit when a double bond is present. 4-chlorobut-1-ene, not 1-chlorobut-3-ene.

#### 5 · Conversions — 3 marks

*Recognise it: "how can the following conversions be carried out"*

1. Count carbons. Choose the route. Write reagents above every arrow — the reagent is what's being marked.

#### 6 · Definitions and stereochemistry — 1-2 marks

*Recognise it: "define the following term", or a pair of structures with "which is chiral?"*

1. For chirality: find the carbon with **four different** groups. Two identical groups anywhere on it and it is not chiral.
2. For definitions, give the definition and **one example** — the example is usually worth half the mark.

### Past year questions · mark slots — *what each type is worth*

| Slot           | What turns up there                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1 mark</b>  | Name the type of halide (vinylic / allylic / benzylic). One-line reason questions. Assertion-reason on reactivity orders. ( <i>inferred</i> )                |
| <b>2 marks</b> | "Give the reason" pairs. Two differences between $S_N1$ and $S_N2$ . Arrange in order of boiling point. One-step "what happens when".                        |
| <b>3 marks</b> | IUPAC naming (three compounds). Three-part "give reason for the following". Conversions. Identify A and B in a two-step scheme.                              |
| <b>5 marks</b> | Multi-part sets: a reactivity comparison plus a halide-type identification plus a polyhalogen recall, or a product-prediction set combined with definitions. |

*The video groups these as short answer type I (2 marks, Q1-10), short answer type II (3 marks, Q11-23) and long answer (5 marks, Q24-25). At 6 marks in your blueprint this chapter will most likely appear as a mix of 1- and 2-markers rather than a 5-marker — but the 5-mark sets are still the best revision, because they bundle four separate ideas into one question.*

### Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2021 Q7 · 2024 Q13(b) · 2026 Q24(a)

Which of a pair reacts faster by  $S_N2$ , with reason. Runs across RI vs RCl (leaving group), tertiary vs primary (steric hindrance), and two primaries differing in where the branch sits. The single most repeated question shape in the chapter.

2026 Q13(a) · 2024 Q22(a)

KCN gives cyanide but AgCN gives isocyanide — the ambident nucleophile question. Ionic versus covalent is the whole answer.

2020 Q4 · 2026 Q10(a) · 2021 Q21

$S_N1$ : write the mechanism, name the most reactive isomer, or give two differences from  $S_N2$ . Carbocation stability is the thread through all three.

2017 Q5(a) · lecture PYQ

Why alkyl halides, though polar, are immiscible with water. Energy of bonds broken versus energy released.

2023 Q8(a) · 2024 Q11(c) · lecture PYQ

Arrange in increasing order of boiling point. Appears in both a chain-length form and a halogen-count form.

2024 Q11(b) · lecture PYQ

Why thionyl chloride is preferred for making alkyl chlorides from alcohols. Escapable gaseous by-products.

2026 Q10(b) · 2020 Q18

Wurtz reaction run backwards — the product is given, find the starting halide. Cut the symmetrical alkane in half.

2024 Q13(c) · lecture PYQ

Why Grignard reagents must be prepared under anhydrous conditions. Water destroys them, giving the alkane.

2025 Q19(b) · 2026 Q25(b) · lecture PYQ

Friedel-Crafts on a haloarene, alkylation or acylation, asking for the **major** product. Always para, always because of symmetry and less crowding.

2025 Q15 · 2026 Q23(b) · 2020 Q18

Multi-step "identify A, B, C, D, E" schemes built from Grignard, Wurtz, alcoholic KOH and HBr addition. Three separate papers, near-identical shape.

**Past year questions · cold practice — *answers only* — *work them before you look***

### **Reactivity and mechanism**

- 2021 Q7 — which shows faster  $S_N2$ : *n*-propyl iodide or *n*-propyl chloride? The iodide —  $I^-$  is a better leaving group because of its large size and weak C-I bond
- 2021 Q7 — faster  $S_N2$ : *tert*-butyl chloride or ethyl chloride? Ethyl chloride — primary, so far less steric hindrance

- 2026 Q13(b) — why is cyclohexylmethyl chloride more reactive to  $S_N2$  than cyclohexyl chloride? The first is primary, the second secondary — less crowding at a primary carbon
- 2026 Q10(a) — most reactive  $C_4H_9Br$  isomer towards  $S_N1$ . 2-bromo-2-methylpropane — it forms the most stable, tertiary carbocation
- 2021 Q21 — faster  $S_N1$ : cyclohexylmethyl chloride or benzyl chloride? Benzyl chloride — the benzyl carbocation is resonance-stabilised by the ring
- 2026 Q24(a) — more reactive to  $S_N1$ : 2-bromo-2-methylbutane or 1-bromopentane? 2-bromo-2-methylbutane — tertiary carbocation beats primary
- 2025 Q20(b) — why is allyl chloride hydrolysed faster than *n*-propyl chloride? The allyl carbocation is resonance-stabilised; the propyl one is not
- 2026 Q23(c) — which of  $S_N1/S_N2$  gives inversion, which gives racemisation?  $S_N2$  inversion (back-side attack);  $S_N1$  racemisation (planar carbocation, attack from either face)

### Reason questions

- Lecture PYQ Q1 — why are haloarenes less reactive to nucleophilic substitution? Resonance gives the C-X bond partial double-bond character, and the  $sp^2$  carbon holds the pair more tightly
- 2024 Q16 — why is the dipole moment of chlorobenzene lower than that of cyclohexyl chloride? The  $sp^2$  carbon is more electronegative, so less charge separation; resonance also shortens the C-Cl bond
- 2020 Q9(b) — why does  $-NO_2$  at ortho/para increase haloarene reactivity to nucleophilic substitution? It withdraws electron density, weakening the C-X bond and activating the ring
- 2026 Q25(a) — chlorine shows -I yet is ortho/para directing. Why? -I deactivates the whole ring, but +R resonance raises electron density specifically at ortho and para
- 2024 Q22(c) — why are d- and l-butan-2-ol hard to separate by fractional distillation? Enantiomers have identical boiling points
- 2026 Q24(a) — why is chloroform stored in dark bottles, filled to the brim? Light and air oxidise it to phosgene,  $COCl_2$ , a poisonous gas

### Products and conversions

- 2023 Q3 — isopropyl alcohol +  $PCl_5$ , then AgCN. Isopropyl chloride, then isopropyl isocyanide ( $(CH_3)_2CHNC$ )
- 2023 Q3 — 1-chloropropane + alcoholic KOH, then HBr. Propene, then 2-bromopropane (Markovnikov)
- 2020 Q9(a) — convert but-1-ene to 1-iodobutane. HBr/peroxide  $\rightarrow$  1-bromobutane, then NaI /acetone (Finkelstein)
- 2025 Q12(a) — convert chlorobenzene to biphenyl. Na/dry ether — Fittig reaction
- 2025 Q12(c) — convert 2-bromobutane to but-2-ene. Alcoholic KOH; but-2-ene is major by Saytzeff
- 2025 Q20(a) — methyl bromide + AgF. Methyl fluoride — Swarts reaction
- 2025 Q20(b) — 2,4,6-trinitrochlorobenzene, hydrolysed. 2,4,6-trinitrophenol, i.e. picric acid

- 2026 Q25(a) — chlorobenzene + conc.  $\text{H}_2\text{SO}_4$ , heat: major product. 4-chlorobenzenesulphonic acid
- 2026 Q25(b) — bromobenzene +  $\text{CH}_3\text{Cl}$  / anhydrous  $\text{AlCl}_3$ : major product. 1-bromo-4-methylbenzene (para major)
- 2025 Q19(a) — a nitro-substituted ethylbenzene heated with  $\text{Br}_2$ : major product. Bromination at the benzylic carbon — that free radical is resonance-stabilised
- 2026 Q25(b) — dehydrohalogenation of 1-bromo-1-methylcyclohexane. 1-methylcyclohexene — the more substituted alkene, by Saytzeff

### **Naming, structure and identification**

- 2026 Q2(a) — draw 4-bromo-3-methylpent-2-ene.  $\text{CH}_3-\text{CH}=\text{C}(\text{CH}_3)-\text{CHBr}-\text{CH}_3$
- 2024 Q11(a) — IUPAC name of a four-carbon chain with a double bond and Cl at the far end. 4-chlorobut-1-ene — the double bond takes numbering priority
- 2017 Q14 — IUPAC names of  $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$ , 1,3-dibromobenzene, and a chloropropene. 2-bromobutane; 1,3-dibromobenzene; 3-chloroprop-1-ene
- 2026 Q23(a) — a hydrocarbon  $\text{C}_5\text{H}_{12}$  gives only one monochlorination product. Identify it. 2,2-dimethylpropane — all twelve hydrogens are equivalent
- 2026 Q10(b) — which  $\text{C}_4\text{H}_9\text{Br}$  isomer gives 2,5-dimethylhexane with Na/dry ether? 1-bromo-2-methylpropane — cut the product in half
- 2026 Q24(a) — halogen present in  $\text{CH}_2=\text{CH}-\text{Cl}$ ? Vinylic — chlorine sits directly on a doubly-bonded carbon
- 2017 Q5(b) — which of the pair is the allylic halide? The one where Cl is on an  $sp^3$  carbon *next to* the double bond, not on it
- 2026 Q17(b) — which of a pair is chiral? The one whose carbon carries four *different* groups; two identical groups anywhere and it isn't
- 2026 Q24(b) — define ambident nucleophile and racemic mixture. Two donor atoms, attacks through either ( $\text{CN}^-$ ,  $\text{NO}_2^-$ ); and a 1:1 mixture of enantiomers, optically inactive because the rotations cancel

Built from Sourabh Raina's Haloalkanes and Haloarenes one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 6 (Rationalised 2022-23). NCERT spellings used throughout: **Zaitsev** (NCERT notes it is "also pronounced as Saytzeff"), **ambident** nucleophile, **vicinal** and **geminal** dihalides, **laevorotatory**. C-Cl bond lengths (177 pm in haloalkanes, 169 pm in haloarenes) are NCERT's. First-contact material: every term marked **[exposure]** is explained from scratch before use.

Two stretches of both transcripts were lost at automatic-transcription chunk boundaries and were recovered by re-transcribing those windows from the source audio: the  $S_N2$  mechanism and ambident-nucleophile block in the lecture, and PYQ questions 12, 13, 17, 18, 22 and 23. All twenty-five PYQ questions are accounted for here.

# Every Chemistry Formula

Class XII CBSE · Chemistry · Chapters 1–6

*Every formula the half-yearly can ask for, each with what its symbols mean, the unit it comes out in, and the one situation that tells you to reach for it. The cue is on the outside and the formula is hidden — so read down the page and say each one before you open it.*

- Formulas: 50
- Recall rules: 18
- Must be instant: 40
- Paper: 70 marks

## How to use this

● means you should be able to say it before the question finishes loading. ○ means a few seconds of thought is fine — these are the ones you derive or reconstruct rather than recall.

Work down a chapter with everything closed. Say the formula out loud, *then* tap to reveal. A formula you can only recognise is not a formula you have.

Units are not decoration. Most lost marks in Solutions and Electrochemistry are unit slips — molality is per **kilogram of solvent**, molarity per **litre of solution**, and  $\kappa$  in  $\text{S cm}^{-1}$  with  $c$  in  $\text{mol L}^{-1}$  is what makes the 1000 appear in molar conductivity.

## Recognise index

*Every cue on one screen. Read the cue, say the formula, tap to check.*

### 1 · Solutions — 15 marks

- C1.1 Concentration by mass, volume, or parts per million
- C1.2 Mole fraction of a component
- C1.3 Molarity, from mass or from moles
- C1.4 Molality, and why it is the one used for colligative properties
- C1.5 A gas dissolving in a liquid under pressure
- C1.6 Vapour pressure of a mixture of two volatile liquids
- C1.7 Vapour pressure when the solute is non-volatile
- C1.8 Relative lowering of vapour pressure
- C1.9 Boiling point raised by a dissolved solute
- C1.10 Freezing point lowered by a dissolved solute
- C1.11 Osmotic pressure across a semipermeable membrane
- C1.12 Molar mass from any one colligative property
- C1.13 The measured effect is bigger or smaller than predicted
- C1.14 Degree of dissociation or association from  $i$

### 2 · Electrochemistry — 14 marks

- C2.1 Standard cell potential from two electrode potentials

- C2.2 Cell potential at non-standard concentrations
- C2.3 Nernst for a single electrode
- C2.4 Free energy change of a cell reaction
- C2.5 Equilibrium constant from cell potential
- C2.6 Conductance, resistance and the cell constant
- C2.7 Conductivity from measured resistance
- C2.8 Molar conductivity from conductivity
- C2.9 Limiting molar conductivity of an electrolyte from its ions
- C2.10 Degree of dissociation of a weak electrolyte
- C2.11 Dissociation constant of a weak acid from conductivity
- C2.12 Mass deposited during electrolysis
- C2.13 Moles of electrons passed

### 3 · Chemical Kinetics — 13 marks

- C3.1 Rate of a reaction from a concentration change
- C3.2 Rate written for a balanced equation with coefficients
- C3.3 Rate law and the order of a reaction
- C3.4 Units of the rate constant for any order
- C3.5 Zero order — concentration against time
- C3.6 Zero order half-life
- C3.7 First order rate constant from concentrations
- C3.8 First order half-life
- C3.9 First order reaction in the gas phase, measured by total pressure
- C3.10 Rate constant against temperature
- C3.11 Activation energy from a straight-line plot
- C3.12 Activation energy from two temperatures

### 4 · d and f Block — 11 marks

- C4.1 Magnetic moment from unpaired electrons, and back again
- C4.2 General electronic configuration of a block

### 5 · Coordination Compounds — 11 marks

- C5.1 Oxidation number of the metal inside a complex
- C5.2 Coordination number when a ligand grips more than once
- C5.3 Tetrahedral splitting compared with octahedral
- C5.4 Crystal field stabilisation energy of an octahedral complex
- C5.5 High spin or low spin — deciding which
- C5.6 Number of geometrical isomers of a complex

## 6 · Haloalkanes and Haloarenes — 6 marks

- C6.1 Rate law that identifies the substitution mechanism
- C6.2 Degree of unsaturation from a molecular formula

Constants — worth the same marks as the formulas

- K Every constant this paper can hand you, with its unit

## CH 1 Solutions — 14 formulas · 15 marks

### ● C1.1 Concentration by mass, by volume, or in parts per million

$$\% w/w = \frac{w_B}{w_A + w_B} \times 100 \quad \% v/v = \frac{V_B}{V_{\text{soln}}} \times 100$$

$$\% w/v = \frac{w_B}{V_{\text{soln in mL}}} \times 100 \quad \text{ppm} = \frac{w_B}{w_{\text{soln}}} \times 10^6$$

| Symbol            | Meaning                  | Unit          |
|-------------------|--------------------------|---------------|
| $w_B$             | mass of solute           | g             |
| $w_A$             | mass of solvent          | g             |
| $V_{\text{soln}}$ | volume of solution       | mL or L       |
| all four          | the concentration itself | dimensionless |

#### Use it when:

The question gives masses or volumes and no moles anywhere. ppm is for trace quantities — pollutants, dissolved gases, hardness of water.

#### Trap:

$\% w/w$  divides by the mass of the **solution**, not the solvent. Adding 10 g of solute to 90 g of water gives 10%, not 11.1%.

### ● C1.2 Mole fraction of a component

$$x_A = \frac{n_A}{n_A + n_B} \quad x_A + x_B = 1$$

| Symbol | Meaning                       | Unit          |
|--------|-------------------------------|---------------|
| $x_A$  | mole fraction of the solvent  | dimensionless |
| $n_A$  | moles of solvent, $= w_A/M_A$ | mol           |
| $n_B$  | moles of solute               | mol           |

#### Use it when:

Raoult's law or Henry's law is involved — both are written in mole fractions, never in molarity.

**Trap:**

For water as solvent,  $M_A = 18$ , so  $n_A$  is large and  $x_B$  is tiny. Do not round  $x_A$  to 1 before subtracting — you will get  $x_B = 0$ .

● **C1.3 Molarity, from moles or straight from mass**

$$M = \frac{n_B}{V_{\text{soln in L}}} = \frac{w_B \times 1000}{M_B \times V_{\text{soln in mL}}}$$

| Symbol | Meaning                   | Unit                |
|--------|---------------------------|---------------------|
| $M$    | molarity                  | $\text{mol L}^{-1}$ |
| $n_B$  | moles of solute           | mol                 |
| $M_B$  | molar mass of solute      | $\text{g mol}^{-1}$ |
| $V$    | volume of <b>solution</b> | L (or mL)           |

**Use it when:**

Volume of solution is given or asked. Also the concentration used in osmotic pressure,  $\pi = CRT$ .

**Trap:**

Molarity **changes with temperature** because volume does. Molality does not. If a question mentions heating or cooling and then asks which concentration term is unaffected, the answer is molality.

● **C1.4 Molality — and the unit that makes it the colligative one**

$$m = \frac{n_B}{w_A \text{ in kg}} = \frac{w_B \times 1000}{M_B \times w_A \text{ in g}}$$

| Symbol | Meaning                     | Unit                 |
|--------|-----------------------------|----------------------|
| $m$    | molality                    | $\text{mol kg}^{-1}$ |
| $w_A$  | mass of <b>solvent</b> only | kg (or g)            |
| $M_B$  | molar mass of solute        | $\text{g mol}^{-1}$  |

**Use it when:**

Anything colligative:  $\Delta T_b$ ,  $\Delta T_f$ . Both are written in molality precisely because it is temperature-independent.

**Trap:**

Per kg of **solvent**, not of solution. The 1000 in the second form converts grams of solvent to kilograms — it is not the same 1000 as in molarity, where it converts mL of solution to L.

### ● C1.5 A gas dissolving in a liquid under pressure

$$p = K_H x$$

| Symbol | Meaning                                        | Unit          |
|--------|------------------------------------------------|---------------|
| $p$    | partial pressure of the gas above the solution | bar (or kbar) |
| $x$    | mole fraction of the gas in solution           | dimensionless |
| $K_H$  | Henry's law constant                           | bar (or kbar) |

#### Use it when:

A gas is being dissolved — soda bottles, deep-sea diving and the bends, oxygen at altitude, aquatic life in warm water.

#### Trap:

$K_H$  goes **up** with temperature, so solubility goes **down** — which is why warm water holds less dissolved oxygen. A higher  $K_H$  means a *less* soluble gas.

### ● C1.6 Vapour pressure of a mixture of two volatile liquids

$$p_{\text{total}} = p_A^\circ x_A + p_B^\circ x_B = p_B^\circ + (p_A^\circ - p_B^\circ) x_A$$

| Symbol      | Meaning                                                            | Unit          |
|-------------|--------------------------------------------------------------------|---------------|
| $p_A^\circ$ | vapour pressure of pure A                                          | bar or mm Hg  |
| $x_A$       | mole fraction of A in the <b>liquid</b>                            | dimensionless |
| $y_A$       | mole fraction of A in the <b>vapour</b> , $= p_A/p_{\text{total}}$ | dimensionless |

#### Use it when:

Both components evaporate. The second form is the equation of a straight line in  $x_A$  — that is the plot they ask you to sketch.

#### Trap:

Vapour composition  $y_A$  is not the same as liquid composition  $x_A$ . The vapour is always richer in the more volatile component.

### ● C1.7 Vapour pressure when the solute is non-volatile

$$p_{\text{soln}} = x_A p_A^\circ$$

| Symbol            | Meaning                             | Unit          |
|-------------------|-------------------------------------|---------------|
| $p_{\text{soln}}$ | vapour pressure of the solution     | bar or mm Hg  |
| $x_A$             | mole fraction of the <b>solvent</b> | dimensionless |
| $p_A^\circ$       | vapour pressure of pure solvent     | bar or mm Hg  |

**Use it when:**

A solid, non-volatile solute (sugar, urea, a salt) is dissolved. Only the solvent contributes vapour.

**Trap:**

It is  $x$  of the **solvent**, not the solute. Using  $x_B$  here is the single commonest slip in this chapter.

● **c1.8 Relative lowering of vapour pressure**

$$\frac{p_A^\circ - p_{\text{soln}}}{p_A^\circ} = x_B = \frac{n_B}{n_A + n_B} \approx \frac{w_B M_A}{M_B w_A}$$

| Symbol | Meaning                                      | Unit                |
|--------|----------------------------------------------|---------------------|
| $x_B$  | mole fraction of solute                      | dimensionless       |
| $M_A$  | molar mass of solvent                        | $\text{g mol}^{-1}$ |
| $M_B$  | molar mass of solute — usually what is asked | $\text{g mol}^{-1}$ |

**Use it when:**

Two vapour pressures are given (pure solvent and solution) and a molar mass is wanted.

**Trap:**

The final approximation assumes  $n_B \ll n_A$  — valid for a dilute solution only. If the question says concentrated, use the exact form.

● **c1.9 Boiling point raised by a dissolved solute**

$$\Delta T_b = K_b m \quad \Delta T_b = T_b - T_b^\circ$$

| Symbol       | Meaning                                           | Unit                   |
|--------------|---------------------------------------------------|------------------------|
| $\Delta T_b$ | elevation of boiling point                        | K                      |
| $K_b$        | molal elevation constant (ebullioscopic constant) | $\text{K kg mol}^{-1}$ |
| $m$          | molality                                          | $\text{mol kg}^{-1}$   |

**Use it when:**

A boiling point shifts. For water  $K_b = 0.52 \text{ K kg mol}^{-1}$ .

**Trap:**

$K_b$  is a property of the **solvent** alone — it never depends on which solute you dissolved.

● **c1.10 Freezing point lowered by a dissolved solute**

$$\Delta T_f = K_f m \quad \Delta T_f = T_f^\circ - T_f$$

| Symbol       | Meaning                                         | Unit                   |
|--------------|-------------------------------------------------|------------------------|
| $\Delta T_f$ | depression of freezing point                    | K                      |
| $K_f$        | molal depression constant (cryoscopic constant) | K kg mol <sup>-1</sup> |
| $m$          | molality                                        | mol kg <sup>-1</sup>   |

#### Use it when:

A freezing point shifts — antifreeze in a radiator, salt on an icy road. For water  $K_f = 1.86 \text{ K kg mol}^{-1}$ .

#### Trap:

Note the order of subtraction is reversed from  $\Delta T_b$ : freezing point goes **down**, so it is pure minus solution. Both  $\Delta T$  values come out positive.

### ● C1.11 Osmotic pressure across a semipermeable membrane

$$\pi = CRT = \frac{n_B}{V}RT \quad \pi V = n_B RT$$

| Symbol | Meaning              | Unit                                           |
|--------|----------------------|------------------------------------------------|
| $\pi$  | osmotic pressure     | bar or atm                                     |
| $C$    | molarity             | mol L <sup>-1</sup>                            |
| $R$    | gas constant         | 0.0821 L atm K <sup>-1</sup> mol <sup>-1</sup> |
| $T$    | absolute temperature | K                                              |

#### Use it when:

A membrane, a molar mass of a **protein or polymer**, or the word isotonic/hypertonic appears. This is the method of choice for large molar masses because  $\pi$  is measurably large even when  $\Delta T_f$  would be tiny.

#### Trap:

Uses molarity, not molality — the only colligative property that does. Match the  $R$  value to the pressure unit you want.

### ● C1.12 Molar mass from any one colligative property

$$M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A} \quad M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A} \quad M_B = \frac{w_B RT}{\pi V}$$

| Symbol | Meaning                  | Unit                |
|--------|--------------------------|---------------------|
| $M_B$  | molar mass of the solute | $\text{g mol}^{-1}$ |
| $w_B$  | mass of solute           | g                   |
| $w_A$  | mass of solvent          | g                   |
| $V$    | volume of solution       | L                   |

**Use it when:**

"Calculate the molar mass of the solute" — the standard 3-marker. All three are just C1.9–C1.11 rearranged; do not memorise them separately.

**Trap:**

The 1000 is there because  $w_A$  is in grams while molality needs kilograms. Drop it and your answer is out by exactly a factor of 1000.

● **C1.13 When the measured effect is bigger or smaller than predicted**

$$i = \frac{\text{observed colligative property}}{\text{calculated colligative property}} = \frac{\text{normal molar mass}}{\text{abnormal molar mass}}$$

$$\Delta T_b = iK_b m \quad \Delta T_f = iK_f m \quad \pi = iCRT$$

| Symbol  | Meaning                                                               | Unit          |
|---------|-----------------------------------------------------------------------|---------------|
| $i$     | van't Hoff factor                                                     | dimensionless |
| $i > 1$ | solute dissociates — e.g. NaCl, $i \rightarrow 2$                     | dimensionless |
| $i < 1$ | solute associates — e.g. benzoic acid in benzene, $i \rightarrow 0.5$ | dimensionless |

**Use it when:**

The solute is ionic, or the question uses the word "abnormal", or the observed  $\Delta T_f$  does not match the calculated one.

**Trap:**

Molar mass and  $i$  move **opposite** ways. Dissociation ( $i > 1$ ) makes the measured molar mass come out *lower* than the true value.

○ **C1.14 Degree of dissociation or association from  $i$**

$$\alpha_{\text{dissociation}} = \frac{i - 1}{n - 1} \quad \alpha_{\text{association}} = \frac{1 - i}{1 - \frac{1}{n}}$$

| Symbol   | Meaning                                                       | Unit          |
|----------|---------------------------------------------------------------|---------------|
| $\alpha$ | degree of dissociation or association                         | dimensionless |
| $n$      | number of particles one formula unit gives (or combines into) | dimensionless |
| $i$      | van't Hoff factor                                             | dimensionless |

### Use it when:

$i$  has been found and the percentage dissociation is asked. For  $\text{K}_2\text{SO}_4$ ,  $n = 3$ ; for a dimerising acid,  $n = 2$ .

### Trap:

Read  $n$  off the actual dissociation equation, not the number of atoms.  $\text{K}_2\text{SO}_4 \longrightarrow 2\text{K}^+ + \text{SO}_4^{2-}$  gives three particles, so  $n = 3$ .

## CH 2 Electrochemistry — 13 formulas · 14 marks

### ● C2.1 Standard cell potential from two electrode potentials

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = E_{\text{right}}^{\circ} - E_{\text{left}}^{\circ}$$

| Symbol                    | Meaning                                               | Unit |
|---------------------------|-------------------------------------------------------|------|
| $E_{\text{cell}}^{\circ}$ | standard cell potential                               | V    |
| $E^{\circ}$               | standard <b>reduction</b> potential of each electrode | V    |

### Use it when:

Two half-cells and their  $E^{\circ}$  values are given. A positive  $E_{\text{cell}}^{\circ}$  means the reaction is spontaneous as written. Daniell cell: 1.1 V.

### Trap:

Both values in the table are **reduction** potentials — subtract, never add, and never flip the sign of the anode value and then also subtract.

### ● C2.2 Cell potential when concentrations are not 1 M

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \log Q \xrightarrow{298 \text{ K}} E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$$

| Symbol | Meaning                                          | Unit                                      |
|--------|--------------------------------------------------|-------------------------------------------|
| $n$    | moles of electrons in the balanced cell reaction | dimensionless                             |
| $Q$    | reaction quotient, products over reactants       | dimensionless                             |
| $F$    | Faraday constant                                 | 96487 C mol <sup>-1</sup>                 |
| $R$    | gas constant                                     | 8.314 J K <sup>-1</sup> mol <sup>-1</sup> |

### Use it when:

Any concentration is given that is not 1 M, or the question says "calculate the emf of the cell".

**Trap:**

NCERT uses **0.059**, not 0.0591. Pure solids and liquids never appear in  $Q$ . Get  $n$  from the balanced equation, not from the charge on one ion.

○ **C2.3 Nernst equation for a single electrode**

$$E_{(M^{n+}/M)} = E_{(M^{n+}/M)}^{\circ} - \frac{0.059}{n} \log \frac{1}{[M^{n+}]}$$

| Symbol     | Meaning                        | Unit                |
|------------|--------------------------------|---------------------|
| $[M^{n+}]$ | concentration of the metal ion | mol L <sup>-1</sup> |
| $n$        | charge on the metal ion        | dimensionless       |
| $E$        | electrode potential            | V                   |

**Use it when:**

Only one half-cell is being asked about, or a concentration cell is given where both electrodes are the same metal at different concentrations.

**Trap:**

The solid metal is taken as unit activity, so it is the 1 in the numerator. For a concentration cell  $E_{\text{cell}}^{\circ} = 0$  and the whole emf comes from the log term.

● **C2.4 Free energy change of a cell reaction**

$$\Delta_r G = -nFE_{\text{cell}} \quad \Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

| Symbol       | Meaning                                  | Unit                      |
|--------------|------------------------------------------|---------------------------|
| $\Delta_r G$ | Gibbs energy change of the cell reaction | J mol <sup>-1</sup>       |
| $n$          | moles of electrons transferred           | dimensionless             |
| $F$          | Faraday constant                         | 96487 C mol <sup>-1</sup> |

**Use it when:**

The question asks for  $\Delta G$ , for maximum work obtainable, or whether a reaction is feasible.

**Trap:**

Answer comes out in joules, not kilojoules — divide by 1000 if the options are in kJ. The minus sign is part of the formula: a positive emf gives a negative  $\Delta G$ .

## ○ C2.5 Equilibrium constant from cell potential

$$\Delta_r G^\circ = -RT \ln K_c \quad \implies \quad E_{\text{cell}}^\circ = \frac{0.059}{n} \log K_c$$

| Symbol                  | Meaning                                   | Unit          |
|-------------------------|-------------------------------------------|---------------|
| $K_c$                   | equilibrium constant of the cell reaction | dimensionless |
| $E_{\text{cell}}^\circ$ | standard cell potential                   | V             |
| $n$                     | moles of electrons                        | dimensionless |

### Use it when:

$K_c$  is asked from  $E^\circ$ , or vice versa. It is just the Nernst equation with  $E_{\text{cell}} = 0$  and  $Q = K_c$  — at equilibrium the cell is dead.

### Trap:

Note this is +, not -: setting  $E_{\text{cell}} = 0$  moves the log term across the equals sign.

## ● C2.6 Conductance, resistance and the cell constant

$$G = \frac{1}{R} \quad G^* = \frac{l}{A} \quad R = \rho \frac{l}{A}$$

| Symbol | Meaning       | Unit                                |
|--------|---------------|-------------------------------------|
| $G$    | conductance   | S (siemens)                         |
| $R$    | resistance    | $\Omega$                            |
| $G^*$  | cell constant | $\text{cm}^{-1}$ or $\text{m}^{-1}$ |
| $\rho$ | resistivity   | $\Omega \text{ cm}$                 |

### Use it when:

A conductivity cell is described by its electrode area and separation, or the cell constant is given directly.

### Trap:

$1 \text{ S} = 1 \Omega^{-1}$ , also written mho. The cell constant has units of **inverse** length because it is length  $\div$  area.

## ● C2.7 Conductivity from a measured resistance

$$\kappa = \frac{1}{\rho} = G \times G^* = \frac{1}{R} \times \frac{l}{A}$$

| Symbol   | Meaning                             | Unit                                    |
|----------|-------------------------------------|-----------------------------------------|
| $\kappa$ | conductivity (specific conductance) | S cm <sup>-1</sup> or S m <sup>-1</sup> |
| $G^*$    | cell constant                       | cm <sup>-1</sup>                        |
| $R$      | measured resistance                 | $\Omega$                                |

#### Use it when:

The question gives a resistance from a conductivity cell.  $\kappa$  is the conductance of a 1 cm cube of solution.

#### Trap:

$\kappa$  **decreases** on dilution (fewer ions per unit volume) while molar conductivity **increases**. Being asked to explain that opposite behaviour is a standing 2-marker.

### ● C2.8 Molar conductivity from conductivity

$$\Lambda_m = \frac{\kappa}{c} = \frac{\kappa \times 1000}{c \text{ in mol L}^{-1}}$$

| Symbol      | Meaning            | Unit                                |
|-------------|--------------------|-------------------------------------|
| $\Lambda_m$ | molar conductivity | S cm <sup>2</sup> mol <sup>-1</sup> |
| $\kappa$    | conductivity       | S cm <sup>-1</sup>                  |
| $c$         | concentration      | mol L <sup>-1</sup>                 |

#### Use it when:

Converting a measured  $\kappa$  into the quantity you actually compare between electrolytes.

#### Trap:

The 1000 exists only because  $\kappa$  is per **cm<sup>3</sup>** while  $c$  is per **litre**. In SI ( $\kappa$  in S m<sup>-1</sup>,  $c$  in mol m<sup>-3</sup>) there is no 1000 and the unit is S m<sup>2</sup> mol<sup>-1</sup>.

### ● C2.9 Limiting molar conductivity built from its ions

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

| Symbol              | Meaning                                         | Unit                                |
|---------------------|-------------------------------------------------|-------------------------------------|
| $\Lambda_m^\circ$   | limiting molar conductivity (infinite dilution) | S cm <sup>2</sup> mol <sup>-1</sup> |
| $\lambda_\pm^\circ$ | limiting molar conductivity of each ion         | S cm <sup>2</sup> mol <sup>-1</sup> |
| $\nu_\pm$           | how many of that ion one formula unit gives     | dimensionless                       |

#### Use it when:

$\Lambda_m^\circ$  of a **weak** electrolyte is wanted — it cannot be measured by extrapolation, so it is assembled from strong electrolytes. This is **Kohlrausch's law of independent migration**

of ions.

**Trap:**

Multiply by  $\nu$ . For  $\text{CaCl}_2$ ,  $\Lambda_m^\circ = \lambda_{\text{Ca}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ$  — the 2 is easy to drop.

● **C2.10 Degree of dissociation of a weak electrolyte**

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

| Symbol            | Meaning                                 | Unit                             |
|-------------------|-----------------------------------------|----------------------------------|
| $\alpha$          | degree of dissociation                  | dimensionless                    |
| $\Lambda_m$       | molar conductivity at concentration $c$ | $\text{S cm}^2 \text{ mol}^{-1}$ |
| $\Lambda_m^\circ$ | limiting molar conductivity             | $\text{S cm}^2 \text{ mol}^{-1}$ |

**Use it when:**

A weak acid or base is given with both conductivity values.

**Trap:**

Only meaningful for a **weak** electrolyte. A strong electrolyte is fully dissociated; its  $\Lambda_m$  rises with dilution because ion-ion interference falls, not because more of it dissociates.

○ **C2.11 Dissociation constant of a weak acid from conductivity**

$$K_a = \frac{c\alpha^2}{1-\alpha} = \frac{c\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}$$

| Symbol   | Meaning                   | Unit                |
|----------|---------------------------|---------------------|
| $K_a$    | dissociation constant     | $\text{mol L}^{-1}$ |
| $c$      | concentration of the acid | $\text{mol L}^{-1}$ |
| $\alpha$ | degree of dissociation    | dimensionless       |

**Use it when:**

Both  $\Lambda_m$  and  $\Lambda_m^\circ$  are given for a weak acid and  $K_a$  or  $\text{p}K_a$  is asked.

**Trap:**

Do not approximate  $1 - \alpha \approx 1$  unless  $\alpha$  is genuinely small (below about 0.05).

● **C2.12 Mass deposited or liberated during electrolysis**

$$Q = It \quad w = ZIt \quad Z = \frac{M}{nF}$$

| Symbol | Meaning                    | Unit              |
|--------|----------------------------|-------------------|
| $Q$    | charge passed              | C                 |
| $I$    | current                    | A                 |
| $t$    | time                       | s                 |
| $w$    | mass deposited             | g                 |
| $Z$    | electrochemical equivalent | $\text{g C}^{-1}$ |

**Use it when:**

A current runs for a time and a mass is asked. Faraday's first law of electrolysis.

**Trap:**

Time must be in **seconds**. A question quoting minutes or hours is testing exactly this.

● **C2.13 Moles of electrons passed, and how much substance that gives**

$$n_{e^-} = \frac{Q}{F} = \frac{It}{96487} \quad \text{moles of product} = \frac{n_{e^-}}{n}$$

| Symbol    | Meaning                                          | Unit                       |
|-----------|--------------------------------------------------|----------------------------|
| $n_{e^-}$ | moles of electrons                               | mol                        |
| $F$       | charge on one mole of electrons                  | $96487 \text{ C mol}^{-1}$ |
| $n$       | electrons needed per ion, from the half-equation | dimensionless              |

**Use it when:**

Several cells are in series, or a gas volume at STP is asked. Faraday's second law: the same charge deposits amounts in the ratio of their equivalent masses.

**Trap:**

Get  $n$  from the half-reaction.  $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$  needs 2 F per mole;  $\text{Al}^{3+}$  needs 3 F.

**CH 3 Chemical Kinetics — 12 formulas · 13 marks**

● **C3.1 Rate of a reaction from a concentration change**

$$r_{\text{av}} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t} \quad r_{\text{inst}} = -\frac{d[R]}{dt}$$

| Symbol | Meaning                     | Unit                               |
|--------|-----------------------------|------------------------------------|
| $[R]$  | concentration of a reactant | $\text{mol L}^{-1}$                |
| $t$    | time                        | s                                  |
| $r$    | rate of reaction            | $\text{mol L}^{-1} \text{ s}^{-1}$ |

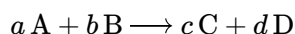
**Use it when:**

A table of concentration against time is given. Instantaneous rate is the slope of the tangent to that curve.

**Trap:**

The minus sign exists so the rate comes out positive — reactant concentration is falling. Never report a negative rate.

● **C3.2 Rate written for an equation that has coefficients**



$$r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = +\frac{1}{c} \frac{d[C]}{dt} = +\frac{1}{d} \frac{d[D]}{dt}$$

| Symbol       | Meaning                         | Unit                              |
|--------------|---------------------------------|-----------------------------------|
| $a, b, c, d$ | stoichiometric coefficients     | dimensionless                     |
| $r$          | the one unique rate of reaction | $\text{mol L}^{-1} \text{s}^{-1}$ |

**Use it when:**

The question gives the rate of disappearance of one species and asks for another.  
 $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$  makes  $\text{NO}_2$  four times as fast as it makes  $\text{O}_2$ .

**Trap:**

You **divide** by the coefficient. Multiplying is the reflex error.

● **C3.3 Rate law, and the order of a reaction**

$$r = k [A]^x [B]^y \quad \text{order} = x + y$$

| Symbol | Meaning                       | Unit                        |
|--------|-------------------------------|-----------------------------|
| $k$    | rate constant                 | depends on order — see C3.4 |
| $x, y$ | order with respect to A and B | dimensionless               |
| order  | overall order                 | dimensionless               |

**Use it when:**

A table of initial rates against initial concentrations is given. Double one concentration, see what the rate does: unchanged  $\rightarrow$  order 0, doubles  $\rightarrow$  1, quadruples  $\rightarrow$  2.

**Trap:**

Order is found by **experiment** and can be zero or fractional. Molecularity comes from the mechanism, is a whole number, and is never zero. A **pseudo first order reaction** is second order on paper but first order in practice because one reactant is in huge excess.

### ● C3.4 Units of the rate constant for any order

$$[k] = \text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$$

| Symbol  | Meaning       | Unit                              |
|---------|---------------|-----------------------------------|
| $n = 0$ | zero order    | $\text{mol L}^{-1} \text{s}^{-1}$ |
| $n = 1$ | first order   | $\text{s}^{-1}$                   |
| $n = 2$ | second order  | $\text{L mol}^{-1} \text{s}^{-1}$ |
| $n$     | overall order | dimensionless                     |

#### Use it when:

The order is asked and only the units of  $k$  are given — this runs backwards perfectly, and it is a favourite one-marker.

#### Trap:

First order  $k$  has no concentration unit at all, which is why a first-order half-life does not depend on concentration.

### ● C3.5 Zero order — concentration against time

$$[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$$

| Symbol  | Meaning                   | Unit                              |
|---------|---------------------------|-----------------------------------|
| $[R]_0$ | initial concentration     | $\text{mol L}^{-1}$               |
| $[R]$   | concentration at time $t$ | $\text{mol L}^{-1}$               |
| $k$     | zero order rate constant  | $\text{mol L}^{-1} \text{s}^{-1}$ |

#### Use it when:

The plot of  $[R]$  against  $t$  is a straight line of slope  $-k$ . Real examples: decomposition of  $\text{NH}_3$  on hot platinum, of  $\text{HI}$  on gold.

#### Trap:

Zero order means the rate is independent of concentration — usually because a catalyst surface is saturated, not because nothing is happening.

### ● C3.6 Zero order half-life

$$t_{1/2} = \frac{[R]_0}{2k}$$

| Symbol    | Meaning               | Unit                                |
|-----------|-----------------------|-------------------------------------|
| $t_{1/2}$ | half-life             | s                                   |
| $[R]_0$   | initial concentration | mol L <sup>-1</sup>                 |
| $k$       | rate constant         | mol L <sup>-1</sup> s <sup>-1</sup> |

**Use it when:**

A zero-order reaction's half-life is asked, or the question tests that  $t_{1/2} \propto [R]_0$  here.

**Trap:**

This one **does** depend on starting concentration — the exact opposite of first order. Getting the two the wrong way round is the standing trap of this chapter.

● **C3.7 First order rate constant from concentrations**

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]} \quad \log[R] = -\frac{k}{2.303}t + \log[R]_0$$

| Symbol  | Meaning                   | Unit                |
|---------|---------------------------|---------------------|
| $k$     | first order rate constant | s <sup>-1</sup>     |
| $[R]_0$ | initial concentration     | mol L <sup>-1</sup> |
| $[R]$   | concentration at time $t$ | mol L <sup>-1</sup> |

**Use it when:**

The commonest numerical in the chapter. The second form says a plot of  $\log[R]$  against  $t$  is a straight line of slope  $-k/2.303$ .

**Trap:**

2.303 converts  $\ln$  to  $\log_{10}$ . If you use  $\ln$ , drop the 2.303 — using both is a factor-of-2.3 error.

● **C3.8 First order half-life**

$$t_{1/2} = \frac{0.693}{k}$$

| Symbol    | Meaning                   | Unit            |
|-----------|---------------------------|-----------------|
| $t_{1/2}$ | half-life                 | s               |
| $k$       | first order rate constant | s <sup>-1</sup> |
| 0.693     | $\ln 2$                   | dimensionless   |

**Use it when:**

Any half-life question on a first-order reaction, including all radioactive decay.

**Trap:**

**Independent of starting concentration.** Being asked to prove that is a standing 2-marker — see the derivations page.

○ **C3.9 First order gas-phase reaction followed by total pressure**

$$k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_t}$$

| Symbol | Meaning                          | Unit            |
|--------|----------------------------------|-----------------|
| $p_i$  | initial pressure of the reactant | bar or atm      |
| $p_t$  | total pressure at time $t$       | bar or atm      |
| $k$    | rate constant                    | $\text{s}^{-1}$ |

**Use it when:**

A gas decomposes into two gas molecules and the data is a table of total pressure against time — e.g. azoisopropane, or  $\text{N}_2\text{O}_5$  decomposition.

**Trap:**

The  $2p_i - p_t$  form holds only when one mole of gas gives two moles of gas. For a different stoichiometry, rebuild the pressure table from scratch.

● **C3.10 Rate constant against temperature**

$$k = A e^{-E_a/RT}$$

| Symbol | Meaning                            | Unit                                      |
|--------|------------------------------------|-------------------------------------------|
| $A$    | frequency (pre-exponential) factor | same as $k$                               |
| $E_a$  | activation energy                  | $\text{J mol}^{-1}$                       |
| $R$    | gas constant                       | $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ |
| $T$    | absolute temperature               | K                                         |

**Use it when:**

Temperature enters the question at all. The Arrhenius equation.

**Trap:**

$e^{-E_a/RT}$  is the **fraction of molecules with energy at least  $E_a$**  — quoting that meaning is often the mark.

○ **C3.11 Activation energy from a straight-line plot**

$$\log k = \log A - \frac{E_a}{2.303 RT}$$

| Symbol    | Meaning       | Unit          |
|-----------|---------------|---------------|
| slope     | $-E_a/2.303R$ | K             |
| intercept | $\log A$      | dimensionless |
| $x$ -axis | $1/T$         | $K^{-1}$      |

**Use it when:**

A graph of  $\log k$  against  $1/T$  is given, or asked to be sketched.

**Trap:**

The slope is **negative** and  $E_a$  is positive — remember to take the magnitude when you convert slope to  $E_a$ .

● **C3.12 Activation energy from two temperatures**

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right] = \frac{E_a}{2.303 R} \left[ \frac{T_2 - T_1}{T_1 T_2} \right]$$

| Symbol     | Meaning                           | Unit                 |
|------------|-----------------------------------|----------------------|
| $k_1, k_2$ | rate constants at $T_1$ and $T_2$ | same units           |
| $E_a$      | activation energy                 | $J \text{ mol}^{-1}$ |
| $T_1, T_2$ | absolute temperatures             | K                    |

**Use it when:**

Two rate constants at two temperatures are given. Also handles "the rate doubles when temperature rises by 10 K" — set  $k_2/k_1 = 2$ .

**Trap:**

Inside the bracket it is  $1/T_1 - 1/T_2$  — the **smaller** temperature first. Reversing it flips the sign of  $E_a$ .

**CH 4 d and f Block Elements — 2 formulas + rules · 11 marks**

*This chapter is mostly trends and reasons rather than calculation. Two things are computable; everything else is recall, collected in the rules table below.*

● **C4.1 Magnetic moment from unpaired electrons — and back again**

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

| Symbol        | Meaning                        | Unit               |
|---------------|--------------------------------|--------------------|
| $\mu$         | spin-only magnetic moment      | BM (Bohr magneton) |
| $n$           | number of unpaired electrons   | dimensionless      |
| $n = 1, 2, 3$ | gives $\mu = 1.73, 2.83, 3.87$ | BM                 |
| $n = 4, 5$    | gives $\mu = 4.90, 5.92$       | BM                 |

### Use it when:

Either direction: count unpaired electrons from the configuration and get  $\mu$ , or read  $\mu$  off the question and work back to  $n$ . Same formula serves Chapter 5.

### Trap:

$n$  is unpaired electrons in the **ion**, not the atom. For a transition metal ion, remove the 4s electrons first, then the 3d.

## ○ C4.2 General electronic configuration of each block

d-block:  $(n-1)d^{1-10} ns^{1-2}$     lanthanoid:  $4f^{1-14} 5d^{0-1} 6s^2$     actinoid:  $5f^{1-14} 6d^{0-1} 7s^2$

| Symbol | Meaning                                             | Unit          |
|--------|-----------------------------------------------------|---------------|
| $n$    | period number                                       | dimensionless |
| Cr     | exception: $3d^5 4s^1$ , half-filled is stabler     | dimensionless |
| Cu     | exception: $3d^{10} 4s^1$ , fully-filled is stabler | dimensionless |

### Use it when:

Writing a configuration, or explaining why Zn, Cd and Hg are not really transition metals — their  $d$  subshell is full in both atom and ion.

### Trap:

Electrons fill 4s before 3d but are **removed from 4s first**.  $\text{Fe}^{2+}$  is  $3d^6$ , not  $3d^4 4s^2$ .

## Rules, not formulas — recall these as facts

| Ask                                           | Answer                                                                                                                  |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Highest oxidation state in 3d series          | +7 in Mn (as $\text{KMnO}_4$ ); Cr reaches +6                                                                           |
| Most stable oxidation state across the series | +2 for most, becoming more stable to the right                                                                          |
| Why coloured                                  | d-d transition of an unpaired $d$ electron; $\text{Sc}^{3+}$ ( $d^0$ ) and $\text{Zn}^{2+}$ ( $d^{10}$ ) are colourless |
| Lanthanoid contraction                        | Steady size decrease across 4f, from poor shielding by $f$ electrons                                                    |
| Its main consequence                          | Zr and Hf (also Nb/Ta) have almost identical radii and are hard to separate                                             |
| Why alloys form readily                       | Similar atomic radii let one metal substitute for another in the lattice                                                |
| Why good catalysts                            | Variable oxidation states and the ability to adsorb reactants on the surface                                            |
| Most common lanthanoid oxidation state        | +3; actinoids show far more variability                                                                                 |

## CH 5 Coordination Compounds — 6 formulas · 11 marks

### ● C5.1 Oxidation number of the metal inside a complex

$$x + \sum (\text{ligand charges}) = \text{charge on the coordination sphere}$$

| Symbol  | Meaning                                                                                                      | Unit          |
|---------|--------------------------------------------------------------------------------------------------------------|---------------|
| $x$     | oxidation number of the central metal                                                                        | dimensionless |
| neutral | $\text{NH}_3$ , $\text{H}_2\text{O}$ , $\text{CO}$ , en — contribute 0                                       | dimensionless |
| anionic | $\text{Cl}^-$ , $\text{CN}^-$ , $\text{OH}^-$ contribute $-1$ ; $\text{C}_2\text{O}_4^{2-}$ contributes $-2$ | dimensionless |

#### Use it when:

Always — it is the first step of every naming, VBT and CFT question. In  $\text{K}_4[\text{Fe}(\text{CN})_6]$ :  $x + 6(-1) = -4$ , so  $x = +2$ .

#### Trap:

In  $[\text{Ni}(\text{CO})_4]$  the ligands are neutral and the sphere is neutral, so nickel is in oxidation state **zero**. This exact complex is the standing trap.

### ● C5.2 Coordination number when a ligand grips more than once

$$\text{CN} = \sum_{\text{ligands}} (\text{denticity}) = \text{number of donor atoms bonded to the metal}$$

| Symbol                                      | Meaning                                  | Unit          |
|---------------------------------------------|------------------------------------------|---------------|
| CN                                          | coordination number                      | dimensionless |
| en                                          | ethane-1,2-diamine — didentate, counts 2 | dimensionless |
| C <sub>2</sub> O <sub>4</sub> <sup>2-</sup> | oxalate — didentate, counts 2            | dimensionless |
| EDTA                                        | hexadentate, counts 6                    | dimensionless |

**Use it when:**

Working out geometry. [Co(en)<sub>3</sub>]<sup>3+</sup> has three ligands but a coordination number of **6**.

**Trap:**

Count **donor atoms**, not ligands. This is the difference between getting octahedral and getting the wrong geometry entirely.

● **c5.3 Tetrahedral splitting compared with octahedral**

$$\Delta_t = \frac{4}{9} \Delta_o$$

| Symbol     | Meaning                                   | Unit                                       |
|------------|-------------------------------------------|--------------------------------------------|
| $\Delta_o$ | octahedral crystal field splitting energy | J mol <sup>-1</sup> (or cm <sup>-1</sup> ) |
| $\Delta_t$ | tetrahedral splitting energy              | J mol <sup>-1</sup> (or cm <sup>-1</sup> ) |

**Use it when:**

Asked to relate the two, or to explain why tetrahedral complexes are essentially always high spin.

**Trap:**

Because  $\Delta_t$  is small, it is almost never bigger than the pairing energy — so **no low-spin tetrahedral complexes** in this syllabus. Also note the splitting is *inverted*: *e* below *t<sub>2</sub>*.

○ **c5.4 Crystal field stabilisation energy of an octahedral complex**

$$\text{CFSE} = (-0.4x + 0.6y) \Delta_o \quad \text{for } t_{2g}^x e_g^y$$

| Symbol   | Meaning                                          | Unit                                          |
|----------|--------------------------------------------------|-----------------------------------------------|
| <i>x</i> | electrons in the lower <i>t<sub>2g</sub></i> set | dimensionless                                 |
| <i>y</i> | electrons in the upper <i>e<sub>g</sub></i> set  | dimensionless                                 |
| CFSE     | crystal field stabilisation energy               | J mol <sup>-1</sup> (or units of $\Delta_o$ ) |

**Use it when:**

Comparing the stability of two configurations. Each *t<sub>2g</sub>* electron is stabilised by 0.4 $\Delta_o$ ; each *e<sub>g</sub>* electron is destabilised by 0.6 $\Delta_o$ .

**Trap:**

The barycentre must balance:  $3 \times 0.4 = 2 \times 0.6$ . If your two numbers do not satisfy that, you have the sets the wrong way round.

● **C5.5 High spin or low spin — deciding which**

$$\Delta_o > P \Rightarrow \text{low spin (pair up)} \quad \Delta_o < P \Rightarrow \text{high spin (stay unpaired)}$$

| Symbol       | Meaning                                                                                                           | Unit                |
|--------------|-------------------------------------------------------------------------------------------------------------------|---------------------|
| $\Delta_o$   | crystal field splitting energy                                                                                    | $\text{J mol}^{-1}$ |
| $P$          | pairing energy                                                                                                    | $\text{J mol}^{-1}$ |
| strong field | $\text{CN}^-$ , $\text{CO}$ , $\text{NH}_3$ — large $\Delta_o$ , low spin                                         | —                   |
| weak field   | $\text{I}^-$ , $\text{Br}^-$ , $\text{Cl}^-$ , $\text{F}^-$ , $\text{H}_2\text{O}$ — small $\Delta_o$ , high spin | —                   |

**Use it when:**

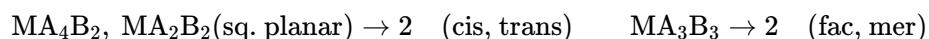
Predicting magnetic behaviour or hybridisation. Only matters for  $d^4$  to  $d^7$ ; below and above that there is no choice to make.

**Trap:**

Spectrochemical series, weakest to strongest:



○ **C5.6 How many geometrical isomers a complex has**



| Symbol | Meaning                                        | Unit |
|--------|------------------------------------------------|------|
| cis    | identical ligands adjacent ( $90^\circ$ )      | —    |
| trans  | identical ligands opposite ( $180^\circ$ )     | —    |
| fac    | three identical ligands on one triangular face | —    |
| mer    | three identical ligands around a meridian      | —    |

**Use it when:**

"What type of isomerism is shown by..." — check the formula shape first, then the geometry.

**Trap:**

**Tetrahedral complexes show no geometrical isomerism** — every position is adjacent to every other, so there is no "trans". Only the **cis** form of  $[\text{M}(\text{en})_2\text{X}_2]$  is optically active.

**CH 6 Haloalkanes and Haloarenes — 2 formulas + orders · 6 marks**

*An organic chapter, so almost everything is a reactivity order rather than an equation. The two computable things come first; the orders that actually earn the marks are collected below*

them.

● C6.1 The rate law that identifies the substitution mechanism



| Symbol        | Meaning                                             | Unit                                     |
|---------------|-----------------------------------------------------|------------------------------------------|
| $S_N2$        | second order, one step, transition state, inversion | $k$ in $\text{L mol}^{-1} \text{s}^{-1}$ |
| $S_N1$        | first order, two steps, carbocation, racemisation   | $k$ in $\text{s}^{-1}$                   |
| $[\text{RX}]$ | concentration of the alkyl halide                   | $\text{mol L}^{-1}$                      |

**Use it when:**

Asked for two differences between the mechanisms, or told how the rate responds to changing the nucleophile's concentration.

**Trap:**

The reactivity orders are **opposite**:  $S_N2$  runs  $1^\circ > 2^\circ > 3^\circ$  (steric hindrance),  $S_N1$  runs  $3^\circ > 2^\circ > 1^\circ$  (carbocation stability). Check which mechanism the question named before you answer.

○ C6.2 Degree of unsaturation from a molecular formula

$$\text{DoU} = \frac{2C + 2 + N - H - X}{2}$$

| Symbol    | Meaning                              | Unit          |
|-----------|--------------------------------------|---------------|
| $C, H, N$ | counts of carbon, hydrogen, nitrogen | dimensionless |
| $X$       | count of halogen atoms               | dimensionless |
| DoU       | rings plus $\pi$ bonds               | dimensionless |

**Use it when:**

A molecular formula is given and you must decide how many isomers or what skeleton is possible. Oxygen is ignored — it does not change the count.

**Trap:**

Halogen counts like hydrogen (subtract it). A benzene ring alone gives DoU = 4: three  $\pi$  bonds plus one ring.

## The orders that earn the marks

| Comparison                       | Order, and the reason                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------|
| Reactivity of RX to substitution | $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$ — C-X bond weakens down the group           |
| $S_N1$ rate                      | $3^\circ > 2^\circ > 1^\circ$ — carbocation stability; benzylic and allylic beat all three    |
| $S_N2$ rate                      | $\text{CH}_3\text{X} > 1^\circ > 2^\circ > 3^\circ$ — steric hindrance                        |
| Alcohol + HX                     | $3^\circ > 2^\circ > 1^\circ$ , and $\text{HI} > \text{HBr} > \text{HCl}$                     |
| Boiling point of isomers         | Falls as branching rises — less surface contact, weaker van der Waals                         |
| Melting point of dihalobenzenes  | para > ortho $\approx$ meta — para packs better in the crystal lattice                        |
| Free radical stability           | benzylic $\approx$ allylic > $3^\circ > 2^\circ > 1^\circ$ — resonance beats hyperconjugation |
| Aqueous vs alcoholic KOH         | Aqueous $\rightarrow$ substitution (alcohol); alcoholic $\rightarrow$ elimination (alkene)    |
| Markovnikov vs peroxide          | Peroxide effect works with HBr <b>only</b>                                                    |
| Saytzeff (NCERT: Zaitsev)        | The more substituted alkene is the major elimination product                                  |

## CONST Constants and conversions — *know the units too*

### ● K Every constant this paper can hand you, with its unit

| Symbol | Meaning                             | Value and unit                                 |
|--------|-------------------------------------|------------------------------------------------|
| $R$    | gas constant, energy form           | $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$      |
| $R$    | gas constant, pressure-volume form  | $0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$ |
| $F$    | Faraday constant                    | $96487 \text{ C mol}^{-1}$                     |
| $K_b$  | ebullioscopic constant, water       | $0.52 \text{ K kg mol}^{-1}$                   |
| $K_f$  | cryoscopic constant, water          | $1.86 \text{ K kg mol}^{-1}$                   |
| $N_A$  | Avogadro constant                   | $6.022 \times 10^{23} \text{ mol}^{-1}$        |
| 2.303  | converts $\ln$ to $\log_{10}$       | dimensionless                                  |
| 0.693  | $\ln 2$ , for first-order half-life | dimensionless                                  |
| 0.059  | $2.303RT/F$ at 298 K, for Nernst    | V                                              |
| $T$    | 0 °C in kelvin                      | 273.15 K                                       |
| STP    | molar volume of an ideal gas        | $22.4 \text{ L mol}^{-1}$                      |

### Trap:

Match  $R$  to the units you want out. Using 8.314 with pressures in atm, or 0.0821 when the answer should be in joules, is a whole-question error and not a rounding one.

Built from Sourabh Raina's one-shot and PYQ videos for Chapters 1-6, and cross-checked against NCERT Class XII Chemistry (Rationalised 2022-23). NCERT conventions used

throughout: **0.059** rather than 0.0591 in the Nernst equation,  $F = 96487 \text{ C mol}^{-1}$ , molar conductivity in  $\text{S cm}^2 \text{ mol}^{-1}$ , and **Zaitsev** (which NCERT notes is "also pronounced as Saytzeff"). The six chapter pages carry the teaching and the past-year questions; this page is only the recall layer. Derivations for the formulas that have them live on the companion page, **Chemistry, Derived**.

# Chemistry, Derived

Class XII CBSE · Chemistry · Chapters 1–3

*Twelve derivations, one algebraic move per line, each with the reason it is allowed. Every one ends in a formula that is already on Every Chemistry Formula — the point of this page is that you can rebuild them when memory fails, and that you can write the proof out when the paper asks for it.*

- Derivations: 12
- Figures: 12
- Chapters: 3
- Marks covered: 42

## How to use this

Chapters 4, 5 and 6 have no derivations worth the name — their marks come from recall and reasoning, and those live on the formula sheet. Everything derivable in this paper is on this page.

Read the **setup** first and draw the figure yourself before you look at the steps. A derivation you can only follow is not one you can write.

The *italic reason* beside a step is what turns it from an assertion into a proof. In the exam, those reasons are usually where the marks are.

## CH 3 Chemical Kinetics — 5 derivations · 13 marks

### D1 Integrated rate law for a zero order reaction — 2-3 marks

A reaction  $R \rightarrow P$  whose rate does not depend on the concentration of R at all. Start from the definition of rate, put the zero-order rate law into it, and integrate from the start of the reaction to time  $t$ . Write  $[R]_0$  for the concentration at  $t = 0$  and  $[R]$  for the concentration at time  $t$ .

**Figure.** Graph of concentration of R against time for a zero order reaction: a straight line starting at the initial concentration on the vertical axis and falling with constant slope minus  $k$ , meeting the time axis when all of R is used up.

*Zero order is the only order whose concentration-time plot is a straight line. Its slope is  $-k$  and its intercept is  $[R]_0$ , which is how both are read off experimental data.*

1. Rate  $= -\frac{d[R]}{dt}$  — (definition; minus sign because  $[R]$  falls)
2. Rate  $= k[R]^0 = k$  — (zero order: anything to the power 0 is 1)
3.  $-\frac{d[R]}{dt} = k$
4.  $d[R] = -k dt$  — (separating the variables)
5.  $\int_{[R]_0}^{[R]} d[R] = -k \int_0^t dt$  — (integrating between the two states)
6.  $[R] - [R]_0 = -k(t - 0)$
7.  $[R] - [R]_0 = -kt$

$$8. [R] = -kt + [R]_0 \text{ — (the straight line in the figure, } y = mx + c \text{)}$$

$$9. kt = [R]_0 - [R] \text{ — (rearranging for } k \text{)}$$

$$10. k = \frac{[R]_0 - [R]}{t}$$

**Result:**  $[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$

*Units check:  $k$  comes out as concentration  $\div$  time,  $\text{mol L}^{-1} \text{ s}^{-1}$  — which is exactly what C3.4 predicts for  $n = 0$ .*

## D2 Half-life of a zero order reaction — 2 marks

Half-life  $t_{1/2}$  is the time at which exactly half the reactant is left. Take the integrated law from D1 and substitute that one condition.

**Shared setup with D1.** Every half-life derivation is the integrated rate law with  $[R] = [R]_0/2$  and  $t = t_{1/2}$  substituted. If you can do D1 and D3, you can do D2 and D4 without learning anything new.

**Figure.** Zero order concentration against time, with the half-life marked: the concentration reaches half its initial value at a time equal to the initial concentration divided by twice  $k$ , and reaches zero at twice that time.

*Because the line is straight, the reaction finishes at exactly  $2t_{1/2}$  — a zero order reaction really does run out, which no first order reaction ever does.*

$$1. k = \frac{[R]_0 - [R]}{t} \text{ — (from D1)}$$

$$2. \text{ At } t = t_{1/2}, [R] = \frac{[R]_0}{2} \text{ — (definition of half-life)}$$

$$3. k = \frac{[R]_0 - \frac{[R]_0}{2}}{t_{1/2}} \text{ — (substituting both)}$$

$$4. k = \frac{\frac{[R]_0}{2}}{t_{1/2}}$$

$$5. k t_{1/2} = \frac{[R]_0}{2}$$

$$6. t_{1/2} = \frac{[R]_0}{2k}$$

**Result:**  $t_{1/2} = \frac{[R]_0}{2k}$  — directly proportional to the initial concentration

## D3 Integrated rate law for a first order reaction — 3 marks

Same reaction  $R \rightarrow P$ , but now the rate is proportional to  $[R]$ . The integration produces a natural logarithm, which is then converted to base 10 because every exam table and graph uses  $\log$ .

**Figure.** Graph of  $\log$  of concentration of  $R$  against time for a first order reaction: a straight line with intercept  $\log$  of the initial concentration and slope minus  $k$  over 2.303.

*A first order reaction is straight only after you take the logarithm. That is the practical test: if  $\log[R]$  against  $t$  is a line, the reaction is first order and  $k$  is  $2.303 \times$  the magnitude of its slope.*

$$1. \text{ Rate} = -\frac{d[R]}{dt} \text{ — (definition)}$$

$$2. \text{ Rate} = k[R]^1 = k[R] \text{ — (first order rate law)}$$

3.  $-\frac{d[R]}{dt} = k[R]$
4.  $\frac{d[R]}{[R]} = -k dt$  — (separating the variables)
5.  $\int_{[R]_0}^{[R]} \frac{d[R]}{[R]} = -k \int_0^t dt$
6.  $\ln[R] - \ln[R]_0 = -kt$  — (integral of  $1/x$  is  $\ln x$ )
7.  $\ln \frac{[R]}{[R]_0} = -kt$  — (law of logarithms)
8.  $\ln \frac{[R]_0}{[R]} = kt$  — (inverting the fraction flips the sign)
9.  $2.303 \log \frac{[R]_0}{[R]} = kt$  — ( $\ln x = 2.303 \log_{10} x$ )
10.  $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$
11. Rearranged for a graph:  $\log[R] = -\frac{k}{2.303}t + \log[R]_0$  — (the straight line drawn above)

**Result:**  $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$

Units check: the log is dimensionless, so  $k$  has units of  $1/\text{time}$ ,  $s_1^-$  — matching C3.4 for  $n = 1$ . That is also why the next derivation comes out independent of concentration.

#### D4 First order half-life, and why it ignores concentration — 2-3 marks

Substitute the half-life condition into D3. Watch what happens to  $[R]_0$  — the whole point of this derivation is that it cancels.

**Figure.** First order decay curve: concentration falls from its initial value to one half after one half-life, to one quarter after two half-lives and one eighth after three, with the three time intervals all equal.

Each successive halving takes the same time, no matter how little is left. That is the visual statement of the algebraic cancellation below, and it is why radioactive decay has a fixed half-life.

1.  $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$  — (from D3)
2. At  $t = t_{1/2}$ ,  $[R] = \frac{[R]_0}{2}$  — (definition of half-life)
3.  $k = \frac{2.303}{t_{1/2}} \log \frac{[R]_0}{[R]_0/2}$
4.  $\frac{[R]_0}{[R]_0/2} = 2$  — ( $[R]_0$  cancels — this is the whole result)
5.  $k = \frac{2.303}{t_{1/2}} \log 2$
6.  $\log 2 = 0.301$
7.  $k = \frac{2.303 \times 0.301}{t_{1/2}} = \frac{0.693}{t_{1/2}}$
8.  $t_{1/2} = \frac{0.693}{k}$

**Result:**  $t_{1/2} = \frac{0.693}{k}$  — independent of the initial concentration

Step 4 is the answer to "show that the half-life of a first order reaction is independent of initial concentration". Do not skip it; it is the mark.

### D5 Activation energy from rate constants at two temperatures — 3 marks

Start from the Arrhenius equation, take logarithms to make it linear, write it once at each of two temperatures  $T_1$  and  $T_2$ , and subtract. The frequency factor  $A$  is unknown and must disappear — that is what the subtraction is for.

**Figure.** Arrhenius plot:  $\log k$  against  $1/T$  is a descending straight line whose intercept is  $\log A$  and whose slope is minus the activation energy divided by  $2.303 R$ .

$T_2 > T_1$ , so  $1/T_2$  sits to the left of  $1/T_1$  and the higher temperature gives the higher  $\log k$ . Reading the axis backwards is the commonest error on this plot.

1.  $k = A e^{-E_a/RT}$  — (Arrhenius equation)
2.  $\ln k = \ln A - \frac{E_a}{RT}$  — (taking natural logs of both sides)
3.  $\log k = \log A - \frac{E_a}{2.303RT}$  — (converting to base 10)
4. At  $T_1$ :  $\log k_1 = \log A - \frac{E_a}{2.303RT_1}$
5. At  $T_2$ :  $\log k_2 = \log A - \frac{E_a}{2.303RT_2}$
6. Subtracting:  $\log k_2 - \log k_1 = -\frac{E_a}{2.303RT_2} + \frac{E_a}{2.303RT_1}$  — ( $\log A$  cancels — it is the same constant at both temperatures)
7.  $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right]$
8.  $\frac{1}{T_1} - \frac{1}{T_2} = \frac{T_2 - T_1}{T_1 T_2}$  — (common denominator)
9.  $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[ \frac{T_2 - T_1}{T_1 T_2} \right]$

**Result:**  $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[ \frac{T_2 - T_1}{T_1 T_2} \right]$

The bracket is  $\frac{1}{T_1} - \frac{1}{T_2}$ , smaller temperature first. If you get a negative  $E_a$ , you reversed it at step 7.

## CH 2 Electrochemistry — 3 derivations · 14 marks

### D6 The Nernst equation — 3 marks

A galvanic cell running the reaction  $aA + bB \rightarrow cC + dD$  with  $n$  electrons transferred. Two facts are combined: the thermodynamic relation between Gibbs energy and the reaction quotient, and the electrochemical relation between Gibbs energy and cell potential.

**Figure.** A galvanic cell: a zinc electrode in zinc sulphate solution on the left as the anode where oxidation occurs, a copper electrode in copper sulphate solution on the right as the cathode where reduction occurs, joined by a salt bridge that dips into both solutions and by an external circuit with a voltmeter through which electrons flow from left to right.

The Daniell cell,  $E_{\text{cell}}^\circ = 1.1 \text{ V}$ . The Nernst equation says what happens to that reading when the two solutions are not at  $1 \text{ M}$  — and the sign of the correction follows from Le Chatelier, not

from memory.

1.  $\Delta_r G = \Delta_r G^\circ + RT \ln Q$  — (thermodynamics: Gibbs energy away from standard state)

2.  $\Delta_r G = -nFE_{\text{cell}}$  — (electrical work done by the cell)

3.  $\Delta_r G^\circ = -nFE_{\text{cell}}^\circ$  — (same relation at standard state)

4.  $-nFE_{\text{cell}} = -nFE_{\text{cell}}^\circ + RT \ln Q$  — (substituting 2 and 3 into 1)

5. Divide throughout by  $-nF$ :

6.  $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q$

7.  $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{2.303RT}{nF} \log Q$  — (converting to base 10)

8. At 298 K:  $\frac{2.303 \times 8.314 \times 298}{96487} = 0.059$  — (substituting  $R$ ,  $T$  and  $F$ )

9.  $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{n} \log Q$

**Result:**  $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{2.303RT}{nF} \log Q \xrightarrow{298 \text{ K}} E_{\text{cell}}^\circ - \frac{0.059}{n} \log Q$

Step 8 is worth writing out in full — being asked where the 0.059 comes from is a common follow-up, and “ $2.303RT/F$  at 298 K” is the whole answer.

## D7 Equilibrium constant from standard cell potential — 2-3 marks

Take the Nernst equation and ask what it says at the one moment the cell stops working. A cell at equilibrium has no driving force, so it reads zero volts — and at that moment the reaction quotient has become the equilibrium constant.

**Shared setup with D6.** This is the Nernst equation with one substitution. Do not learn it as a separate formula.

**Figure.** Graph of cell potential against  $\log Q$ : a descending straight line of slope minus 0.059 over  $n$ , with intercept the standard cell potential at  $\log Q$  equals zero, crossing the horizontal axis at  $\log$  of the equilibrium constant where the cell is dead.

The cell potential falls linearly as products build up. Where the line crosses zero,  $Q$  has reached  $K_c$  — the battery is flat. Reading the intercept and the zero-crossing off this one line gives both results.

1.  $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{n} \log Q$  — (from D6)

2. At equilibrium the cell delivers no work, so  $E_{\text{cell}} = 0$  — (a dead cell)

3. At equilibrium  $Q = K_c$  — (definition of the equilibrium constant)

4.  $0 = E_{\text{cell}}^\circ - \frac{0.059}{n} \log K_c$

5.  $E_{\text{cell}}^\circ = \frac{0.059}{n} \log K_c$  — (moving the log term across — this is why the sign is now positive)

6.  $\log K_c = \frac{n E_{\text{cell}}^\circ}{0.059}$  — (rearranged for  $K_c$ )

7. Equivalently, from  $\Delta_r G^\circ = -nFE_{\text{cell}}^\circ$  and  $\Delta_r G^\circ = -RT \ln K_c$ :

8.  $nFE_{\text{cell}}^\circ = RT \ln K_c$  — (equating the two expressions for  $\Delta_r G^\circ$ )

**Result:**  $E_{\text{cell}}^\circ = \frac{0.059}{n} \log K_c \quad \log K_c = \frac{n E_{\text{cell}}^\circ}{0.059}$

## D8 Molar conductivity from conductivity — where the 1000 comes from — 2 marks

A conductivity cell of electrode area  $A$  and separation  $l$ . Conductivity  $\kappa$  is defined for a 1 cm cube of solution; molar conductivity  $\Lambda_m$  is defined for however much solution contains exactly one mole. The 1000 is nothing but a  $\text{cm}^3$ -to-litre conversion, and this derivation is really a units argument.

**Figure.** A conductivity cell: two parallel electrodes of area  $A$  separated by a distance  $l$ , with the cell constant defined as  $l$  divided by  $A$ , beside a unit cube of solution of side one centimetre whose conductance is the conductivity.

*Two different reference volumes:  $\kappa$  is per cubic centimetre,  $\Lambda_m$  is per mole. Converting between them is the entire content of the formula.*

1.  $\Lambda_m = \kappa \times V$  — ( $V$  = volume of solution containing exactly 1 mole of electrolyte)
2. Let the concentration be  $c$  mol per litre — (the unit every exam question uses)
3.  $c$  moles occupy 1 L, so 1 mole occupies  $\frac{1}{c}$  L
4. 1 L = 1000  $\text{cm}^3$  — (the conversion — this is the 1000)
5. 1 mole occupies  $\frac{1000}{c} \text{cm}^3$
6.  $\Lambda_m = \kappa \times \frac{1000}{c}$
7. Units:  $\text{S cm}^{-1} \times \text{cm}^3 \text{mol}^{-1} = \text{S cm}^2 \text{mol}^{-1}$  — (confirming the standard unit)

**Result:**  $\Lambda_m = \frac{\kappa \times 1000}{c}$  in  $\text{S cm}^2 \text{mol}^{-1}$

*In strict SI ( $\kappa$  in  $\text{S m}^{-1}$ ,  $c$  in  $\text{mol m}^{-3}$ ) step 4 disappears and  $\Lambda_m = \kappa/c$  in  $\text{S m}^2 \text{mol}^{-1}$ . The 1000 is a unit artefact, not physics.*

## CH 1 Solutions — 4 derivations · 15 marks

### D9 Relative lowering of vapour pressure equals the solute's mole fraction — 3 marks

A non-volatile solute B dissolved in a volatile solvent A. Only A can evaporate, so the vapour above the solution is pure A — but there is less of it, because solute particles occupy part of the surface. Start from Raoult's law and rearrange.

**Figure.** Two beakers side by side: pure solvent on the left with many molecules escaping from its surface, and a solution on the right whose surface is partly occupied by solute particles so fewer solvent molecules escape, giving a lower vapour pressure.

*Solute particles (filled circles) take up surface sites, so fewer solvent molecules can leave. The fraction of the surface still available to solvent is  $x_A$  — which is exactly what Raoult's law asserts.*

1.  $p_A = x_A p_A^\circ$  — (Raoult's law; the solute is non-volatile so  $p_{\text{soln}} = p_A$ )
2.  $p_A^\circ - p_A = p_A^\circ - x_A p_A^\circ$  — (subtracting both sides from  $p_A^\circ$ )
3.  $p_A^\circ - p_A = p_A^\circ(1 - x_A)$
4.  $x_A + x_B = 1$ , so  $1 - x_A = x_B$  — (mole fractions sum to one)
5.  $p_A^\circ - p_A = p_A^\circ x_B$
6.  $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B$  — (dividing by  $p_A^\circ$  — the left side is the relative lowering)

7.  $x_B = \frac{n_B}{n_A + n_B}$  — (definition of mole fraction)
8. For a dilute solution  $n_B \ll n_A$ , so  $n_A + n_B \approx n_A$
9.  $\frac{p_A^\circ - p_A}{p_A^\circ} \approx \frac{n_B}{n_A} = \frac{w_B/M_B}{w_A/M_A}$
10.  $\frac{p_A^\circ - p_A}{p_A^\circ} = \frac{w_B M_A}{M_B w_A}$  — (rearranged to give molar mass)

**Result:**  $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B \quad M_B = \frac{w_B M_A p_A^\circ}{w_A (p_A^\circ - p_A)}$

Steps 1–6 are exact. Step 8 is the only approximation, and it is only valid for a dilute solution — say so if you use it.

### D10 Elevation of boiling point is proportional to molality — 3 marks

A liquid boils when its vapour pressure equals the external pressure. A non-volatile solute lowers the vapour pressure at every temperature (D9), so the solution's curve sits below the solvent's — and has to be taken to a higher temperature before it reaches the 1 atm line.

**Figure.** Vapour pressure against temperature for pure solvent and for solution: the solution curve lies below the solvent curve everywhere, so it meets the one atmosphere line at a higher temperature, and the gap between the two crossing points is the elevation of boiling point.

The dashed curve is the solution. It never reaches 1 atm at the solvent's boiling point, so the temperature must be raised by  $\Delta T_b$  — and for a dilute solution the two curves are near enough parallel that the shift is proportional to how much vapour pressure was lost.

1. For a dilute solution, the lowering of vapour pressure is proportional to the elevation of boiling point:  $\Delta T_b \propto (p_A^\circ - p_A)$  — (the two curves are almost straight and almost parallel over this small range)
2.  $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B$  — (from D9)
3. So  $\Delta T_b \propto x_B$  — ( $p_A^\circ$  is a constant for a given solvent)
4.  $x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A}$  — (dilute solution)
5.  $\Delta T_b \propto \frac{n_B}{n_A}$
6.  $n_A = \frac{w_A}{M_A}$ , so  $\frac{n_B}{n_A} = \frac{n_B M_A}{w_A}$
7.  $M_A$  is a constant for a given solvent, so  $\Delta T_b \propto \frac{n_B}{w_A}$
8.  $\frac{n_B}{w_A \text{ in kg}} = m$ , the molality — (definition of molality)
9.  $\Delta T_b \propto m$
10.  $\Delta T_b = K_b m$  — (introducing the proportionality constant)
11.  $m = \frac{w_B \times 1000}{M_B w_A}$  — (molality written from masses,  $w_A$  in grams)
12.  $\Delta T_b = \frac{K_b w_B \times 1000}{M_B w_A}$
13.  $M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A}$

**Result:**  $\Delta T_b = K_b m \quad M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A}$

$K_b$  is the elevation for a 1 molal solution, and depends only on the solvent — for water  $0.52 \text{ K kg mol}^{-1}$ .

### D11 Depression of freezing point is proportional to molality — 3 marks

A solution freezes when the vapour pressure of the liquid equals that of the pure solid solvent. Adding solute lowers the liquid's vapour pressure, so the two curves now meet at a lower temperature. Structurally identical to D10 — only the meeting point changes.

**Shared setup with D10.** Both start from "vapour pressure is lowered by  $x_B$ " and end in "proportional to molality". If you can write one, change the meeting condition and you have written the other.

**Figure.** Vapour pressure against temperature near the freezing point: the solid solvent line crosses the pure solvent curve at the normal freezing point on the right, and crosses the lower solution curve at a lower temperature on the left, the gap between the two crossings being the depression of freezing point.

*Freezing happens where the liquid curve crosses the solid curve. Lowering the liquid curve moves that crossing to the left, and the shift is  $\Delta T_f$ . Note the solid curve is unchanged — solute does not dissolve in the solid solvent.*

1. A solution freezes when the vapour pressure of the liquid equals that of the pure solid solvent — (definition of freezing point)
2. For a dilute solution,  $\Delta T_f \propto (p_A^\circ - p_A)$  — (same near-linear argument as D10)
3.  $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B$  — (from D9)
4.  $\Delta T_f \propto x_B$
5.  $x_B \approx \frac{n_B}{n_A}$  — (dilute solution)
6.  $\Delta T_f \propto \frac{n_B}{w_A}$  — ( $M_A$  constant for a given solvent)
7.  $\Delta T_f \propto m$  — (definition of molality)
8.  $\Delta T_f = K_f m$
9.  $m = \frac{w_B \times 1000}{M_B w_A}$
10.  $M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A}$

**Result:**  $\Delta T_f = K_f m$        $M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A}$

$K_f$  for water is  $1.86 \text{ K kg mol}^{-1}$  — larger than  $K_b$ , which is why freezing-point depression is the more sensitive method for finding a molar mass.

### D12 Degree of dissociation from the van't Hoff factor — 2-3 marks

One mole of an electrolyte  $A_n$  that breaks into  $n$  particles, with a fraction  $\alpha$  actually dissociating. Colligative properties count *particles*, so build the particle total after dissociation and compare it with the total before.

**Figure.** Particle count before and after dissociation: one mole of solute starts as one particle, and after a fraction alpha dissociates into n particles the total becomes one minus alpha plus n alpha, whose ratio to the original is the van't Hoff factor.

Only the fraction  $\alpha$  splits; the rest stays whole. Counting both contributions is the entire derivation —  $i$  is just the ratio of the two particle totals.

1.  $A_n \longrightarrow n A$ , starting from 1 mole with a fraction  $\alpha$  dissociating
2. Moles of  $A_n$  left undissociated =  $1 - \alpha$
3. Moles of  $A$  produced =  $n\alpha$  — (each dissociated mole gives  $n$  particles)
4. Total moles of particles after =  $(1 - \alpha) + n\alpha$
5.  $i = \frac{\text{total particles after}}{\text{total particles before}} = \frac{1 - \alpha + n\alpha}{1}$  — (definition of the van't Hoff factor)
6.  $i = 1 - \alpha + n\alpha$
7.  $i - 1 = \alpha(n - 1)$  — (collecting the  $\alpha$  terms)
8.  $\alpha = \frac{i - 1}{n - 1}$
9. For **association**,  $n$  molecules combine into 1, so the total after is  $1 - \alpha + \frac{\alpha}{n}$
10.  $i = 1 - \alpha + \frac{\alpha}{n} = 1 - \alpha \left(1 - \frac{1}{n}\right)$
11.  $\alpha = \frac{1 - i}{1 - \frac{1}{n}}$

**Result:**  $\alpha_{\text{dissoc}} = \frac{i - 1}{n - 1} \quad \alpha_{\text{assoc}} = \frac{1 - i}{1 - \frac{1}{n}}$

Dissociation gives  $i > 1$  and association  $i < 1$ ; both formulas return a positive  $\alpha$  because the numerator flips with them.

Built from Sourabh Raina's one-shot and PYQ videos for Chapters 1-3, and cross-checked against NCERT Class XII Chemistry (Rationalised 2022-23) for constants and conventions —  $F = 96487 \text{ C mol}^{-1}$ ,  $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ ,  $K_b = 0.52$  and  $K_f = 1.86 \text{ K kg mol}^{-1}$  for water, and **0.059** rather than 0.0591 in the Nernst equation.

Chapters 4, 5 and 6 are not represented here because they contain no derivations — their marks come from trends, reasons and reactivity orders, all of which are on the companion page, **Every Chemistry Formula**.